Recall and the Scarring Effects of Job Displacement

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Abstract:

This paper studies the scarring effect of job displacement by explicitly considering workers who return to their previous employer. Using German administrative data I show that these "recalled" workers face larger earnings losses than non-recalled workers, especially in the short run. I develop a model that accounts for these heterogeneous effects by recall status. The estimated model shows that recalled workers' larger short-run losses are explained by differences in nonemployment duration. Long-term losses are similar because recalled workers' increased likelihood of repeated job loss is offset by the benefits of initially retaining their match and reduced skills depreciation during nonemployment.

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Workers who lose their job are exposed to a large and persistent loss in earnings compared to similar workers who did not lose their job (see e.g. Jacobson et al., 1993). The estimation of this "scarring" effect of job loss generally focuses on so-called "displaced workers", workers who lose their job through a mass layoff and do so permanently. However, it has also been documented that a large fraction of workers who lose their job will return to their former employer rather than move to a new employer.[1]

In this paper, I explore the long-run effects of displacement on earnings, estimating and explaining the effects separately for workers who are recalled to their previous employer and those who were not. As recalled workers account for approximately 13.8% of full-time workers separating from their job (and 7.3% of displaced full-time workers), and these workers are generally abstracted from in existing work, exploring these long-run effects by recall status both empirically and through a model contributes to a more complete understanding of the long-run effects of job loss. Furthermore, since recalled workers are not subject to all channels highlighted by the literature as explaining earnings losses, and may in particular not lose firm-specific human capital, earnings losses experienced by recalled workers are also informative on the sources of non-recalled workers' earnings losses.

To estimate the scarring effects of displacement by recall status, I use administrative employer-employee data from Germany. These data allow me to reliably identify individuals who are recalled to their former employing establishment. I separately estimate the scarring effect of displacement for recalled and non-recalled workers, rather than restricting the sample to omit recalls as the existing literature has generally done.[2] My baseline estimation uses the interaction-weighted estimator, as proposed by Sun and Abraham (2021), which allows me to account for the fact that workers displaced in different years may face different effects compared to the control group of never-displaced workers. Nevertheless, it should be noted that since the definition of recall is based on observations after the displacement, it is likely that there is selection into recall

[1] For example, Fujita and Moscarini (2017) found that these workers account for over 40% of U.S. employed workers who move into unemployment upon losing their job. For German workers, Mavromaras and Rudolph (1998) find slightly lower numbers, whereas Burda and Mertens (2001) find an incidence of more than 50% in their sample of (imputed) displacement. In Section 2.1 I show that I find recall rates ranging from 11 to 17% in my data after imposing full-time status, but before conditioning on displacement, which brings the rates down to 4 to 12%.

[2] For example, Lachowska et al. (2020) focused on "permanent separations", and Schmieder et al. (2023) omitted any worker who returns to work for the same employer in the first 10 years after displacement. While the empirical exercise in this paper focuses on displacement, in line with this existing work on the consequences of job loss, I discuss in Appendix C.3.3 how expanding the treatment group to all workers separating from their job (where the recall rate is higher) affects the results.

both along observable and unobservable dimensions. As such, one should be careful in interpreting the resulting empirical estimates.

Given that human capital losses have previously been argued to play a large role in explaining displaced workers' earnings losses, and some of this human capital is likely to be firm- or job-specific, one might expect recalled workers to experience lower earnings losses than non-recalled workers. I estimate that recalled workers instead experience larger earnings losses in the short run than displaced workers who are not recalled, both compared to a control group of never-displaced workers. While this difference shrinks over time, it does not fully disappear. In particular, 1 year after displacement a recalled worker earns approximately 35.5% less, whereas a non-recalled worker earns approximately 23.5% less, while 10 years after displacement the recalled and non-recalled worker earn approximately 21% and 25% less respectively. Estimating the effect on the employment fraction (fraction of the year spent in an employment relation) reveals that the larger short-run earnings loss is primarily driven by employment. Furthermore, I find that a recalled worker is more likely than a non-recalled worker to be separated from their job again in the first few years following the initial displacement (approximately 10 percentage points in the first year). Finally, I show that the estimated earnings losses of recalled and non-recalled workers mask a substantial amount of heterogeneity within each group. In particular, if one estimates the earnings losses only for workers who transition to a new job within 30 days, non-recalled workers suffer lower earnings losses than recalled workers (in the short run and in the long run), whereas conditioning on a transition period of more than 30 days yields the opposite result.

Next, I develop a rich model that can explain the differences in earnings losses experienced by recalled and non-recalled workers, while still also matching the average scarring effect of displacement. In particular, I explicitly distinguish between workers who are expecting to be recalled and workers who are not (interpreting them as two different states). In addition, I use elements that have been used in existing models to explain the average scarring effect of displacement, such as the aforementioned human capital and heterogeneity of firms by productivity and separation rates (as in Jarosch, 2023). By allowing recalled workers to follow a different path than non-recalled workers (starting before the recall materializes), I capture several possible explanations for the observed severity of the scarring effect of displacement for recalled workers, some of which may not be directly observable in the data. Furthermore, separating the paths at the time of displacement also allows me to incorporate a number of worker-driven selection channels.

I show that the model, estimated to match moments obtained from the data, accounts for the results from the empirical section despite not explicitly targeting them. I then use model simulations to decompose the difference in earnings loss between recalled and non-recalled displaced workers. I find that, in line with the empirical results, the recalled workers' earnings losses in the long run are primarily exacerbated by the instability of the job to which the worker is recalled, whereas an important factor in the short run is that non-recalled workers often transition immediately into their next job instead of spending some time in unemployment. On the other hand, recalled workers experience a lower human capital depreciation rate, do not "fall off the job ladder" at the time of their initial displacement, and may involuntarily select into the non-recall group after entering nonemployment, which offsets the negative influence of the aforementioned channels on the recalled workers' earnings loss (relative to that of non-recalled workers).

Comparing the decomposition of a recalled worker's earnings loss to that of a non-recalled worker's earnings loss I find that, while human capital depreciation may be important to explain the long-run earnings loss for non-recalled workers, its impact for recalled workers is close to zero. This is because recalled workers experience a low probability of human capital depreciation during their initial nonemployment spell. This, in turn, suggests that while a policy targeting a worker's loss of human capital during nonemployment (such as a retraining program) may be successful in bringing down the earnings losses for non-recalled workers, it will not be effective in alleviating recalled workers' earnings losses. Since these recalled workers already face a larger relative earnings loss on average than the non-recalled workers in the short run, it can therefore be concluded that such a policy would not be desirable.

In empirically investigating the impact of recalls on the long-term consequences of job loss (and mass layoffs in particular), I contribute to a substantial existing literature. This literature goes back to Jacobson et al. (1993), and generally finds that workers who are displaced suffer a large and persistent earnings loss, and highlights the important role of working hours in the short run and wages in the long run for explaining these average losses (Lachowska et al., 2020). Burda and Mertens (2001), Nedelkoska et al. (2015), and Schmieder et al. (2023) have found similar results in the context of Germany, whereas Bertheau et al. (2023) illustrate how effects differ between countries (although they do not include Germany). Recently, the focus of this literature has shifted towards investigating heterogeneity in this scarring effect of displacement. In particular, some recent work using data from Austria (Gulyas and Pytka, 2025) and Sweden (Athey et al., 2026) uses machine learning methods to consider a large number of dimensions at once, finding a particularly large role for worker age and schooling as well as pre-displacement firm characteristics such as the size and wage premium in explaining earnings losses. A related body of research focuses on one dimension at a time, thereby placing it closer to this paper's empirical sections, and explores how the consequences of displacement differ by pre-displacement earnings (Guvenen et al., 2017, Leenders, 2025), gender (Illing et al., 2024), age (Leenders and Wallenius, 2024), skills (Seim, 2019), and aggregate conditions (Davis and Von Wachter, 2011, Schmieder et al., 2023), among others. The group of subsequently recalled workers is generally omitted from this analysis, either by focusing on establishment closures (e.g. Athey et al., 2026) or by explicitly excluding any worker who returns within a certain number of years (e.g. 10 years in Gulyas and Pytka, 2025, Schmieder et al., 2023, and Illing et al., 2024). By focusing on ex-post recall status, and thus focusing on a group of workers that is traditionally omitted from the analysis, my paper therefore enriches this literature beyond adding another dimension of heterogeneity underlying the commonly estimated average cost of job loss.

Additionally, the empirical section of this paper contributes to the growing literature analyzing the incidence and consequences of recalls. The topic of recall has been studied quite extensively, going back to studies such as Feldstein (1976) and Katz (1986), with recent work in Nekoei and Weber (2015) and Nekoei and Weber (2020) distinguishing between the expectation of recall and the actual materialization of recall. The topic has gained additional attention during the Covid-19 pandemic, with studies like Hall and Kudlyak (2022) and Forsythe et al. (2022) highlighting the unusually large role recalls played in labour market dynamics during early months of the pandemic. In the context of Germany, the incidence of recall has been analyzed by Mavromaras and Rudolph (1998), who focus in particular on the distinction between recall after a layoff and recall after an employment interruption (during which the connection between employer and employee is not legally severed), finding that the former tends to be more common among smaller firms. Using a subset of the data used in my paper, Jost (2022) highlights the interplay of recall and fixed-term contracts, finding that recalls are especially common for workers coming off a fixed term contract. However, while the incidence of recall has therefore been studied in some detail before, the literature has been limited in its discussion of how recalled workers differ from non-recalled workers in terms of their subsequent earnings and other related outcomes. In this paper, I contribute to this literature by focusing on providing a reliable estimate of the consequences of displacement for subsequently recalled workers.

This paper also contributes to the literature providing theoretical analysis of the long-term consequences of displacement, and does so by distinguishing between recalled and non-recalled workers. The theoretical analysis of the long-term consequences of displacement has only recently started gaining more attention, after Pries (2004) and Davis and Von Wachter (2011) noted that a standard job search model cannot generate the large losses observed in the data. Some recent work has attempted to resolve this issue with some success. The paper closest to mine in terms of the model is Jarosch (2023), who proposes a model in which firms differ not only in terms of productivity, but also in their separation rate, thereby allowing for workers to experience several subsequent displacements after the initial one (as observed earlier by Stevens, 1997). This, combined with depreciation of human capital, enables the model to reproduce the average earnings loss after displacement. Other models that have been successful in replicating the average earnings loss after displacement include Krolikowski (2017), Huckfeldt (2022), Jung and Kuhn (2019), Burdett et al. (2020), and Gregory et al. (2025).[3]

Finally, I contribute to the theoretical analysis of recalls by explicitly distinguishing between nonemployed workers expecting a recall and not expecting a recall. The existing body of literature that builds the possibility of recall into a model goes back to early work such as Feldstein (1976), Pissarides (1982), and Katz and Meyer (1990). More recently, recall has been explicitly modeled in Fujita and Moscarini (2017), Albertini et al. (2023), and Gertler et al. (2022).[4] However, the existing work generally focuses exclusively on the impact of recall on labour market flows, and refrains from commenting on how workers' earnings are affected. Furthermore, the way recall is often modeled is by considering the current job to be "paused" while the worker is unemployed.[5] In my model, this is not quite the case, as I make a sharp distinction between workers with a potential recall and other unemployed workers, thus allowing for divergence between eventually recalled and non-recalled workers before the recall actually materializes.

The rest of this paper is organized as follows: Section 1 describes the data and method-

[3]Huckfeldt (2022) shows that workers who switch occupations suffer larger losses than workers who stay in their former occupation. This may seem to contradict my result that recalled workers face similar earnings losses as non-recalled workers, but this is not quite the case. In particular, for workers to be considered recalled in my setting, they do not necessarily have to return to the same occupation (although most of them do, as I show in Table 1). Indeed, as I do not explicitly consider occupational switching after displacement, this can be considered as one of the dimensions of heterogeneity that are still masked by some of my estimated effects (including those by ex-post recall status).

[4]In the context of the Covid-19 pandemic, the possibility of recall is also explicitly modeled in Gregory et al. (2020) and Gallant et al. (2020).

[5]The exception to this is the model in Gertler et al. (2022), which separately considers workers in temporary unemployment. My model differs from theirs in many dimensions. In particular, their model focuses primarily on the decisions made on the firm side, and therefore incorporates endogenous separations as well as the firm's choice of whether to place a worker on temporary layoff over the business cycle. On the other hand, my model focuses primarily on the worker side, and therefore does not feature the decision by the firm (or a business cycle), but does explicitly include the worker's choice between unemployment types and allows for the worker to search while expecting a recall.

ology used to generate the empirical results, which are presented in Section 2. Section 3 then presents the model, while Section 4 discusses its calibration. Section 5 presents the quantitative results from the model, after which Section 6 concludes.

1 Data and Estimation Methods

Throughout this paper I use administrative data from the German Federal Employment Agency's (BA) Institute for Employment Research (IAB). In particular, I use the Sample of Integrated Labour Market Biographies, SIAB (Antoni et al., 2019b), which draws a 2% random sample of private sector worker histories (for workers employed between 1975 and 2017), whose observations are matched with the relevant establishment data.[6] Each observation in the original data represents one spell of employment or non-employment, and is marked by a start and end date. From this spell-level data I construct a yearly and quarterly linked employer-employee dataset, in which the establishment information is used from the establishment at which the individual was employed on the first day of the year/quarter.[7] Further restricting observations to individuals aged between 25 and 60 leads to a large dataset which nevertheless has some gaps in some workers' time series. These gaps mainly occur because individuals are not observed if they are employed for the government, self-employed, or not receiving any social security benefits during nonemployment.[8] The resulting (yearly) analysis dataset includes roughly 24.4 million observations. Further summary statistics on both workers and establishments, both across the entire sample and by ex-post recall status, are presented in Section 2.1, as well as in Appendix C.1 and C.2.

For the purpose of estimating the specification described below, I define a worker as separated in some period t if the worker's employment spell with their establishment ends in period t. This means that the worker either no longer works for the same establishment in period $t + 1$ or returned to the establishment after being away for more than 31 days. Throughout, I focus in particular on workers whose social security notification indicates that employment at the estab-

[6] In the data, an establishment is defined as all locations of a firm within a municipality (Gemeinde) that belong to the same economic class.

[7] To be specific, the yearly dataset is used to obtain the empirical results from Section 2, whereas I use the quarterly dataset to obtain the moment values used to estimate the model in Section 4. If the individual is non-employed at the start of the year/quarter (or employed at multiple establishments), the information is used for the establishment from which the individual has the highest earnings in that period.

[8] Other reasons for not observing an individual include working (and moving) abroad, and being short-time employed. See Antoni et al. (2019a) for a detailed description of missing observations, sampling procedures, and variables included in the dataset.

lishment was ended for a reason that could point to displacement.[9] I then define such a worker as displaced if the establishment either closes or experiences a mass layoff.[10] Following the literature, an establishment is defined to experience a mass layoff if the employment at the establishment in the next period is at most 80% of the establishment's maximum employment over the previous five years, and the establishment has a net outflow of at least 20% of its workforce in the displacement year.

In order to determine whether a worker was recalled to their previous establishment, I look ahead at most 5 years after separation[11]. If the worker's first full-time employing establishment after separation is the same as her (full-time) employing establishment before displacement, I define the worker as recalled. Alternatively, a worker is also considered recalled if the worker started at a different establishment within 31 days, but returned to their pre-separation establishment (full time) either in the same year as the separation or in the next year (with the recalling establishment being the worker's main employer in the next year).[12] This should be thought of as a fairly narrow definition of a recall, as a number of potentially relevant cases are not considered. First of all, since I cannot identify whether two different establishments belong to the same firm, I do not consider workers who move to a different establishment within the same firm as recalled. Secondly, since I condition on full-time status before and after the separation, I miss workers who come back on a part-time contract or moved to a part-time status prior to separation. Finally, due to my definition of separation, I will miss direct recalls with nonemployment spells of less than 31 days. As workers with such a spell are not marked as separated, they can also not be marked as recalled.

Traditionally, the literature focuses on displacements rather than all job separations in

[9]For example, I exclude "separations" that are caused by maternity leave or sick pay. In contrast, in order for a worker to be "separated" I only require the notification to indicate that the employment spell was terminated (temporarily or permanently, for any reason). Throughout the analysis I drop trainees, casual workers, and partially retired workers.

[10]I use an extension file that clarifies the reason for an establishment leaving the sample. In particular, I do not consider an establishment to be closed if a large portion of the workers at the establishment finds employment at a common establishment after the closure. See appendix C.2 for more details.

[11]While I look ahead for up to 5 years in order to determine whether a worker was recalled, the majority of the recalls take place within the first year: 81.9% of all recalls within these 5 years materialize in the first year. This suggests that looking ahead for slightly less (or more) than 5 years is not likely to substantially alter the sample of recalled workers.

[12]I consider this second group to be (indirectly) recalled, as these workers will likely have taken employment at a different establishment only for the purpose of bridging the gap until the recall materializes. A worker who is separated from a closing establishment will always be in the non-recalled group of workers. As I show in Appendices C.3.1 and C.3.3, however, excluding these workers does not substantially alter the recall rate or empirical estimation results.

order to avoid estimates being influenced by selection of workers into separation, e.g. due to poor performance (which may carry over into their next job and therefore influence earnings). While using displacements alleviates this particular concern of selection effects, it is important to note that separating the sample of displaced workers by recall status may re-introduce selection effects along a number of dimensions. For example, it may be the case that recalled workers were the best-performing workers among the set of displaced workers. On the other hand, it may be the case that by the time the recall materializes only the lower quality workers remain available, and recalled workers are therefore selected on their short-term experience after the displacement. Similarly, there may be characteristics of the establishment (observed or unobserved) that influence how likely it is that an establishment recalls workers and which type of worker they would recall. Since many of these selection channels are along unobservable dimensions, it is unlikely that these could be fully addressed empirically. Furthermore, since the selection channels mentioned here go in different directions, it is unclear in which direction any possible bias would go. As such, one should be careful not to interpret empirical estimation results based on these groups as providing any causal evidence of the effect of a recall (conditional on displacement). Rather, they should be interpreted as an indication of how the experiences these two groups of displaced workers differ, keeping in mind that the composition of the two groups is influenced by selection. In a model such as the one presented in Section 3, on the other hand, I can incorporate some of these selection channels, thus allowing me to account for this selection when explaining the data through the lense of the model.

The empirical estimation results presented in the next section are based on the interaction-weighted estimator from Sun and Abraham (2021). This estimator allows the average effect of displacement to be different for workers displaced in different years, while all effects are still estimated in a single estimation (so that displaced workers do not appear in the control group in a different year). In practice, I estimate the average effect of displacement using equation (1), whereas equation (1') is used to estimate the effect of displacement by recall status:

$$e_{it} = \alpha_i + \gamma_t + \beta X_{it} + \sum_{C \neq 0} \sum_{\substack{k=-4 \\ k \neq -2}}^{K} \delta_k^C D_{it}^{C,k} + u_{it} \tag{1}$$

$$e_{it} = \alpha_i + \gamma_t + \beta X_{it} + \sum_{r=\{0,1\}} \sum_{C \neq 0} \sum_{\substack{k=-4 \\ k \neq -2}}^{K} \delta_k^{C,r} D_{it}^{C,r,k} + u_{it} \tag{1'}$$

In equation (1), i refers to the individual and t to the year. The dependent variable, e_{it}, refers to the outcome variable of interest. In most cases, this is the individual's yearly earnings or the fraction of the year spent in an employment relationship. Other outcome variables include the (yearly) job loss rate and the (yearly) average daily wage.[13] Explanatory variables include an individual fixed effect α_i and a time fixed effect γ_t, as well as an error term u_{it}. Additionally, I follow the literature by controlling for a quartic in age βX_{it}, where β is a vector containing the corresponding four coefficients. The variables of interest are a series of indicator variables $D_{it}^{C,k}$, which equal 1 if individual i from cohort C was displaced in period $t-k$. A cohort C is defined as the calendar year of displacement, with $C = 0$ corresponding to the cohort of individuals who I do not observe being displaced at all. The set of dummy variables for this "never-treated" group is omitted, implying that this group acts as the control group. I follow the discussion in Borusyak et al. (2024) by omitting two values of k. This is because generally the set of relative time indicators $D_{it}^{C,k}$ is collinear with itself as well as with the time fixed effect. In order to allow for anticipation one period ahead, the first period I omit is $k = -2$. The second omitted period is the earliest period, $k = -5$, thereby maximizing the distance between the two omitted periods and making the results less sensitive to fluctuations in these periods. For treated individuals ($C \neq 0$), observations for $k \notin [-5, K]$ are dropped.

When I analyze the effects of displacement by recall status, I estimate equation (1'). This equation is an adjusted version of equation (1), in which I allow the effect of displacement to be different for workers with a different ex-post recall status, as denoted by superscript r on the indicator variable $D_{it}^{C,r,k}$ and its corresponding coefficient $\delta_k^{C,r}$.

Estimation of equation (1) yields a set of estimates $\hat{\delta}_k^C$ for all $C \neq 0$ and $k \neq \{-5, -2\}$, which indicate the absolute gain (or loss) in the outcome variable of interest compared to the control group. In order to interpret these coefficients, I divide them by the average value of the outcome variable for the control group in the corresponding year t.[14] These relative coefficients are then averaged over C, using a weighted average that assigns each pair (C, k) a weight equal to the number of observations with (C, k) divided by the number of observations of relative time period k (across cohorts). Since all coefficients δ_k^C are estimated in a single estimation procedure,

[13]Note that my measure of earnings includes only earnings subject to social security. Among others, this does generally not include severance payments.

[14]Note that I only do this for nonbinary outcome variables such as earnings, employment fraction, and daily wage. For binary variables, such as job loss, I skip this step and present the averaged raw coefficients instead.

I can also form (point-wise) confidence intervals for the resulting values $\hat{\delta}_k$.[15]

When estimating equation (1) or (1'), I follow the literature by restricting my sample of displaced workers to individuals with an establishment tenure (prior to displacement) of at least 6 years (to ensure attachment to the job), and working at establishments with at least 50 employees (to avoid classifying a job loss as a mass layoff when only a limited amount of workers lose their job).[16] Furthermore, I drop individuals who were displaced more than once according to this restricted definition, and omit workers from the control group if I never see them satisfy these restrictions.[17] My estimation differs from the existing literature in using the interaction-weighted estimator rather than a "standard" two-way fixed effects estimation. As I show in Appendix C.3.2 (and briefly discuss below), this leads me to estimate larger and more persistent earnings and employment losses, but it does not have a major effect on the comparison between recalled and non-recalled workers.

2 Empirical Results

2.1 The Incidence of Displacement and Recall

Before analyzing the effect of displacement on earnings, it is worth investigating how common a separation or displacement event (as well as subsequent recall) is. This subsection presents separation, displacement, and recall rates for the entire sample. Unless specified otherwise, all graphs and tables in this subsection are generated using a "full sample", on which I have not applied the restrictions on pre-displacement establishment tenure and establishment size used to generate the "estimation sample" on which equation (1) is estimated.

First of all, Figure 1 displays the separation and displacement rates over time for the full sample and the estimation sample. For the full sample, the separation rate averages roughly 10.3%

[15] After estimating equation (1'), the same procedure is followed for recalled and non-recalled workers separately, thus resulting in two sets of values, $\hat{\delta}_k^r$ for $r = \{0, 1\}$.

[16] Note the difference between establishment and firm tenure. Establishment tenure is constructed using observed spells in the data and does not reset if a worker returns after being away. Firm tenure, which I do not observe since I cannot link different establishments of the same firm, informs eligibility for severance pay (among others) and may reset after being away from the firm. Because I observe all (private-sector) establishments at which any of the sampled workers were ever employed (in the years covered by the data), the data is likely to be biased towards larger establishments. The restriction on establishments having at least 50 employees further amplifies this bias.

[17] In Appendix C.3.3, I show how the results presented below are affected by requiring only 1 year of pre-displacement establishment tenure. Results including workers displaced from slightly smaller establishments or workers with multiple displacements (considering their first displacement as the treatment) are virtually identical to the main results, and are available upon request.

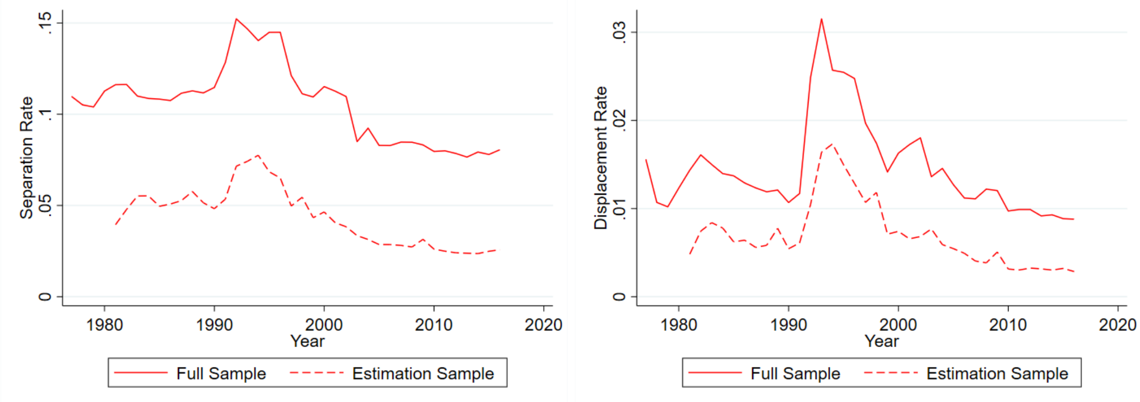

Figure 1: *The incidence of separation (left) and displacement (right) over time, plotted separately for the full sample of employed workers (solid) and the regression estimation sample (dashed).*

whereas the displacement rate is approximately 1.4% on average. For the estimation sample the separation and displacement rates are much lower, averaging roughly 4.2% and 0.7% respectively. All rates display substantial variation over time, and in particular the aftermath of the German reunification in 1990 is clearly visible.[18] While separation and displacement rates peak around recessions, these peaks are relatively mild in magnitude. For example, separation and displacement rates increased during the Great Recession but remained below pre-2005 levels.

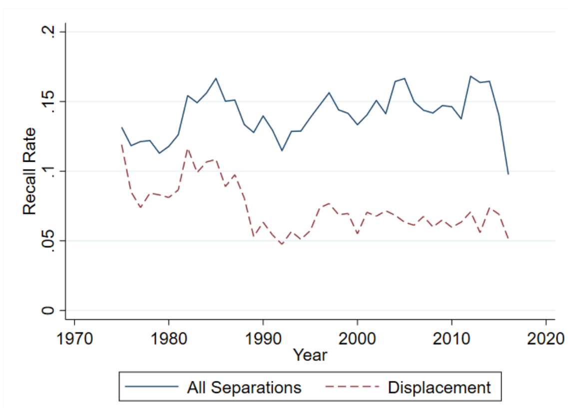

Figure 2: *The incidence of recall within 5 years of job loss over time, for all separations (solid) and conditional on displacement (dashed).*

Next, Figure 2 shows the incidence of recall within 5 years, unconditionally (for all separations) or conditional on displacement. The unconditional incidence of recall generally fluctuates

[18]Note that workers from East Germany are generally not included in the data before the reunification, so therefore the jump in separation and displacement rates can also partially be explained as a composition effect.

between 11% and 17%, which is much lower than the incidence found in the US by Fujita and Moscarini (2017), and also roughly 10 percentage points below the incidence reported for Germany in the 1980s by Mavromaras and Rudolph (1998). This is likely to be explained partially by my fairly narrow definition of a recall, in which I require both pre- and post-separation spells at the firm to be full-time, do not include recalls that materialize within 30 days, and cannot identify recalls to a different establishment of the same company. The recall rate conditional on displacement is lower, and fluctuates between 4% and 8% in recent decades. This indicates that generally roughly 6 to 7 percent of the workers who are displaced (including workers whose employing establishment shut down) return to their previous employer. In Appendix C.3.1, I discuss how displacement and recall rates vary over the earnings distribution, and show how observed rates are affected by removing establishment closures or indirect recalls.

	Recalled		Non-Recalled		Full Sample	
	Mean	(Std.Dev.)	Mean	(Std.Dev.)	Mean	(Std.Dev.)
Gender (female)	0.298	(0.46)	0.439	(0.50)	0.464	(0.50)
Age	39.53	(9.36)	38.66	(9.00)	41.25	(9.93)
Education (university)	0.063	(0.24)	0.119	(0.32)	0.155	(0.36)
Establishment size	309.9	(1,721)	611.0	(2,211)	1,147	(4,610)
Establishment tenure (days)	1,057	(1,301)	1,325	(1,694)	2,242	(2,253)
Yearly earnings (2015 Euros)	18,891	(14,383)	22,821	(18,849)	27,349	(20,621)
Occupational Change (3-digit)	0.151	(0.36)	0.367	(0.48)	-	-

Table 1: *Summary statistics by displacement and recall status. The table shows the estimated mean and standard deviation of a number of important variables, measured the year before displacement takes place (if relevant).*[19]

Table 1 provides an indication of how different recalled and non-recalled workers are, compared to each other and compared to the full sample.[20] The table shows that recalled workers are more likely than non-recalled workers to be male, have a lower education level, come from a smaller establishment, and tend to have lower yearly earnings from employment. Furthermore, non-recalled workers are more likely to switch occupations (at the 3-digit level) upon being displaced than recalled workers, indicating that recalled workers often (but not always) return to a

[19]The number of observations underlying the statistics depicted in Table 1 varies within each column because observations may be missing in specific variables for a variety of reasons (such as nonemployment). Specifically, the underlying number of observations for each of the columns varies between 11,158 and 15,155, between 160,656 and 200,014, and between 20,117,996 and 24,413,668.

[20]Note that the "Full Sample" in Table 1 also includes full-year nonemployed workers, with the exception of the summary statistics on establishment-related variables, which are missing for these workers. The "Full Sample" thus also includes workers who will not be in either of the treatment and control group in the regression-based analysis.

similar job when being recalled. This continues to hold if I focus on 1-digit occupations, where the occupational mobility rate is 7.9% for recalled and 20.5% for non-recalled workers.

	Time to re-employment				
Time to re-separation	< 1mo	1-3mo	3-6mo	6-12mo	> 12mo
< 1 month	0.018	0.023	0.018	0.020	0.008
1-3 months	0.031	0.043	0.039	0.056	0.018
3-6 months	0.049	0.040	0.058	0.061	0.022
6-12 months	0.056	0.057	0.172	0.042	0.038
> 12 months	0.004	0.002	0.029	0.048	0.048

Table 2: *Distribution of displaced and (ex-post) recalled workers over different combinations of time to re-employment and time to re-separation (in months). The table lists the fraction of workers falling into the relevant combination of categories, relative to the total number of displaced and subsequently recalled workers for whom both re-employment and re-separation is observed.*

One potential concern when discussing the incidence (and consequences) of recall is that these recalls may be driven by seasonal workers. In Table 2, I document how displaced and subsequently recalled workers are spread across different categories of time to re-employment and time to re-separation (measured from the point of re-employment).[21] A comparable tabulation for non-recalled workers can be found in Appendix C.3.1. If the recalls reflect seasonality, I would expect the time to re-employment and the time to re-separation to add up to approximately a year, with time to re-employment being at least 3 months and at most 12 months. Based on Table 2, this applies for up to a third of recalled workers. I show in Appendix C.3.3 that the estimation results discussed below are not driven by these potentially seasonal workers.

2.2 The Average Scarring Effect of Job Loss

Figure 3 shows the estimates of the average scarring effect of displacement on earnings and employment (fraction of the year employed), obtained using the interaction-weighted estimator from Sun and Abraham (2021), as described in Section 1. Workers who are displaced lose roughly 20-25% of their earnings in the short run. This earnings loss is shown to be quite persistent, with these displaced workers still earning almost 20% less 10 years after the job loss took place. For employment fraction a similar pattern arises, with short-run losses of approximately 20% and long-run losses of more than 10%. Comparing these estimates to those obtained in the existing literature, generally using two-way fixed effects estimation, reveals that these results are consistent with the

[21] Workers who are listed as being re-employed within one month are by definition workers who were indirectly recalled. Using time to recall instead of time to re-employment yields similar conclusions, as seen in Appendix C.3.1.

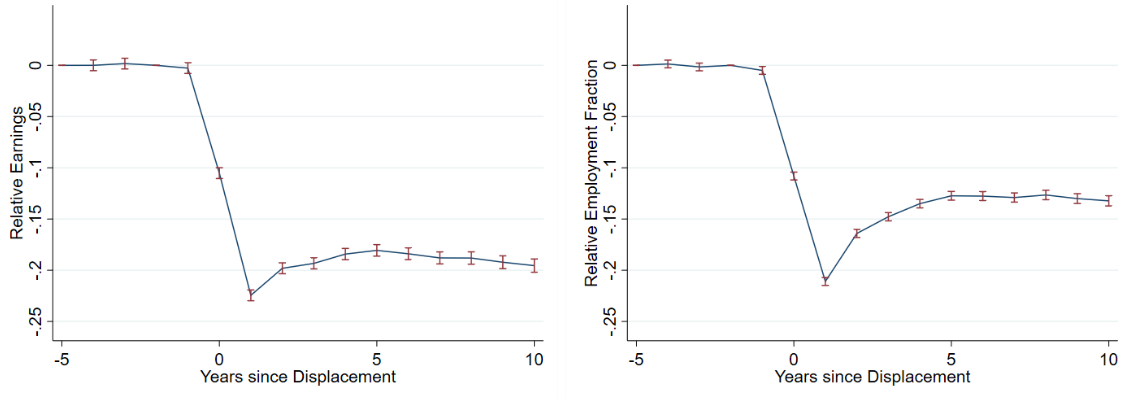

Figure 3: *The effect of displacement on earnings (left) and employment fraction (right), relative to the control group of never-displaced workers, using estimated coefficients from equation* (1). *The error bars correspond to 95% pointwise confidence intervals*

literature (e.g. Schmieder et al., 2023) for short-run losses, but not for long-run losses: While existing estimates generally suggest that employment fully recovers after roughly 15 years and earnings recover substantially, the interaction-weighted estimator reveals that the recovery stagnates after roughly 5 years, with both earnings and employment remaining substantially below that of the control group. This difference is quite striking, but should be interpreted carefully given the difference in the estimation method. In particular, while the interaction-weighted estimation used in this paper implies that the control group consists of never-displaced workers, estimates using a standard TWFE estimation method can be interpreted as relative to workers who were not displaced *in a particular year of interest*. As some of those workers may be displaced in a later year, this leads to a stronger estimated recovery, as pointed out in Krolikowski (2018). In Appendix C.3.2, I show that using an alternative estimation method in which year-specific datasets of displaced and control workers are stacked (as done in Flaaen et al., 2019, Schmieder et al., 2023, and Jarosch, 2023, among others) before estimating the effects of displacement earnings and employment indeed leads to estimated long-run losses roughly in line with those found in the literature.

2.3 The Scarring Effect of Displacement for Recalled Workers

In Figure 4, I show how the effects of displacement on employment and earnings differs by ex-post recall status.[22] The results show that workers who are recalled suffer from larger earnings

[22] As I do not observe whether a worker expects to be recalled, I divide workers according to whether or not a recall materializes within 5 years of the job loss. This may not exactly line up with whether a worker expected to be recalled, but given the correlation between the recall rate and the recall expectations (see e.g. Nekoei and Weber, 2015) it serves as a good proxy.

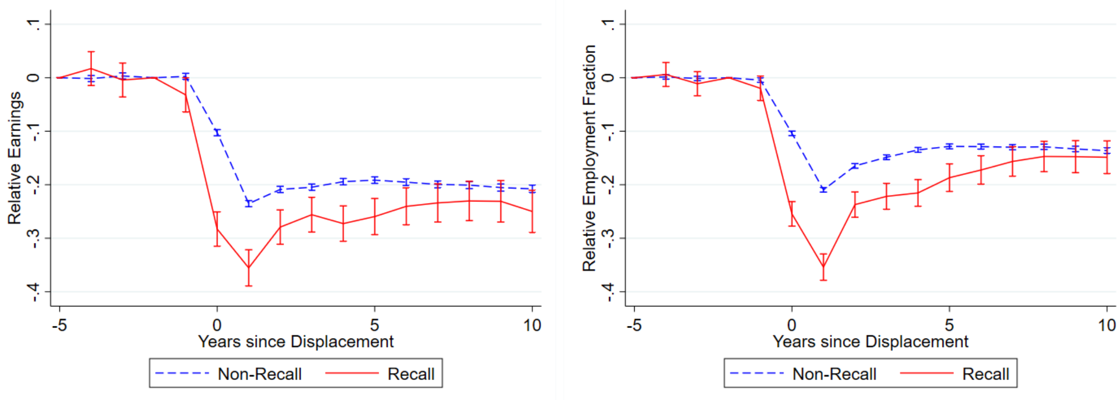

Figure 4: *The effect of displacement on earnings (left) and employment fraction (right) by ex-post recall status, relative to the control group of never-displaced workers, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals.*

losses compared to non-recalled (displaced) workers, especially in the short run. Similarly, recalled workers tend to do worse in the short run when it comes to days employed in the year. In fact, the short-run difference between recalled and non-recalled workers is slightly larger for employment than it is for earnings, thus suggesting that this difference is partially offset by recalled workers doing better in terms of wages in the short run. In Appendix C.3.4, I show the estimated effect on daily wages as well as a number of other variables not discussed in this section (such as the occupational mobility rate). Furthermore, I show that the result from Figure 4 continues to hold when I estimate it separately for a number of subsamples in which workers are similar along the dimensions highlighted in Table 1. This is particularly relevant in light of potential concern about selection into recall affecting the results, as discussed in Section 1. Finally, I show in Appendix C.3.3 that the result is robust to changes in the definition of displacement or recall.

In Figure 5, I show the estimated effect of ex-post recall status on subsequent job loss rates. Recalled workers are more likely to be separated again shortly after being recalled. While non-recalled workers are roughly 8.1 percentage points more likely to be separated than the control group one year after their initial displacement (and 6.3 and 6.2 percentage points more likely two and three years after displacement), recalled workers are more than 18 percentage points more likely than the control group to be separated again in the first year after displacement (and 10.5 and 7.9 percentage points more likely two and three years after displacement). This seems to indicate that recalled workers return to an unstable job.

One might wonder whether recalled workers seem to be doing worse because they spend

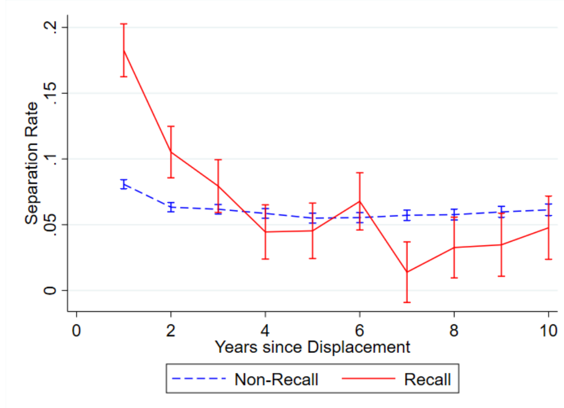

Figure 5: *The effect of displacement on separation rates by ex-post recall status, relative to the control group of never-displaced workers, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. The estimation allows for multiple displacements per individual (classifying the worker according to their first displacement). Only results from period $k = 1$ onwards are displayed here.*

some time in nonemployment (or alternative employment) by definition, whereas some of the non-recalled workers may transition into a new job immediately, e.g. because they anticipated their impending layoff and already searched for a new job prior to the materialization of the layoff.[23]

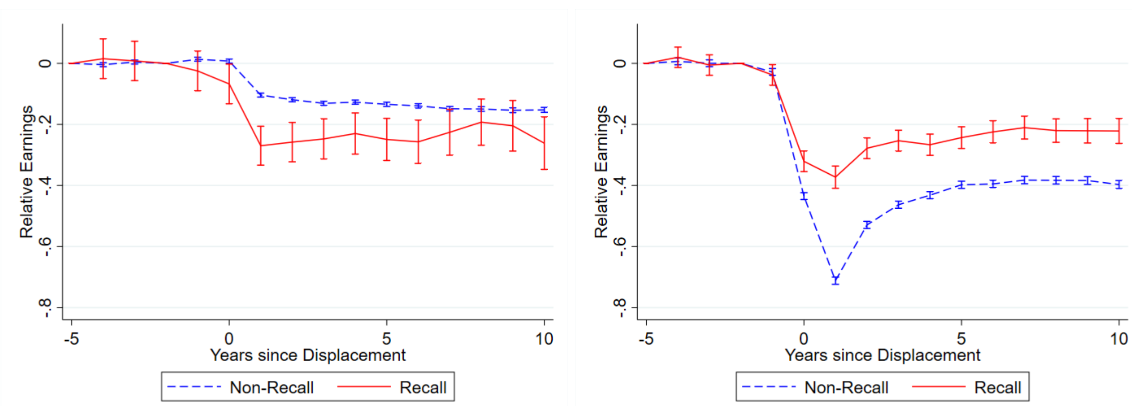

Figure 6: *The effect of displacement on earnings by ex-post recall status, relative to the control group, using estimated coefficients from equation* (1'). *Effects are estimated separately for workers who transition directly (within 30 days, left) and workers who spend some time in nonemployment (right) after their displacement. The error bars correspond to 95% pointwise confidence intervals.*

Figure 6 shows that conditioning on transitioning within 30 days of the layoff or condi-

[23] See Simmons (2025), who investigates this issue in the context of the United States, United Kingdom, and other European countries.

tioning on being nonemployed for at least 30 days does indeed yield very different results.[24] This is primarily driven by the earnings losses of non-recalled workers differing substantially between the two groups, whereas the earnings losses of the recalled workers do not differ much. As a result, the earnings losses for recalled workers are larger than those experienced by non-recalled workers when conditioning on experiencing at most 30 days of nonemployment, whereas the opposite is true when conditioning on experiencing more than 30 days of nonemployment.

3 Model

In this section, I develop a model of the labour market, with the aim of explaining the heterogeneity I observed in Section 2. In order to do so, the discrete-time model discussed below explicitly features the possibility of recall, as a separate state, reflecting my observation that workers who are recalled face a potentially different earnings path. By allowing workers who may be recalled to be in a different state, I can account for differences between recalled and non-recalled workers, both after their non-employment spell and during their non-employment spell, some of which are not explicitly observable in the data. Furthermore, by incorporating a number of worker decisions in favour or against the possibility of a recall, I can also account for a number of the selection issues discussed in Section 1.

3.1 Environment

The economy is populated by workers and firms, both of which are heterogeneous. In line with Jarosch (2023), firms differ in productivity y and baseline separation risk δ, as summarized by vector $\theta = [y, \delta]$. Workers differ in human capital s, and can be either employed, unemployed, or non-employed with a potential future recall. Human capital s evolves over time, increasing by Δ_s with probability ψ_e when employed, and decreasing by Δ_s when non-employed, with probability ψ_u if unemployed or $\psi_r \psi_u$ if non-employed with a potential recall.[25]

[24] Note that due to the definition of recall, recalled workers who transition within 30 days are necessarily workers who were recalled indirectly. In other words, this "transition within 30 days" refers to their transition to a different establishment, rather than their eventual recall (which nevertheless materializes by the end of the following year).

[25] The worker's human capital cannot go below s_{min}, so technically the probability ψ_u depends on s: If $s = s_{min}$, then $\psi_u = 0$. In practice, s_{min} is set sufficiently low such that workers will only reach s_{min} in very rare instances.

3.1.1 Firms

Each firm can hire at most one worker. If a firm is matched to a worker, production takes place according to the log-linear production function $p(s,y) = e^{s+y}$, and the firm pays a wage w to the worker. The match faces a separation shock with probability δ_a, where a refers to the worker's attachment to the job. Whenever the worker matches with a new firm, the attachment is low ($a = L$), and the separation risk equals $\delta_L = \delta + \delta_{nh}$. Each period, the worker becomes highly attached to the firm with probability ξ^a, and upon becoming highly attached the separation risk reduces to the value of the baseline (match-specific) separation risk: $\delta_H = \delta$.

If the separation shock materializes, this separation is permanent with probability $(1 - \phi^f)$, in which case the worker and firm return to an unemployed and unmatched status. However, with probability ϕ^f the job destruction may only be temporary and the worker can choose a potential recall.[26] Nevertheless, the productivity of the match is reduced by $c^f \geq 0$ upon recall, such that the recalled match produces $p(s,y') = p(s,y) - c^f$ (where y' is restricted to be in the range of y). Furthermore, the worker returns as a worker with low attachment, although the separation rate attached to the firm (and therefore to the match) is increased by $c^\delta \geq 0$ instead of δ_{nh}. As such, I refer to this as a third possible value of attachment ($a = R$), even though these workers transition to the high attachment state $a = H$ at the same rate as other workers with low attachment. The intuition behind the recall productivity penalty is that the firm is likely to incur costs for firing and re-hiring the worker and may have undergone restructuring to survive the circumstances that lead to the layoff in the first place, which it will prefer to earn back (e.g. by lowering the worker's wage).[27] The penalty on the separation rate directly reflects the observation in Section 2.3 that recalled workers are more likely to be separated again within a year of being re-employed, which I interpret as evidence of the worker returning to an unstable firm.[28] As such, I expect the separation penalty c^δ to be larger than δ_{nh} (but I do not impose this restriction in the model estimation).

[26] With probability $\phi^{rg}\phi^r$, this recall takes place in the same model period as the initial displacement.

[27] Instead of explicitly lowering the wage, I choose to lower the productivity of the firm. Since each instance of firing and re-hiring comes at a new cost, the production penalty c^f can compound if the worker is displaced and recalled more than once. In contrast, the separation penalty c^δ does not compound.

[28] This interpretation of the penalties is further supported in Appendix C.3.3, where I show that non-displaced workers from these firms tend to experience a downward trajectory in their earnings in the years after the (pseudo-)displacement as well.

3.1.2 Workers

Workers are assumed to be infinitely-lived, and unable to transfer resources between periods. I assume a log utility function, with a discount factor β. Each worker enters the market as newly unemployed and with human capital s_0. An unemployed worker meets a firm with probability λ^u, and this firm is drawn from the distribution $G(\theta)$. Since firms and jobs are identical in this model, this distribution $G(\theta)$ can also be interpreted as the job offer distribution. Additionally, a separated worker moving into regular unemployment finds a new job in the same period with probability λ^{ug}.[29] If the worker meets a firm, the worker decides whether or not to accept the job. If the worker accepts, she becomes employed and receives wage w. If the worker does not accept, or does not receive an offer, the worker receives $b(s, \varepsilon)$, which can be interpreted as the one-period value of being unemployed (and is related to the unemployment benefit), and the value of which depends on whether the worker is eligible to receive the unemployment benefit (as indicated by ε). Whenever a worker enters nonemployment, she is assumed to be eligible ($\varepsilon = 1$), but each period she loses eligibility with probability ξ^{ε}. For eligible workers, the value of the benefit is set equal to a fraction of the lowest possible production a worker could produce in a match: $b(s, 1) = bp(s, y^{min})$. This proxies a setting in which the unemployment benefit depends on the last earned wage, while not ruling out the scenario where unemployed workers reject some job offers.[30] Workers who are no longer eligible for the benefit ($\varepsilon = 0$) instead receive a value $b(s, 0) = \bar{b} = 0.1p(s_0, \bar{y})$, where $\bar{y}$ is the mean value of y in $G(\theta)$. This proxies the welfare benefit, which in reality equals roughly 10 to 12% of the average earnings subject to social security.

Naturally, an employed worker faces the same job destruction and recall shocks as the firm, and receives the wage w. Additionally, an employed worker meets another firm with probability λ^e, and if she does the offer is again drawn from distribution $G(\theta)$. Upon receiving such an offer, the employed worker can decide to switch to the new firm or to reject the offer. If the worker decides to reject the offer, it can be used to re-bargain with the current employer.

While the worker is non-employed with a potential recall, she keeps track of two human capital stocks. Human capital s_r is relevant only for the case in which the worker is recalled, whereas s_u

[29] These workers finding a new job in the same period can be thought of as the model equivalent of workers moving to a new job within 31 days, as discussed in Section 2.3. Alternatively, it can be interpreted as a simplified way to capture that workers may anticipate the impending layoff and may therefore search (and find) a new job before the layoff actually materializes, as pointed out in Simmons (2025).

[30] In the case where $b = 1$, this unemployment benefit is very similar to the one seen in Bagger et al. (2014). In particular, the lower the value of the parameter b is, the lower the value of being unemployed is, and therefore the more job offers will be accepted.

is relevant for all other cases, as well as for the unemployment benefit, which equals $b(s_u, \varepsilon)$ in line with the regular unemployed worker discussed above. The worker's recall human capital s_r decreases by Δ_s with probability $\psi_r \psi_u$, reflecting that the recall human capital may either depreciate faster ($\psi_r > 1$) or slower ($\psi_r < 1$) than s_u, which moves in line with the human capital of a regularly unemployed worker. In particular, one might expect that the depreciation of s_r is slower, since the worker already knows whom she might be employed by in the future, and therefore can keep her job-specific knowledge from depreciating.
The worker is recalled to her previous match with probability ϕ^r every period. When the recall materializes, the firm characteristics change as described in subsection 3.1.1.[31] If the recall does not materialize in a period, the worker meets a new employer with probability $\lambda^r \lambda^u$, where λ^r is expected to be below 1 (but not restricted as such). If the worker meets a new employer, this employer is again drawn from distribution $G(\theta)$, and the worker can decide whether to accept the offer (leading to a wage w). Finally, if the worker does not get recalled and also does not meet a new firm (or rejects the offer from the new firm), she moves to the regular state of unemployment at a rate ξ^r or by choice, thus losing the potential recall. This exogenous probability ξ^r of moving to the regular unemployment state can be interpreted as a loss-of-recall shock in the spirit of Gertler et al. (2026). Among others, this reflects that some firms may close down in the years following the initial layoff, thus excluding the possibility of a later recall.

3.1.3 Wage Setting

In determining the wages, I follow a similar procedure to Bagger et al. (2014). At the time of bargaining the worker and firm agree on a piece-rate $R = e^r$, and the worker receives a wage of $w = Rp(s, y) = e^{r+s+y}$ until either the match is destroyed (because of separation or because the worker switches firms) or the worker receives an offer that triggers re-bargaining.[32]

When the worker and the firm meet, the piece rate is determined using the maximum surplus a worker could extract from the match and the maximum surplus that could be extracted from the outside option. In practice, this maximum surplus equals the value function of the worker if the piece-rate R is set equal to 1 (or $r = 0$), and I denote this value as W^{max}. The piece-rate is set such that the surplus extracted by the worker (W) equals the maximum surplus she could

[31] The loss of the outside option is a simplifying assumption, but is justified by the fact that the worker did not exercise this outside option upon being displaced, so that the firm may no longer consider the threat of leaving to accept this outside offer to be credible.

[32] Additionally, the piece-rate changes when the worker's attachment level changes, but the worker uses the same outside option as before in this case. This can be interpreted as the wage change being previously negotiated.

extract from her outside option, plus a constant fraction of the excess maximum surplus of the pending match. This fraction, κ, is interpreted as the bargaining power of the worker. Denoting the maximum surplus from the outside option by W^{oo}:

$$W(s, s, \theta, \hat{\theta}, a, \hat{a}) = W^{oo} + \kappa \left(W^{max}(s, \theta, a) - W^{oo}\right) \tag{2}$$

Here, it is explicitly taken into account that in general the match value for the worker, W, depends on the value of the firm characteristics θ, the outside option firm characteristics $\hat{\theta}$, the worker attachment to the firm (a) and their attachment to the outside option firm ($\hat{a}$), and the worker's human capital, both current (s) and when the worker and firm last bargained ($\hat{s}$).[33] Equation (2) can take four distinct forms. First, if the worker is coming out of (regular) unemployment, the outside option value W^{oo} equals the value of unemployment, $U(s, \varepsilon)$, and $\hat{\theta} = u$. Then, denoting by x the characteristics of the worker's new firm, and with some abuse of notation (putting ε in the place of the now undefined $\hat{a}$), equation (2) can be rewritten as equation (3):

$$W(s, s, x, u, L, \varepsilon) = U(s, \varepsilon) + \kappa \left(W^{max}(s, x, L) - U(s, \varepsilon)\right) \tag{3}$$

$$W(s, s, x, \theta, L, a) = W^{max}(s, \theta, a) + \kappa \left(W^{max}(s, x, L) - W^{max}(s, \theta, a)\right) \tag{4}$$

$$W(s, s, \theta, x, a, L) = W^{max}(s, x, L) + \kappa \left(W^{max}(s, \theta, a) - W^{max}(s, x, L)\right) \tag{5}$$

$$W(s_r, s_r, \theta, r, R, \varepsilon) = \tilde{T}(s_r, s_r, \theta, \varepsilon) + \kappa \Big(W^{max}(s_r, \theta', R) - \tilde{T}(s_r, s_r, \theta, \varepsilon)\Big) \tag{6}$$

If the worker is moving between two jobs, from a firm with characteristics θ to a firm with characteristics x, the outside option W^{oo} equals the maximum surplus that could have been obtained at her previous job, $W^{max}(s, \theta, a)$, so that equation (2) can be rewritten as equation (4). If the worker is using a job offer from a firm with characteristics x to extract more value from her current employer, the outside option W^{oo} equals the maximum surplus that could have been obtained from this job offer, $W^{max}(s, x, L)$, and equation (2) can be rewritten as equation (5). Finally, if the worker is being recalled, the determination of the worker's surplus is very similar to that of a worker being hired from unemployment (equation 3), but the recalled worker uses the continuation value of non-employment with a potential recall, $\tilde{T}(s_r, s_r, \theta, \varepsilon)$, which reflects that upon rejecting the offer, the worker can choose to (or be forced to) give up the potential recall and move to regular

[33] At the time of bargaining, the human capital "when the worker and firm last bargained" ($\hat{s}$) is set equal to the current human capital (s), so in equations (2) to (6) I set $\hat{s} = s$.

unemployment.[34] Furthermore, since the maximum value obtained from the match changed due to the penalty on production, the firm characteristic used for the determination of the maximal surplus obtained from the recall is not quite the same as the previous characteristic (as denoted by using θ' rather than θ).[35]

3.2 Timing and Value Functions

To summarize the setup of the model, every model period can be divided into 4 stages. At the start of the period, in the first stage, the human capital level of the workers, the employed worker's attachment, and the non-employed worker's eligibility for unemployment benefits are updated. Then, in the second stage, recall materialization, separation, and recall choice take place.[36] In the third stage, workers may receive an offer from a firm (unless they moved to or from the potential recall state in the previous stage), after which they choose to accept or reject it, (re-)bargaining takes place, and non-employed workers with a potential recall may move to permanent unemployment. Finally, production and wages (or unemployment or welfare benefits) are realized at the end of the period.

Using the above description, I can write out the value functions of the worker and the firm. In particular, I write out these value functions from the viewpoint of a worker/firm at the end of the period (before the start of the production stage). The value of unemployment U for a worker with human capital s and benefit eligibility ε can be written out as follows:

$$\begin{aligned} U(s,\varepsilon) = \ln(b(s,\varepsilon)) + \beta \mathbb{E}_{s',\varepsilon'|s,\varepsilon,u} \Bigg\{ \lambda^u \int_{x\in\Theta^u(s',\varepsilon')} W(s',s',x,u,L,\varepsilon') dG(x) \\ + \left(1 - \lambda^u \int_{x\in\Theta^u(s',\varepsilon')} dG(x)\right) U(s',\varepsilon') \Bigg\} \end{aligned} \tag{7}$$

Here, the set $\Theta^u(s,\varepsilon)$ is the set of firm characteristics of the firms from whom the worker would accept an job offer if her current human capital level is s. Using equation (3), this set can be specified as $\Theta^u(s,\varepsilon) = \{x \in [0,1] \times \mathbb{R}_+ : W^{max}(s,x,L) \geq U(s,\varepsilon)\}$. As shown in Appendix A.4,

[34] Note that I make a simplifying assumption that the value of s used in bargaining always reflects only the relevant human capital level. For example, if the worker is being recalled, she uses $\tilde{T}(s_r, s_r, \theta, \varepsilon)$ rather than $\tilde{T}(s_u, s_r, \theta, \varepsilon)$.

[35] Note that if the worker finds a new job while still having a potential recall, the corresponding value $W(s_u, s_u, x, f, L, \varepsilon)$ follows equation (3), replacing the value of the outside option by $\tilde{T}(s_u, s_u, \theta, \varepsilon)$.

[36] Note that by recall choice, I mean only the choice a worker faces when confronted with a separation shock that may not be permanent. As I assume that the worker cannot choose to transition to permanent unemployment from the temporary unemployment state until the recall materialization shock ϕ^r and job search is realized, this second type of recall choice does not take place until the end of the third stage.

equation (3) can be used to rewrite (7) in terms of W^{max}, U, and parameters only:

$$U(s,\varepsilon) = \ln(b(s,\varepsilon)) + \beta\mathbb{E}_{s',\varepsilon'|s,\varepsilon,u}\left\{\lambda^u \int_{x\in\Theta^u(s',\varepsilon')} \kappa\left(W^{max}(s',x,L) - U(s',\varepsilon')\right) dG(x) + U(s',\varepsilon')\right\} \tag{8}$$

Similarly, the value function T for a worker with human capital s_u and s_r and benefit eligibility ε, non-employed with a potential recall to a job of (former) type $\theta = [\delta, y]$, is as follows:

$$\begin{aligned}
T(s_u,s_r,\theta,\varepsilon) = {} & \ln(b(s_u,\varepsilon)) + \beta\mathbb{E}_{s_u',s_r',\varepsilon'|s_u,s_r,\varepsilon,r}\left\{\phi^r W(s_r',s_r',\theta',r,R,\varepsilon')\right. \\
& +(1-\phi^r)\lambda^r\lambda^u \int_{x\in\Theta^r(s_u',\theta,\varepsilon')} W(s_u',s_u',x,f,L,\varepsilon')dG(x) \\
& \left. +(1-\phi^r)\left(1-\lambda^r\lambda^u\int_{x\in\Theta^r(s_u',\theta,\varepsilon')} dG(x)\right)\tilde{T}(s_u',s_r',\theta,\varepsilon')\right\} \qquad (9)\\
\tilde{T}(s_u',s_r',\theta,\varepsilon') = {} & \xi^r U(s_u',\varepsilon') + (1-\xi^r)\max\{T(s_u',s_r',\theta,\varepsilon'), U(s_u',\varepsilon')\} \qquad (10)
\end{aligned}$$

Here, $W(s_r',s_r',\theta',r,R,\varepsilon')$ and $W(s_u',s_u',x,f,L,\varepsilon')$ are as defined above. Since the worker loses her outside option upon separating (even if the separation is temporary), the value function T does not depend on $\hat{s}$ or $\hat{\theta}$. As a worker finding a new job while still having the potential of a recall may use either the value of unemployment or the value of non-employment with a potential recall as their outside option, as reflected by equation (10) with $s_r' = s_u'$, the set of accepted offers $\Theta^r(s_u',\theta,\varepsilon')$ is slightly different from the corresponding set for an unemployed worker $\Theta^u(s',\varepsilon')$. Further, note that $\theta' = [\delta, y']$, where y' is the maximum of y^{min} (the lower bound of the range of y) and y' such that $p(s,y') = p(s,y) - c^f$. Just like value function $U(s,\varepsilon)$, this value function $T(s_u,s_r,\theta,\varepsilon)$ can be rewritten using the bargaining equations (3) and (6):

$$\begin{aligned}
T(s_u,s_r,\theta,\varepsilon) = {} & \ln(b(s_u,\varepsilon)) + \beta\mathbb{E}_{s_u',s_r',\varepsilon'|s_u,s_r,\varepsilon,r}\left\{\phi^r\kappa W^{max}(s_r',\theta',R) + \phi^r(1-\kappa)\tilde{T}(s_r',s_r',\theta,\varepsilon')\right. \\
& +(1-\phi^r)\left(\lambda^r\lambda^u\int_{x\in\Theta^r(s_u',\theta,\varepsilon')}\left(\kappa W^{max}(s_u',x,L) + (1-\kappa)\tilde{T}(s_u',s_u',\theta,\varepsilon')\right)dG(x)\right. \\
& \left.\left. +\left(1-\lambda^r\lambda^u\int_{x\in\Theta^r(s_u',\theta,\varepsilon')} dG(x)\right)\tilde{T}(s_u',s_r',\theta,\varepsilon')\right)\right\} \qquad (11)
\end{aligned}$$

The value of employment W for a worker with human capital s, matched with attachment a to a firm of type θ, is as specified below:

$$
\begin{aligned}
W(s,\hat{s},\theta,\hat{\theta},a,\hat{a}) &= \ln(R(\hat{s},\theta,\hat{\theta},a,\hat{a})p(s,y)) \\
&\quad + \beta\mathbb{E}_{s',a'|s,a,e}\left\{\delta_{a'}\left[\phi^f \max\left\{\hat{T}(s',\theta),\hat{U}(s')\right\} + (1-\phi^f)\hat{U}(s')\right]\right. \\
&\quad +(1-\delta_{a'})\left[\lambda^e\left(\int_{x\in\Theta^1(s',\theta,a')} W(s',s',x,\theta,L,a')dG(x) + \int_{x\in\Theta^2(s',\hat{s},\theta,\hat{\theta},a',\hat{a})} W(s',s',\theta,x,a',L)dG(x)\right)\right. \\
&\quad \left.\left. + \left(1-\lambda^e\int_{x\in\Theta^1(s',\theta,a')\cup\Theta^2(s',\hat{s},\theta,\hat{\theta},a',\hat{a})} dG(x)\right)W(s',\hat{s},\theta,\hat{\theta},a',\hat{a})\right]\right\}
\end{aligned} \tag{12}
$$

Here, I denote by $\hat{s}$ the value of human capital at the time of the most recent bargaining. Similarly, $\hat{\theta} \in \{[0,1]\times\mathbb{R}_+, u, r, f\}$ represents the firm characteristics corresponding to the job offer that was used for bargaining and $\hat{a}$ corresponds to the attachment to that firm.[37] The set $\Theta^1(s,\theta,a)$ is the set of characteristics of firms from whom the worker (with human capital s) would accept an job offer if she is currently employed with attachment a at a firm with characteristics θ, and $\Theta^2(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$ is the set of characteristics of firms whose offers would be used by this worker to trigger re-bargaining at her current match. Using equations (4) and (5), these sets can be specified as $\Theta^1(s,\theta,a) = \{[0,1]\times\mathbb{R}_+ : W^{max}(s,x,L) \geq W^{max}(s,\theta,a)\}$ and $\Theta^2(s,\hat{s},\theta,\hat{\theta},a,\hat{a}) = \{x \in [0,1]\times\mathbb{R}_+ : W^{max}(s,\theta,a) > W^{max}(s,x,L) \geq W^{max}(\hat{s},\hat{\theta},\hat{a})\}$.[38] Note that $\hat{T}$ and $\hat{U}$ correspond to the value of a newly separated worker who chose to either potentially be recalled or move into unemployment, with some abuse of notation (since these are not separate states). These values reflect the possibility of these workers being re-employed in the same period, and therefore relate to value functions (11) and (8) as follows:

$$\hat{T}(s',\theta) = \phi^{rg}\phi^r W(s',s',\theta',r,R,1) + (1-\phi^{rg}\phi^r)T(s',s',\theta,1) \tag{13}$$

$$\hat{U}(s') = \lambda^{ug}\int_{x\in\Theta^u(s',1)} W(s',s',x,u,L,1)dG(x) + \left(1-\lambda^{ug}\int_{x\in\Theta^u(s',1)} dG(x)\right)U(s',1) \tag{14}$$

[37] If a worker comes out of unemployment, she does not have such a job offer to use for bargaining, and uses the value of unemployment instead. With some abuse of notation, I denote this by setting $\hat{\theta} = u$ and $\hat{a} = \varepsilon$. Similarly, I denote the setting for workers being recalled as $\hat{\theta} = r$ and workers finding a new job while still having a potential recall as $\hat{\theta} = f$.

[38] Note that the two sets $\Theta^1(s,\theta,a)$ and $\Theta^2(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$ do not overlap. Further, together they do not cover all possible values of $x \in [0,1]\times\mathbb{R}_+$, revealing the third possible result of receiving an outside offer: if the offer is not good enough for the worker to use to trigger re-bargaining, the worker discards the offer and remains employed under her previously bargained piece-rate.

Using equation (12), the value for W^{max} can be deduced for every combination of s and θ, by setting $R(\hat{s}, \theta, \hat{\theta}, a, \hat{a}) = 1$ and using bargaining equations (4) and (5). The resulting expression, which is derived in Appendix A.4, no longer depends on the bargaining benchmark, as the outcome of the bargaining (which is the piece-rate) is already known:

$$\begin{aligned}W^{max}(s,\theta,a) = \ln(p(s,y)) + \beta\mathbb{E}_{s',a'|s,a,e}\Big\{\delta_{a'}\Big[\phi^f \max\Big\{\hat{T}(s',\theta), \hat{U}(s')\Big\} + (1-\phi^f)\hat{U}(s')\Big] \\ +(1-\delta_{a'})\Big[\lambda^e\int_{x\in\Theta^1(s',\theta,a')}\kappa\Big(W^{max}(s',x,L) - W^{max}(s',\theta,a')\Big)\,dG(x) + W^{max}(s',\theta,a')\Big]\Big\}\end{aligned} \tag{15}$$

One could also set up a value function of a producing firm, $J(s, \hat{s}, \theta, \hat{\theta}, a, \hat{a})$. However, since the above equations suffice to solve the model for a given set of parameters, these value functions, the flow equations, and the equilibrium definition are deferred to Appendices A.1 to A.3.

3.3 Discussion

Before calibrating the model, it is worth briefly reflect on the different channels through which the model may generate differences between (ex-post) recalled and non-recalled workers such as those observed in Section 2.

A number of channels are explicitly associated with one of the parameters of the model. This includes the penalty that the worker takes on their match productivity c^f, which negatively affects the recalled worker. Similarly, given that I expect the additional separation risk of a newly recalled worker c^δ to be higher than the additional risk faced by a newly hired worker δ^{nh}, this penalty on the separation rate is also expected to have a negative impact on the recalled worker. On the other hand, if the worker loses less human capital in case of being recalled ($\psi^r < 1$), this would have a positive effect on the recalled worker. Finally, the transition rates faced by workers in the potential recall state are modeled to be different, but upon impact (λ^{ug} versus $\phi^{rg}\phi^r$) and in the nonemployment state (λ^u versus $\phi^r + (1-\phi^r)\lambda^r\lambda^u$).

Related to the channels discussed above, there are two additional channels associated with a parameter which may indirectly affect the comparison between (ex-post) recalled and non-recalled workers in the model. These are the loss of unemployment benefits (with associated probability ξ^ε) and the loss-of-recall shock (with associated probability ξ^r). The former of these affects both nonemployment states equally, and is therefore not necessarily a clear positive or negative channel for the recalled worker. The loss-of-recall shock, on the other hand, is expected to have a negative impact on non-recalled workers as it forces workers who are already in nonemployment (and therefore have a worse path than the worker entering nonemployment, who may be re-employed immediately) into the non-recall group.

Aside from these channels that are associated with specific parameters, the model includes a few other ways through which (ex-post) recalled and non-recalled workers may be different. One such channel arises at the time of bargaining, where a worker hired from regular unemployment uses only the value of unemployment as the outside option, whereas a recalled worker uses the maximum of the value of unemployment or the value of the potential recall state. Since the worker is in the potential recall state by choice, the latter is expected to be higher, making this a positive channel for the recalled worker.
Finally, one of the main differences between the recalled and non-recalled worker is the fact that the recalled worker does not lose the match to their previous firm. This introduces an element of selection into the comparison between the ex-post recalled and non-recalled worker, which is active at the specific times: upon being offered the choice to enter the potential recall state, and while in the potential recall state upon receiving an offer for a new job and upon making the choice to move to the regular unemployment state. Whether or not the worker chooses to (continue to) wait for a potential recall depends on the relative value of their potentially retained match through all of the above channels, and this consideration may change as shocks (fail to) materialize.

4 Calibration

For the purpose of the calibration, I set up the distribution of firms $G(\theta)$ as a combination of marginal distributions of productivity y and separation rate δ, and make parametric assumptions on these marginal distributions. In particular, I assume that the marginal distribution of δ is a Beta distribution with parameters η_δ and μ_δ, reshaped to the $[0, 0.25]$ interval, whereas the marginal distribution of y is a Pareto distribution with scale parameter μ_y and shape parameter η_y. I then follow Jarosch (2023) in combining the two marginal distributions into the bivariate distribution $G(\theta)$ using Frank's copula with parameter ρ (thereby allowing for correlation between the two variables). These assumptions yield a total of 26 parameters that need to be identified. Of these 26 parameters, I set 5 exogenously. In particular, as one model period corresponds to one quarter, I set the discount factor $\beta = 0.95^{1/4}$ to reflect an annual interest rate of 5%. Furthermore, I normalize the human capital grid by setting $s_0 = 0$ and $\Delta_s = 0.1$. Finally, I fix both ξ^a and ξ^ε at 0.25, such that the worker on average takes a year to lose benefit eligibility or become highly attached to their job. The remaining 21 parameters are estimated using indirect inference (Gourieroux et al., 1993), operationalized using the TikTak global optimization algorithm (Arnoud et al., 2022;Guvenen et al., 2021). In the next subsection, I describe which moments I use to identify these remaining parameters.

4.1 Calibration Moments

I identify 24 moments that together identify the values of the 21 parameters that I estimate using the indirect inference method from Gourieroux et al. (1993). While the parameters are estimated simultaneously, I divide the parameters into six groups, and argue that each of these groups are largely identified by a corresponding group of moments.

The first set of moments consists of transition rates from employment to non-employment, and these moments are used to calibrate parameters governing the marginal distribution of δ, and the separation penalty for recalled and newly hired workers (c^δ and δ^{nh}). Since workers on average reach the high attachment state one year after being recalled or newly hired, after which c^δ and δ^{nh} no longer apply, I use the average separation rate into non-employment for workers with an establishment tenure of 1-3.5, 3.5-6, 6-9, and 9+ years respectively to identify the parameters of the marginal distribution of δ (μ_δ and η_δ). Then, to discipline the additional separation risk faced by newly hired workers (δ^{nh}) and recently recalled workers (c^δ), I use the average job loss rate and the subsequent separation rate after re-employment following a recall relative to the subsequent separation rate after re-employment following any displacement.

The second set of moments focuses on the average wage level and its variance. Given the direct link between production and wages in the model, I use these moments to identify the marginal distribution of firm productivity y. In particular, I use the median-p25 and p75-p25 ratio of wages to aid in identifying the shape parameter η_y and scale parameter μ_y of the marginal distribution of y.

The third set of moments provides information regarding job finding probabilities, both on-the-job and from non-employment. In particular, the fraction of job-to-job transitions that coincided with a displacement helps to identify the meeting probability for newly unemployed workers (λ^{ug}). After all, such direct transitions to a new job will be observed as job-to-job transitions. The overall quarterly job-to-job transition rate therefore also contributes to identifying this parameter, while also informing the value of the on-the-job meeting rate λ^e. Similarly, the average job finding rate closely corresponds to the job finding rate of unemployed workers, λ^u.

The next set of moments focuses on wage growth within and between job spells. The specific moments used here include the net replacement rate in unemployment, which closely relates to the parameter b included in the expression for the instantaneous value of non-employment $b(s, \varepsilon)$ for eligible workers. I follow Gregory (2023) in taking this moment from OECD (2020). The average yearly wage growth, conditional on full-year full-time employment, helps to identify the human capital on-the-job transition rate ψ_e. Since human capital depreciation during unemployment is a driver of increasing wage losses by unemployment duration, I use the gradient in the average difference between pre- and post-layoff wages, conditional on different non-employment duration bins (up to 0.5, 0.5 to 1, or 1 to 2 years) to identify the human capital transition rate during unemployment (ψ_u). Similarly, to identify the human capital transition rate for workers who are non-employed with a potential recall (ψ_r) and the production penalty associated with recall c^f, I use the average difference between pre- and post-recall wages, conditional on non-employment duration (0.25 to 0.5, and 0.5 to 1 year). Specifically, the level in the first duration bin is used to identify the value of c^f, whereas the gradient between the two bins informs the value of ψ_r.

As the model allows for a choice between unemployment and potential recall upon separation, the potential recall offer probability ϕ^f and the recall materialization probabilities ϕ^r and $\phi^{rg}\phi^r$ are likely different from the observed recall and recall materialization probabilities. However, given the close relation between the two, I can use the observed probabilities as calibration targets. I use two sets of recall materialization rate, derived from the observed recall materialization rate within two years and within one year, in order to tease out the difference between ϕ^r and $\phi^{rg}\phi^r$. Similarly, I use information on the fraction of workers expecting a recall (and re-employed within a year) who find a new job instead to inform the probability of meeting a new employer, $\lambda^r\lambda^u$, whereas the fraction of workers finding a new job doing so before the expected recall date informs the rate at which workers transition out of the potential recall state, thus helping to identify the loss-of-recall rate ξ^r.[39]

[39]The moments using workers expecting a recall cannot be estimated from my data since I do not observe recall expectations. Instead, the data equivalent of these moments are based on results in Nekoei and Weber (2015) and Nekoei and Weber (2020). For the model equivalent, I assume that workers expecting a recall are the non-employed workers with a potential recall, and finding a job before the expected recall date corresponds to finding a job from the potential recall state.

The final group consists of all remaining parameters (κ and ρ), which are identified using information on workers' starting wages and the observed correlation between wages and separation rates. In particular, I use the average wage of a new worker (hired out of unemployment) relative to the average wage to identify the bargaining power κ. Finally, for the identification of the copula parameter ρ, I follow a strategy similar to that in Jarosch (2023) by targeting the regression coefficient γ in the estimation equation (16) below:

$$D^{\delta}_{i,t} = \alpha_i + \gamma \log(w_{it}) + u_{i,t} \tag{16}$$

In equation (16), the variable $D^{\delta}_{i,t}$ is a dummy variable that is only filled if the worker i is employed in period t and still observed in period $t+1$. It equals 1 if the worker is separated from their job between t and $t+1$. The explanatory variables include an individual fixed effect α_i and the natural logarithm of the worker's wage in period t, $w_{i,t}$.

4.2 Calibration Results and Model Fit

Table 3 summarizes the estimated moment values and their model counterparts. Generally speaking, the model fits the moments quite well. Nevertheless, the model has trouble matching a few moments. For example, it can be observed that the model tends to have trouble matching separation rates at low tenure, while overshooting the job finding rate and slightly undershooting the recall rate.

The parameter estimates in Table 3 governing the (new) job meeting rates are largely comparable to Jarosch (2023). One exception to this is the on-the-job meeting probability, which is fairly low in my calibration ($\lambda^e = 0.053$). This difference can be explained by the fact that I allow for displaced workers to find a job in the same period as being displaced, and this happens with a fairly high probability, as indicated by the calibrated value of $\lambda^{ug} = 0.544$. However, even with this additional channel, the estimated model struggles to reach the job-to-job transition rate observed in the data.

The calibration implies that a non-employed worker is much less likely to lose human capital if she has a potential recall, as the human capital depreciation rate ($\psi_r \psi_u \approx 0.007$) is close to zero and much lower than the depreciation rate faced by a regular unemployed worker ($\psi_u = 0.048$).[40] However, the recall also comes with substantial negative consequences in the form

[40]Generally speaking, the human capital transition rates are quite low compared to the literature. For example, the monthly (rather than quarterly) values for ψ_u and ψ_e in Jarosch (2023), which are fairly (though not entirely)

Description of Moment(s)	Data	Model	Parameters
Average rate of job loss, tenure 1-3.5y	0.030	0.014	$\eta_\delta = 3.82$
Average rate of job loss, tenure 3.5-6y	0.016	0.009	$\mu_\delta = 68.6$
Average rate of job loss, tenure 6-9y	0.012	0.008	$c^\delta = 0.255$
Average rate of job loss, tenure>9y	0.008	0.007	$\delta^{nh} = 0.054$
Average rate of job loss	0.023	0.012	
Relative subsequent separation, recall	0.109	0.076	
p75-p25 ratio of wages	1.644	1.975	$\eta_y = 8.71$
Median-p25 ratio of wages	1.263	1.386	$\mu_y = 3.24$
Job-to-job transition rate	0.028	0.020	$\lambda^e = 0.053$
Displacement among job-to-job transitions	0.424	0.311	$\lambda^{ug} = 0.544$
Average job finding rate	0.140	0.221	$\lambda^u = 0.226$
Replacement rate	0.6	0.652	$b = 1.278$
Yearly wage growth	0.013	0.006	$\psi_e = 0.033$
Pre- to post-layoff wage,	-0.058	-0.051	$\psi_u = 0.048$
duration gradient	-0.027	-0.024	$\psi_r = 0.144$
Pre- to post-recall wage, duration 0.25-0.5y	-0.019	-0.020	$c^f = 0.194$
Pre- to post-recall wage, duration gradient	0.010	0.011	
Recall rate	0.058	0.042	$\phi^f = 0.542$
Recall materialization rate	0.299	0.341	$\phi^r = 0.154$
	0.296	0.369	$\phi^{rg} = 0.714$
New job finding rate, workers expecting a recall	0.293	0.207	$\lambda^r = 0.174$
New job finding rate, before expected recall	0.25	0.301	$\xi^r = 0.152$
Wage of newly hired worker	0.682	0.748	$\kappa = 0.852$
Coefficient $\hat{\gamma}$ in equation (16)	-0.020	0.001	$\rho = -19.1$

Table 3: *A summary of calibration moments, their values in the data and in the calibrated model, and corresponding parameter values.*

of a production penalty c^f (which is relatively mild)[41] and a substantial penalty on the separation rate c^δ, which implies that after recall the worker's separation rate increases by more than 25 percentage points, whereas this increased separation risk for new hires δ^{nh} is only 5.4 percentage points.

It is worth noting that the recall offer rate ϕ^f is higher than the observed recall rate in the data and model simulation. This is because not all workers choose in favour of a potential recall when offered to do so. Intuitively, this is because while the potential recall state is attractive due to the lower human capital depreciation rate, the recall penalties and lower transition rate out of nonemployment $((1-\phi^r)\lambda^r\lambda^u + \phi^r < \lambda^u)$ offset some of these benefits. This is particularly true for workers whose job already has a low productivity y, as they are likely to draw a new job with a higher productivity y, and such draws are more likely to occur in the regular unemployment state, as the new job meeting probability is much lower (17.4%) in the potential recall state (as illustrated by the value of λ^r).

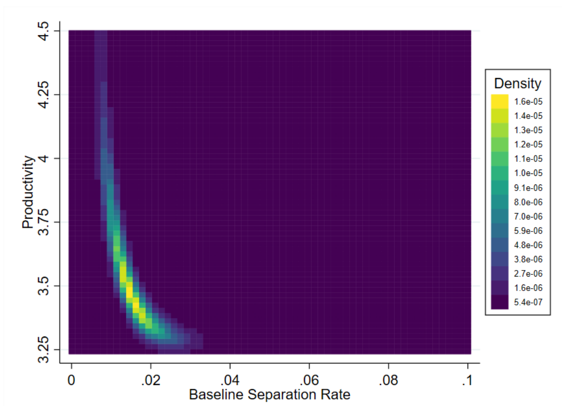

Figure 7: *The joint distribution of firm types faced by workers. A lighter colour in this chart corresponds to a higher density.*

The firm distributions the workers draw from upon receiving an offer are best illustrated in a diagram. Figure 7 visualizes the joint distribution of firms $G(\theta)$. The bulk of the density is located in the bottom left corner of the graph (which corresponds to low productivity and low separation rates), thus illustrating that both marginal distributions of y and especially δ are heavily right-skewed.

comparable in their interpretation, are 0.26 and 0.133 respectively.

[41] To provide context to the magnitude of the production penalty, note that the value of the lowest possible value of production for a worker at the starting level of human capital, $p(s_0, y^{min})$ equals $e^{0+3.24} \approx 25.5$.

5 Simulation Results

In this section, I present the results of the model simulation, using calibrated parameter values from the previous section. I start by comparing the predictions of the model regarding the scarring effects of displacement to the data. These patterns were not explicitly targeted in the calibration, although the average difference between pre- and post-layoff wages for recalled workers can be thought of as indirectly targeting the short-run effects for recalled workers. I then use the model to illustrate the importance of taking into account the possibility of recall, by showing that a policy targeting human capital depreciation will have opposite effects for recalled and non-recalled workers.

5.1 Heterogeneity in the scarring effects of displacement

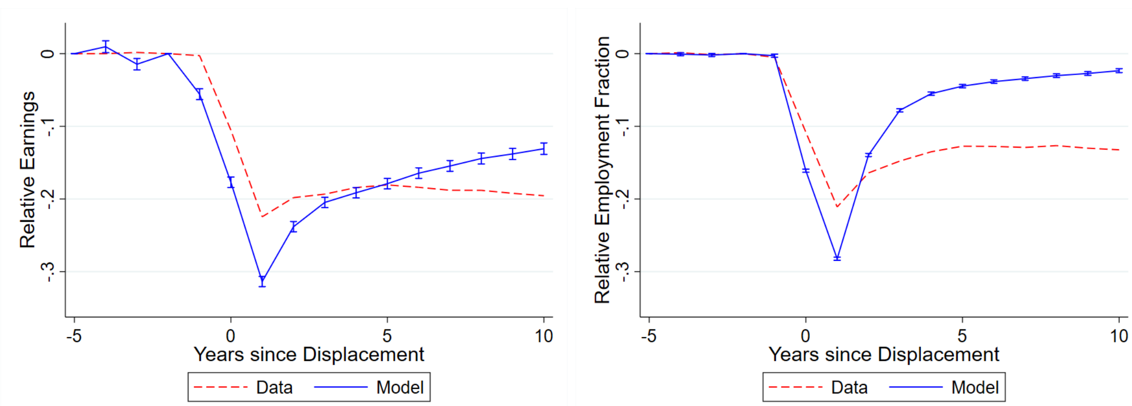

Figure 8: *The effect of displacement on earnings (left) and employment (fraction of the year spent in an employment spell, right), relative to the control group, using model simulation data (solid) and using the data (dashed, corresponding to Figure 3).*

Figure 8 compares the average effect of displacement on earnings and employment in the model and the data (both estimated using equation 1). While the shapes of the graphs are quite similar, the model generates too much recovery in earnings and especially employment (and slightly overshoots the initial impact) compared to the data. This is likely caused in part by the model implying a lower average separation rate than the data, thus leading to more recovery in employment, which in turn leads to more recovery in earnings.

Figure 9 compares the model-predicted effect of displacement on earnings and employment fraction (defined as the fraction of the year spent in an employment spell) by ex-post recall status with the data. The model matches the observation that recalled workers do worse than non-recalled workers after displacement in terms of their earnings and employment in the short run,

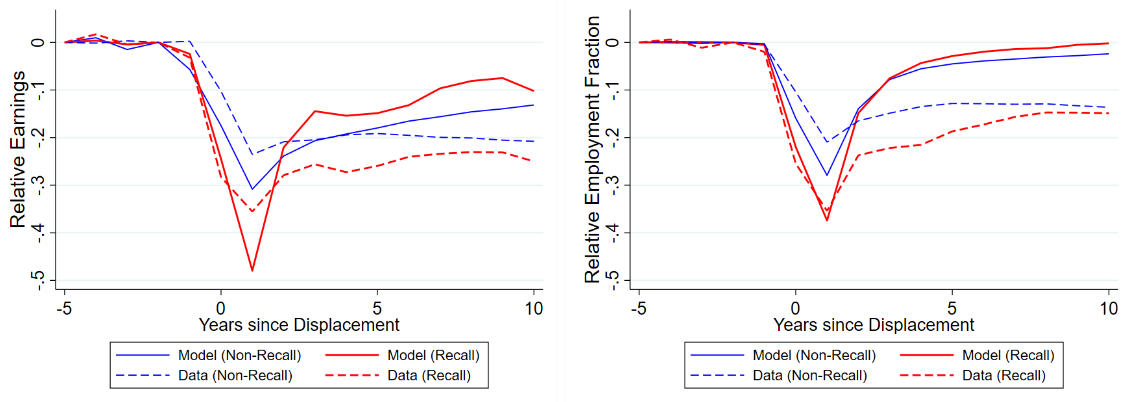

Figure 9: *The effect of displacement on earnings (left) and employment (fraction of the year spent in an employment spell, right) relative to the control group, by ex-post recall status (materialization of recall within 5 years), using model simulation data (solid) and using the data (dashed, corresponding to Figure 4).*

while the effects are fairly similar in the long run.[42] In particular, the excessive recovery observed in Figure 8 is observed for both non-recalled and recalled workers in Figure 9, though slightly more so for recalled workers. This reflects the fact that (as seen in Table 3) in addition to the estimated average separation rate being lower than in the data, the model estimation also implies a lower separation rate penalty than observed in the data.

In Figure 10, I decompose the differences in estimated post-displacement earnings between recalled and non-recalled workers. In particular, I consider all channels (discussed below, and earlier in Section 3.3) through which the ex-post recalled worker is potentially different from a non-recalled worker in my model, and switch these channels off one by one (cumulatively) in order to generate counterfactual earnings differences between recalled and non-recalled workers.[43] I find that long-run differences between recalled and non-recalled workers are negatively driven by the post-recall penalties on match characteristics. In particular, the negative difference is largely driven by the worker going back to a more unstable job (as represented in the model by the separa-

[42] While the model-generated earnings employment losses are estimated to be slightly smaller for (ex-post) recalled workers, in contrast to the findings from the data, the conclusions drawn from the simulation results discussed in this section are not affected by this mismatch. In Appendix B.4, I discuss how parameter estimates and results change if I were to directly target the regression results from both panels of Figure 4 when calibrating the model.

[43] The order in which the channels are switched off is shown in reverse order in the legend. As such, the "Bargaining" channel is the first to be shut down, followed by the "Match Productivity" channel, and so on. For a detailed explanation of this decomposition exercise, see Appendix B.2. This Appendix also includes results of an alternative decomposition in which I use a direct counterfactual simulation (introduced in Appendix B.1) to estimate earnings differences instead of estimating Equation (1') on model-simulated data, as I do for the decompositions depicted here.

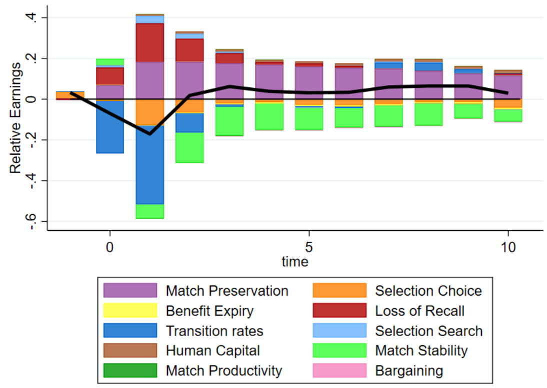

Figure 10: *A decomposition of the difference in the scarring effect of displacement on earnings between (ex-post) recalled and non-recalled workers. The black line represents the total difference, calculated as the difference between the solid red and blue lines in the left panel of Figure 9. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Table B.1.*

tion rate penalty c^δ, and in the figure by "Match Stability"), whereas the impact of the productivity penalty c^f is small (and the corresponding element "Match Productivity" is barely visible in the figure). Essentially, the fact that the worker has a much higher probability of being separated again shortly after being re-employed implies that the worker is likely to be set back in her recovery multiple times, both in terms of human capital and in terms of repeated loss of outside option.

A few other channels are shown in Figure 10 to have a negative impact. A major channel negatively affecting the difference in short-run relative earnings losses is that of the "Transition rates". The effect of this channel is strongly negative, reflecting primarily the higher immediate transition probabilities λ^{ug} of the worker who chose to move to the regular unemployment state. The "Bargaining" channel reflects that recalled workers may use the value of the potential recall state as their outside option, but this turns out to have a very minor (and negative) impact. Similarly, the potential expiry of unemployment benefits ("Benefit Expiry") has a small but negative impact. Finally, the "Selection Choice" channel reflects the impact of allowing the worker to choose between the regular unemployment state and the state of non-employment with a potential recall. This channel has a negative impact because workers who choose against a potential recall (and therefore self-select into the non-recall group) tend to be the workers experiencing the smallest earnings loss. Removing this choice forces some of these workers into the recall group, thus reducing the average earnings loss experienced by recalled workers relative to non-recalled workers.

The aforementioned negative drivers are largely offset by a set of positive drivers. Particularly notable is the positive impact of the loss-of-recall shock. This shock forces some potentially recalled workers to become non-recalled workers (with certainty), and does so after entering nonemployment, thus leading these workers to forego the benefits of the aforementioned higher immediate transition probabilities. As such, workers are likely to have larger than average earnings losses, pushing them away from the group of recalled workers has a positive impact on the comparison of the estimated earnings losses between recalled and non-recalled workers. The "Selection Search" channel, which reflects that the worker is allowed to search while holding a potential recall (as indicated by $\lambda^r > 0$), is shown to have a small positive effect. The finding that the worker holding a potential recall is much less likely to lose any human capital ("Human Capital") has a small but increasingly positive effect, reflecting the low estimated human capital transition probabilities discussed in Section 4. Finally, the element named "Match Preservation" is the residual element. This channel reflects the difference between if all parameters are the same, and the states only differ by whether or not the worker returns to the same match. The fact that this channel has a positive impact reflects that while the displaced workers are negatively selected towards workers who are in a worse match (in terms of productivity and separation rate), the forces of the job ladder are still sufficiently strong such that the match they would find when drawing a (random) new employer from the joint distribution $G(\theta)$ is on average worse than the match they separated from.

While the discussion above focuses on the decomposition of the difference in earnings losses for recalled and non-recalled workers, the setup of the model, and the fact that it is able to match the (short-run) difference in employment fraction as well, also allows me to further decompose earnings losses into employment and wage components. In Figure 11, I use the results of the model-estimated effect of displacement (on earnings, employment, and wages) to decompose the earnings loss into employment and wages. Corresponding to my findings in the data, I find that the short-run difference is partially driven by the employment margin. The wage margin goes in the opposite direction, suggesting that conditional on employment the recalled workers earn more. Note that the simple combination of employment and wages does not fully explain the estimated earnings loss, as indicated by the "Other" category in Figure 11. This residual can be thought of as reflecting a correlation term, as discussed in Schmieder et al. (2023).

In the right panel of Figure 11, I further decompose the employment differences into

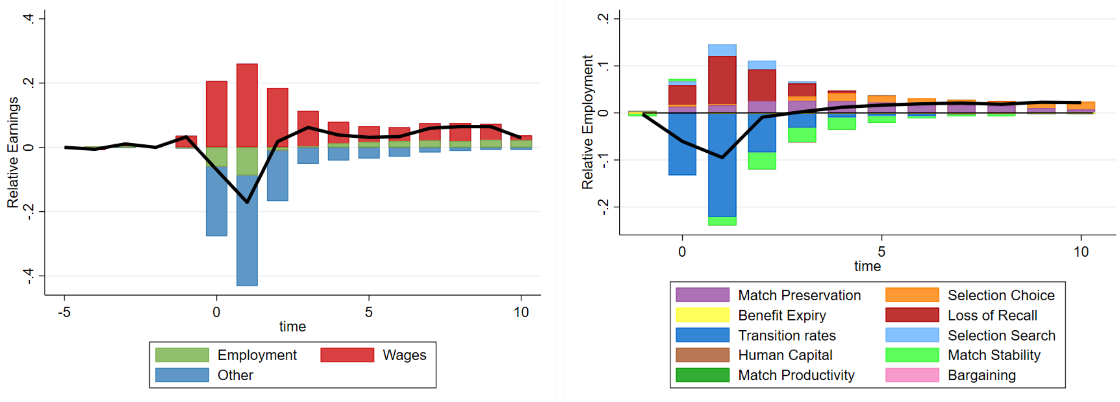

Figure 11: *A decomposition of the difference in the scarring effect of displacement on earnings between (ex-post) recalled and non-recalled workers, into earnings and employment (left), and a decomposition of the difference in the scarring effect of displacement on employment between (ex-post) recalled and non-recalled workers (right). The black line represents the total difference, calculated as the difference between the solid red and blue lines in Figure 9. The decomposition on the left is generated by using the estimation for employment and wages, and backing out the remaining effect as a residual. The decomposition on the right is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Table B.2.*

the same 10 channels used for the earnings decomposition above. A similar decomposition for wages can be found in Appendix B.2. For employment, the main negative drivers are again related to the higher separation rate ("Match Stability") and lower immediate transition rates faced by the recalled worker. Absent these factors, the (ex-post) recalled worker would likely be doing better in terms of employment. This is primarily driven by the loss-of-recall shock pushing already nonemployed workers away from the group of potentially recalled workers. Aside from this, I find a role for workers being allowed to search while holding a potential recall ("Selection Search"), and the fact that while displaced workers in general come from matches of a higher separation rate than the average in the economy, they face an even higher expected separation rate when drawing a new match from the joint distribution $G(\theta)$. This is important especially in the long run, as these workers start separating again, and this is reflected by the positive "Match Preservation" channel in the decomposition.

5.2 Policy Implications

Throughout this paper, I have illustrated that workers who return to their employer after being laid off tend to experience larger earnings losses compared to their non-recalled (but still laid off) counterparts, especially in the short run. In the previous subsections, I have illustrated that this can be explained using the model I developed in Section 3. Given that I have highlighted the importance of accounting for these two sets of workers (recalled and non-recalled) in a model, this might leave one wondering about the policy implications of this result. Unfortunately, the setup of the model is such that it is not necessarily informative to think of a social planner. After all, the model stresses the viewpoint of the worker, and thereby abstracts from other elements that a social planner may wish to take into account, such as the firm choosing who to (potentially) recall and congestion externalities in the labour market (which I abstract from by setting exogenous offer arrival rates). Nevertheless, some lessons can still be drawn from the results in the previous subsections.

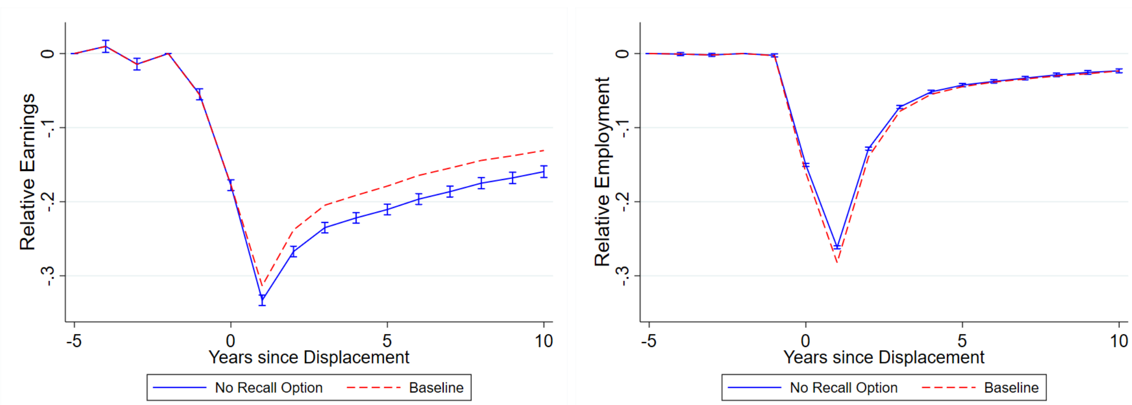

Figure 12: *The effect of displacement on earnings (left) and employment (right), relative to the control group, using model simulation data from the baseline model (dashed) and a version without the recall option (solid).*

A first question that may come to mind, especially based on the empirical results, is whether recalls may be a bad. The fact that the model is able to generate the larger short-run losses among recalled workers with a set of reasonably valued parameters and without having to incorporate any forced (or mistaken) choice for a potential recall provides an indication that the recall option is a positive for the workers who choose it. This is confirmed in Figure 12, in which I show how the estimated average earnings and employment losses change if I shut down the recall option completely (by setting $\phi^f = 0$). The left panel of Figure 12 confirms that the average earnings losses would have been higher in the absence of the recall option. This is in line with the decomposition in Section 5.1, where I found a positive effect of the match preservation channel, and indicates that the workers with the larger expected earnings loss from unemployment select into the potential recall option. The employment losses, on the other hand, do not change substantially. This is also reflected in the fact that the average time to re-employment conditional on entering nonemployment is x in the baseline estimate and only slightly changes to x in the version without the recall option. Similarly, the nonemployment rate in the two versions is fairly similar, at x and x respectively. As such, I can conclude that the recall option has clear positive value from the viewpoint of the worker.

Thinking about more general policies, it is not uncommon for models of job displacement to find a large role for skill losses in explaining average earnings losses in the medium or long run (e.g. Jung and Kuhn, 2019;Jarosch, 2023), thus suggesting scope for a policy targeting human capital depreciation experienced during nonemployment. However, the decomposition of the differences between recalled and non-recalled workers' earnings loss after displacement in the previous section highlighted the depreciation of human capital as a channel that works in favour of the recalled worker. This is driven by workers holding a potential recall facing a human capital depreciation probability of only 0.7%, whereas this probability is 4.8% for a regularly unemployed worker. As such, one might expect such a policy to affect recalled and non-recalled workers differently.

In order to assess the impact of a policy targeting human capital depreciation during nonemployment, I start by separately decomposing the earnings loss experienced by recalled and non-recalled workers. This exercise is similar to the decomposition in Section 5.1, but decomposes the magnitude of the earnings losses rather than the difference between the two magnitudes. As such, the channels considered (and their order) are different. In particular, since I am particularly interested in the effect of human capital transitions, I shut down the corresponding transition rates first, by setting $\psi_r = 0$ ("Human Capital (T)"), $\psi_u = 0$ ("Human Capital (U)"), and $\psi_e = 0$ ("Human Capital (E)"). Other channels include (in order) the impact of the worker losing their outside option, losing eligibility for unemployment benefits, and facing a loss-of-recall shock. The "Frictions" channel measures the impact of spending time in nonemployment (beyond any impact coming through the aforementioned channels), whereas the "Experience Loss" channel measures the impact of the worker becoming a low-attachment worker in their next job. Finally, the effect of the two match penalties c^δ and c^f is grouped into a single channel ("Match Penalties"), after which any remaining earnings loss is argued to be driven by the worker drawing a new match ("Match Loss"). Further details on how the contribution of these channels is calculated is deferred to Appendix B.3.

Figure 13 shows the results of this decomposition. Since the estimated human capital

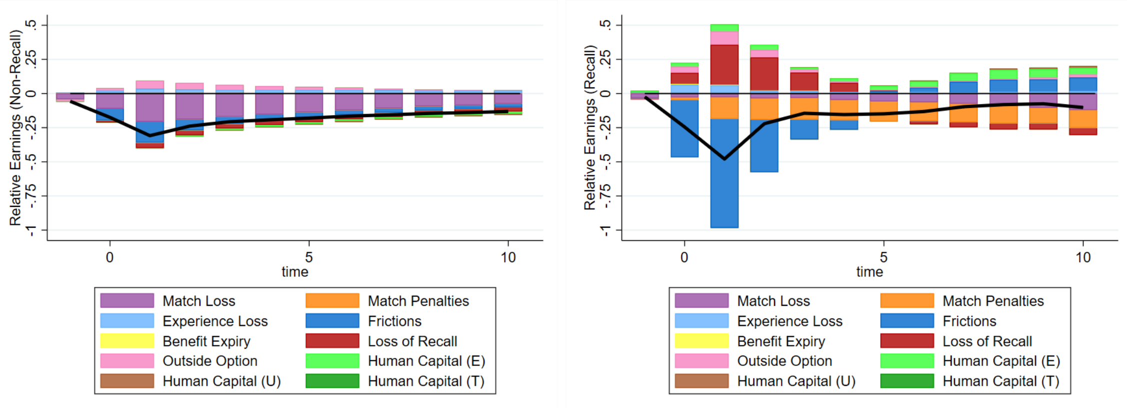

Figure 13: *A decomposition of the scarring effect of displacement on earnings for (ex-post) non-recalled workers (left) and recalled workers (right). The black line represents the total earnings loss, and corresponds to the solid blue and red lines in the left panel of Figure 9. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Tables B.4 and B.5.*

transition probabilities are fairly low, it is natural that their contribution to earnings losses is fairly limited. Nevertheless, the combination of the three human capital channels still accounts for almost 20% of the earnings losses for non-recalled workers 10 years after displacement. For recalled workers, on the other hand, the net contribution is positive, implying that their earnings losses would be larger in the absence of these human capital channels. This is largely driven by the positive impact of the on-the-job human capital appreciation, which can be explained by the timing in the model. In particular, a subset of eventually recalled workers gained human capital at the start of the period in which they were displaced, but this increase is not reflected in their earnings due to the displacement. As a result, these workers return to their previous job with a higher human capital than they had when their earnings were last recorded, leading to an increase in earnings.

Given that the group of non-recalled workers is larger than the group of recalled workers, the decomposition of the average scarring effect of displacement largely resembles the decomposition for non-recalled workers, thus pointing towards human capital as one of the drivers of the long-run losses. Therefore, it is natural to expect that a policy aimed at helping displaced workers may be targeted at bringing down the human capital depreciation probability for non-employed workers. The result in Figure 14 shows the effects of the case where such a policy would completely eliminate any human capital depreciation (i.e. $\psi_u = 0$, and therefore $\psi_r \psi_u = 0$). The figure shows that such a policy would help the non-recalled worker, but not the recalled worker,

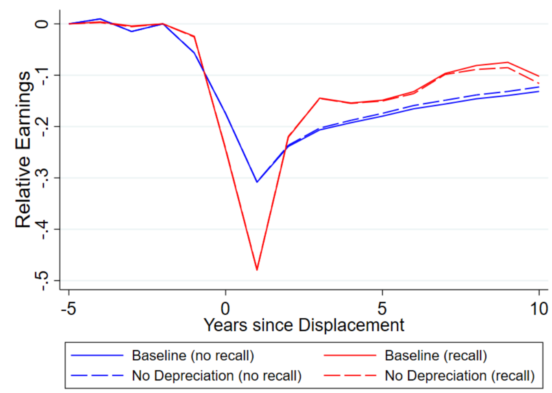

Figure 14: *The effect of displacement on earnings relative to the control group, by ex-post recall status, using model simulation data from the baseline model (solid) and a counterfactual in which non-employed workers do not lose human capital during non-employment ($\psi_u = \psi_r \psi_u = 0$, dashed).*

for whom losses in fact slightly increase. Therefore, keeping in mind that these recalled workers already face larger losses in the short run, such a policy would have the unintended consequence of further disadvantaging the recalled worker.

6 Conclusion

In this paper, I study the scarring effect of displacement on earnings and employment, focusing in particular on how these effects differ based on whether workers are recalled to their previous employer. Using detailed administrative data from Germany, I find that while recalled workers tend to experience slightly larger earnings and employment losses than non-recalled workers in the short run, their average losses are similar in the long run. However, the similarity of these long-run earnings losses mask differences that are uncovered when further distinguishing workers that transition to a new job within 30 days (in which case non-recalled workers experience smaller losses) and workers that take more than 30 days to transition (in which case recalled workers experience smaller losses). Furthermore, I find that recalled workers experience a higher probability of subsequent job loss than non-recalled displaced workers.

As the existing theoretical models cannot account for these observations, I develop a model of the labour market in which I explicitly allow for recall by dividing newly non-employed workers into two separate states, according to whether or not the workers may be recalled in the future. Further adding elements that have been successful in explaining the average scarring effect of displacement, such as human capital which evolves over time according to the worker's employment status, I find that this model, calibrated to the German data, is able to account for the heterogeneity I observe in the data.

I then use the calibrated model to study the main drivers of the heterogeneity in the scarring effects of displacement by ex-post recall status. In particular, I find that if the only difference between recalled and non-recalled worker was their next employer, the recalled worker would experience lower earnings losses (as they would not lose their position on the job ladder). In addition, the human capital depreciation channel is almost nonexistent for the recalled worker, thus further alleviating the earnings losses experienced by the recalled worker. On the other hand, recalled workers face a higher separation rate after the recall materializes (and, less importantly, lower productivity), which plays a large negative role in the long run, as a subsequent separation sets back the worker in their path of recovery, especially if such a subsequent separation no longer comes with (the expectation of) a future recall.

When decomposing the long-run effect of displacement on earnings for recalled and non-recalled workers separately, I find that whereas human capital depreciation plays a large role for non-recalled workers, its role for recalled workers does not start to materialize until a few years later (when their subsequent separations start occurring). This implies that a policy designed to dampen the depreciation of human capital (traditionally identified as one of the key drivers of the long-run effects of displacement on earnings) will likely be much less effective for recalled workers, especially in the short run, thus further widening the long-run gap between recalled and non-recalled workers. For a recalled worker, a more beneficial policy might be one that aids the rehiring firm after the recall takes place, thus reducing the probability that the worker will be displaced again shortly after being rehired.

Based on the results of this paper, one can think of various avenues for future research, and I will highlight a few of those possibilities here. First of all, this paper focuses in particular on the dimension of ex-post recall status, but given the right data it would be interesting to further look into the differences between recall expectations and recall materialization (as emphasized by Nekoei and Weber, 2015), and its consequences for worker's earning paths after job loss.

When it comes to the model, there are also are a number of ways in which one might imagine expanding the analysis presented in this paper. Given the importance of the different human capital depreciation rates of recalled and non-recalled workers, a natural extension would be to explain these differences by distinguishing between firm-specific and general human capital (e.g. as proposed in Audoly et al., 2024). Second of all, the selection channels operating in the model in this paper are partially exogenous, and it would be worth exploring models in which these channels are endogenous instead. The main extension in this direction would incorporate decision making on the firm side of the model, which I largely abstract from in this paper. Similarly, given the research that has pointed out the potential importance of workers involuntarily losing their recall option (e.g. Gertler et al., 2022), including such a "loss-of-recall" channel into the model could be valuable. In order to do so, one would require a reasonable counterpart in the data that could be used to discipline such a channel. To the best of my knowledge, there currently does not exist any research looking into this in the context of the German (or similar) labour market, but conditional on the existence of such data this could be a promising avenue for future research. Finally, when extending the framework in this paper to one with cyclical variation, and especially when doing so in the context of the German labour market, it will be important to explicitly add in the possibility of using short-time employment rather than an explicit layoff when facing an economic downturn. This will be particularly important given the widespread usage of short-time employment policies throughout Europe during the Covid-19 pandemic.

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For Online Publication: Online Appendix

A Model Appendix

A.1 Further Value Functions

As mentioned in Section 3, the model can be solved using value functions from the worker side only. However, the value function J for a firm of type θ, employing a worker with human capital s and attachment a, can still be set up, and looks as follows:

$$\begin{aligned} J(s,\hat{s},\theta,\hat{\theta},a,\hat{a}) &= \Big(1-R(\hat{s},\theta,\hat{\theta},a,\hat{a})\Big)\, p(s,y) \\ &+ \beta\mathbb{E}_{s',a'|s,a,e}\left\{\delta_{a'}\bar{\phi}^f(s',\theta)\hat{J}^f(s',\theta) + (1-\delta_{a'})\left[\lambda^e \int_{x\in\Theta^2(s',\hat{s},\theta,\hat{\theta},a',\hat{a})} J(s',s',\theta,x,a',L)dG(x)\right.\right. \\ &+ \left.\left.\left(1-\lambda^e\int_{x\in\Theta^1(s',\theta,a')\cup\Theta^2(s',\hat{s},\theta,\hat{\theta},a\,\hat{a})} dG(x)\right) J(s',\hat{s},\theta,\hat{\theta},a',\hat{a})\right]\right\} \end{aligned} \tag{A.1}$$

Here, $\bar{\phi}^f(s',\theta) = \phi^f \mathbb{1}_{\hat{T}(s',\theta)>\hat{U}(s')}$, capturing that the worker may choose to forego the option of recall. The value of an unmatched firm is set to $V=0$. Finally, $\hat{J}^f(s',\theta)$ is the value for a firm newly potentially recalling, which can be decomposed into the separation period-specific part and a general value for a firm potentially recalling their former worker:

$$\hat{J}^f(s',\theta) = \phi^{rg}\phi^r J(s',s',\theta',r,R,1) + (1-\phi^{rg}\phi^r)J^f(s',s',\theta,1) \tag{A.2}$$

$$\begin{aligned} J^f(s_u,s_r,\theta,\varepsilon) &= \beta\mathbb{E}_{s'_u,s'_r,\varepsilon'|s_u,s_r,\varepsilon,r}\left\{\phi^r J(s'_r,s'_r,\theta',r,R,1)\right. \\ &\left.+(1-\xi^r)(1-\phi^r)\left(1-\lambda^r\lambda^u\int_{x\in\Theta^r(s'_u,\theta,\varepsilon')} dG(x)\right)\mathbb{1}_{T(s'_u,s'_r,\theta,\varepsilon')>U(s'_u,\varepsilon')} J^f(s'_u,s'_r,\theta,\varepsilon')\right\} \end{aligned} \tag{A.3}$$

A.2 Worker Flows

The description of the model in the main text (Section 3) can be used to construct a number of worker flow equations. In what follows, denote by $d(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$ the density of employed workers with current human capital s, negotiation benchmark human capital $\hat{s}$, matched to a firm with characteristics $\theta\in[0,1]\times\mathbb{R}_+$ and attachment a, and benchmark characteristics $\hat{\theta}\in[0,1]\times\mathbb{R}_+$ and attachment $\hat{a}$, and denote by $d(s,\hat{s},\theta,u,a,\varepsilon)$, $d(s,\hat{s},\theta,r,a,\varepsilon)$, and $d(s,\hat{s},\theta,f,a,\varepsilon)$ the equivalents if this worker used unemployment as the outside option at the time of bargaining, was recently recalled to their current job, or found the current job while holding a potential recall, with ε referring to the worker's benefit eligibility at the time of the corresponding transition out of non-employment. Further, let $d^f(s_u, s_r, \theta, \varepsilon)$ be the density of workers with current human capital s_u and s_r and benefit eligibility ε holding a potential recall to a firm with (pre-recall) characteristics θ, and let $u(s, \varepsilon)$ be the density of unemployed workers with human capital s and benefit eligibility ε. First, define the following densities, defined after the first stage of the model period, in which human capital accumulation (or depreciation) as well as updates of the attachment and benefit eligibility take place:

$$
\begin{aligned}
\bar{d}(s, \hat{s}, \theta, \hat{\theta}, a, \hat{a}) &= (1 - \mathbb{1}_{a \neq H} \xi^a) \Big[(1-\psi_e) d(s, \hat{s}, \theta, \hat{\theta}, a, \hat{a}) + \psi_e d(s - \Delta_s, \hat{s}, \theta, \hat{\theta}, a, \hat{a}) \Big] \\
&\quad + \mathbb{1}_{a=H} \xi^a \Big[(1-\psi_e) d(s, \hat{s}, \theta, \hat{\theta}, L, \hat{a}) + \psi_e d(s - \Delta_s, \hat{s}, \theta, \hat{\theta}, L, \hat{a}) \Big] \\
&\quad + \mathbb{1}_{a=H} \xi^a \Big[(1-\psi_e) d(s, \hat{s}, \theta, \hat{\theta}, R, \hat{a}) + \psi_e d(s - \Delta_s, \hat{s}, \theta, \hat{\theta}, R, \hat{a}) \Big] \\
\bar{d}^f(s_u, s_r, \theta, \varepsilon) &= (1 - \mathbb{1}_{\varepsilon=1} \xi^\varepsilon) \Big[(1 - \psi_r \psi_u)(1-\psi_u) d^f(s_u, s_r, \theta, \varepsilon) + \psi_r \psi_u (1-\psi_u) d^f(s_u, s_r + \Delta_s, \theta, \varepsilon) \\
&\quad + (1 - \psi_r \psi_u) \psi_u d^f(s_u + \Delta_s, s_r, \theta, \varepsilon) + \psi_r \psi_u \psi_u d^f(s_u + \Delta_s, s_r + \Delta_s, \theta, \varepsilon) \Big] \\
&\quad + \mathbb{1}_{\varepsilon=0} \xi^\varepsilon \Big[(1 - \psi_r \psi_u)(1-\psi_u) d^f(s_u, s_r, \theta, 1) + \psi_r \psi_u (1-\psi_u) d^f(s_u, s_r + \Delta_s, \theta, 1) \\
&\quad + (1 - \psi_r \psi_u) \psi_u d^f(s_u + \Delta_s, s_r, \theta, 1) + \psi_r \psi_u \psi_u d^f(s_u + \Delta_s, s_r + \Delta_s, \theta, 1) \Big] \\
\bar{u}(s, \varepsilon) &= (1 - \mathbb{1}_{\varepsilon=1} \xi^\varepsilon) \Big[(1-\psi_u) u(s, \varepsilon) + \psi_u u(s + \Delta_s, \varepsilon) \Big] \\
&\quad + \mathbb{1}_{\varepsilon=0} \xi^\varepsilon \Big[(1-\psi_u) u(s, 1) + \psi_u u(s + \Delta_s, 1) \Big]
\end{aligned}
$$

The flow equations are then as follows:[44]

$$
\begin{aligned}
d'(s, \hat{s}, \theta, \hat{\theta}, a, \hat{a}) &= (1-\delta_a) \left(1 - \lambda^e \int_{x \in \Theta^1(s,\theta,a) \cup \Theta^2(s', \hat{s}, \theta, \hat{\theta}, a, \hat{a})} dG(x) \right) \bar{d}(s, \hat{s}, \theta, \hat{\theta}, a, \hat{a}) \\
&\quad + \mathbb{1}_{a=L} \mathbb{1}_{s=\hat{s}} \lambda^e g(\theta) \left[\iint \sum_{\tilde{a}} (1 - \hat{\delta}_{\hat{a}}) \left(\mathbb{1}_{\theta \in \Theta^1(s, \hat{\theta}, \hat{a})} \bar{d}(s, x, \hat{\theta}, y, \hat{a}, \tilde{a}) \right) dx dy \right] \\
&\quad + \lambda^e \left[g(\hat{\theta}) \iint \sum_{\tilde{a}} (1 - \delta_a) \left(\mathbb{1}_{\hat{\theta} \in \Theta^2(s, x, \theta, y, a, \tilde{a})} \bar{d}(s, x, \theta, y, a, \tilde{a}) \right) dx dy \right] \Bigg\} \qquad \text{(A.4)}
\end{aligned}
$$

[44]Note that when I integrate over y in equation (A.4), I include all possible values for $\hat{\theta}$, including u, r, and f, in this integration. The same holds for the integration over $\hat{x}$ in equations (A.5), (A.6), (A.8), and (A.9). Furthermore, note that I distinguish between different values of δ such that $\delta \in \theta$, $\hat{\delta} \in \hat{\theta}$, and $\delta_x \in x$.

$$
\begin{aligned}
d'(s,\hat{s},\theta,u,a,\varepsilon) &= (1-\delta_a)\left(1-\lambda^e\int_{x\in\Theta^1(s,\theta,a)\cup\Theta^2(s',\hat{s},\theta,u,a,\varepsilon)} dG(x)\right)\bar{d}(s,\hat{s},\theta,u,a,\varepsilon) \\
&\quad + g(\theta)\mathbb{1}_{a=L}\mathbb{1}_{\varepsilon=1}\mathbb{1}_{s=\hat{s}}\mathbb{1}_{\theta\in\Theta^u(s)}\left(\lambda^{ug}\iiint\sum_{\tilde{a},\hat{a}}\delta_{x,\tilde{a}}(1-\bar{\phi}^f(s,x))\bar{d}(s,\tilde{s},x,\hat{x},\tilde{a},\hat{a})d\tilde{s}dxd\hat{x}\right) \\
&\quad + g(\theta)\mathbb{1}_{a=L}\mathbb{1}_{s=\hat{s}}\mathbb{1}_{\theta\in\Theta^u(s,\varepsilon)}\lambda^u\bar{u}(s,\varepsilon) \qquad \text{(A.5)}
\end{aligned}
$$

$$
\begin{aligned}
d'(s,\hat{s},\theta,r,a,\varepsilon) &= (1-\delta_a)\left(1-\lambda^e\int_{x\in\Theta^1(s,\theta,a)\cup\Theta^2(s',\hat{s},\theta,r,a,\varepsilon)} dG(x)\right)\bar{d}(s,\hat{s},\theta,r,a,\varepsilon) \\
&\quad + \mathbb{1}_{a=R}\mathbb{1}_{s=\hat{s}}\int\mathbb{1}_{\theta'\in\Theta^f(\theta)}\left(\sum_{\tilde{a},\hat{a}}\delta_{\theta',\tilde{a}}\bar{\phi}^f(s,\theta')\phi^{rg}\phi^r\iint\bar{d}(s,\hat{s},\theta',\hat{x},\tilde{a},\hat{a})d\hat{s}d\hat{x}\right)d\theta' \\
&\quad + \mathbb{1}_{a=R}\mathbb{1}_{s=\hat{s}}\iint\mathbb{1}_{\theta'\in\Theta^f(\theta)}\phi^r\bar{d}^f(s_u,s,\theta',\varepsilon)d\theta'ds_u \qquad \text{(A.6)}
\end{aligned}
$$

$$
\begin{aligned}
d'(s,\hat{s},\theta,f,a,\varepsilon) &= (1-\delta_a)\left(1-\lambda^e\int_{x\in\Theta^1(s,\theta,a)\cup\Theta^2(s',\hat{s},\theta,f,a,\varepsilon)} dG(x)\right)\bar{d}(s,\hat{s},\theta,f,a,\varepsilon) \\
&\quad + g(\theta)\mathbb{1}_{a=L}\mathbb{1}_{s=\hat{s}}\iint\mathbb{1}_{\theta\in\Theta^r(s,x,\varepsilon)}(1-\phi^r)\lambda^r\lambda^u\bar{d}^f(s,s_r,x,\varepsilon)dxds_r \qquad \text{(A.7)}
\end{aligned}
$$

$$
\begin{aligned}
d^{f\prime}(s_u,s_r,\theta,\varepsilon) &= (1-\xi^r)(1-\phi^r)\left(1-\lambda^r\lambda^u\int_{x\in\Theta^r(s_u,\theta,\varepsilon)} dG(x)\right)\mathbb{1}_{T(s_u,s_r,\theta,\varepsilon)>U(s_u,\varepsilon)}\bar{d}^f(s_u,s_r,\theta,\varepsilon) \\
&\quad + \mathbb{1}_{\varepsilon=1}\mathbb{1}_{s_u=s_r}\iint\sum_{a,\hat{a}}\delta_a\bar{\phi}^f(s_u,\theta)(1-\phi^{rg}\phi^r)\bar{d}(s_u,\hat{s},\theta,\hat{x},a,\hat{a})d\hat{s}d\hat{x} \qquad \text{(A.8)}
\end{aligned}
$$

$$
\begin{aligned}
u'(s,\varepsilon) &= \left(1-\lambda^u\int_{x\in\Theta^u(s,\varepsilon)} dG(x)\right)\bar{u}(s,\varepsilon) \\
&\quad + \mathbb{1}_{\varepsilon=1}\int\sum_a\delta_a(1-\bar{\phi}^f(s,\theta))\left(1-\lambda^{ug}\int_{x\in\Theta^u(s,1)} dG(x)\right)\iint\sum_{\hat{a}}\bar{d}(s,\hat{s},\theta,\hat{x},a,\hat{a})d\hat{x}d\hat{s}d\theta
\end{aligned}
$$

(A.9)

where

$$
\Theta^f(\theta) = \left\{[\delta, y'] \in [0,1]\times\mathbb{R}_+ : y = \max(y^{min},\hat{y});\quad \hat{y} : p(s,\hat{y}) = p(s,y') - c^f\right\}
$$

A.3 Equilibrium

In this model economy, an equilibrium consists of value functions $U(s,\varepsilon)$, $W(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$, $T(s_u,s_r,\theta,\varepsilon)$, $J(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$, and a piece-rate function $R(\hat{s},\theta,\hat{\theta},a,\hat{a})$, such that, given distribution $G(\theta)$ and parameters, the value functions $W(s,\hat{s},\theta,\hat{\theta},a,\hat{a})$ and $U(s,\varepsilon)$ satisfy equations (3) to (6), the value functions and the piece-rate function satisfy equations (7) to (15) and equation (A.1), and the distribution of workers across different states evolves according to equations (A.4) to (A.9).

A.4 Derivation of $W^{max}(s,\theta,a)$ and simplified $U(s,\varepsilon)$

Below, I derive the function $W^{max}(s,\theta,a)$, which is interpreted as the value the worker would derive from a match if they were to receive the entire surplus (i.e. $w(s,\hat{s},\theta,\hat{\theta},a,\hat{a}) = p(s,y)$). I derive this function by rewriting equation (12), and setting $R(\hat{s},\theta,\hat{\theta},a,\hat{a}) = 1$. First, note that one can rewrite the value of non-employment with a potential recall, equation (9) in terms of itself, W^{max}, and U only. In order to do so, I use the bargaining equations (3) and (6), leading to equation (11). Given a guess for W^{max} and U, one can (iteratively) solve equation (11) for the corresponding T. Furthermore, given that the values for T and U are then known (for a given value of W^{max} and U), I can also directly calculate the corresponding values for a newly non-employed worker (either holding a potential recall or not), $\hat{T}(s,\theta)$ and $\hat{U}(s)$. Using these calculations (and leaving in $\hat{T}$ and $\hat{U}$), I can then start to rewrite equation (12), by plugging in $R(\hat{s},\theta,\hat{\theta},a,\hat{a}) = 1$ and rewriting:

$$
\begin{aligned}
W^{max}(s,\theta,a) = \ln(p(s,y)) + \beta\mathbb{E}_{s',a'|s,a,e}\Bigg\{&\delta_{a'}\left[\phi^f \max\{\hat{T}(s',\theta),\hat{U}(s')\} + (1-\phi^f)\hat{U}(s')\right] \\
&+(1-\delta_{a'})\Bigg[\lambda^e \int_{x\in\Theta^1(s',\theta,a')} \Big(W(s',s',x,\theta,L,a') - W(s',\hat{s},\theta,\hat{\theta},a',\hat{a})\Big)\,dG(x) \\
&+\lambda^e \int_{x\in\Theta^2(s',\hat{s},\theta,\hat{\theta},a',\hat{a})} \Big(W(s',s',\theta,x,a',L) - W(s',\hat{s},\theta,\hat{\theta},a',\hat{a})\Big)\,dG(x) + W(s',\hat{s},\theta,\hat{\theta},a',\hat{a})\Bigg]\Bigg\}
\end{aligned}
$$

To simplify the equation above, note that if the worker already is in the position of receiving all the surplus, there is no more room to re-bargain the piece-rate at the current employer. As such, the re-bargaining set $\Theta^2(s',\hat{s},\theta,\hat{\theta})$ is an empty set and the corresponding integral cancels out. Furthermore, assuming the worker gets all the surplus, $W(s',\hat{s},\theta,\hat{\theta},a',\hat{a}) = W^{max}(s',\theta,a')$. Finally, to arrive at equation (15), I simplify the term inside of the integral by using the bargaining equation $W(s,s,x,\theta,L,a) = W^{max}(s,\theta,a) + \kappa\left(W^{max}(s,x,L) - W^{max}(s,\theta,a)\right)$:

$$
\begin{aligned}
W^{max}(s,\theta,a) = \ln(p(s,y)) + \beta\mathbb{E}_{s',a'|s,a,e}\Bigg\{&\delta_{a'}\left[\phi^f \max\{\hat{T}(s',\theta),\hat{U}(s')\} + (1-\phi^f)\hat{U}(s')\right] \\
&+(1-\delta_{a'})\Bigg[W^{max}(s',\theta,a') \\
&+\lambda^e \int_{x\in\Theta^1(s',\theta,a')} \Big(W^{max}(s',\theta,a') + \kappa\left(W^{max}(s',x,L) - W^{max}(s',\theta,a')\right) - W^{max}(s',\theta,a')\Big)\,dG(x)\Bigg]\Bigg\}
\end{aligned}
$$

$$
= \ln(p(s,y)) + \beta \mathbb{E}_{s',a'|s,a,e} \left\{ \delta_{a'} \left[\phi^f \max\{\hat{T}(s',\theta), \hat{U}(s')\} + (1-\phi^f)\hat{U}(s') \right] \right.
$$

$$
\left. + (1-\delta_{a'}) \left[\lambda^e \int_{x \in \Theta^1(s',\theta,a')} \kappa \left(W^{max}(s',x,L) - W^{max}(s',\theta,a') \right) dG(x) + W^{max}(s',\theta,a') \right] \right\}
$$

Similarly, I can use the bargaining equation (3) to remove the value function W from the value function U, equation (7):

$$
U(s,\varepsilon) = \ln(b(s,\varepsilon)) + \beta \mathbb{E}_{s',\varepsilon'|s,\varepsilon,u} \left\{ \lambda^u \int_{x \in \Theta^u(s',\varepsilon')} W(s',s',x,u,L,\varepsilon') dG(x) \right.
$$

$$
\left. + \left(1 - \lambda^u \int_{x \in \Theta^u(s',\varepsilon')} dG(x) \right) U(s',\varepsilon') \right\}
$$

$$
U(s,\varepsilon) = \ln(b(s,\varepsilon)) + \beta \mathbb{E}_{s',\varepsilon'|s,\varepsilon,u} \left\{ \lambda^u \int_{x \in \Theta^u(s',\varepsilon')} \left(W(s',s',x,u,L,\varepsilon') - U(s',\varepsilon') \right) dG(x) \right.
$$

$$
\left. + U(s',\varepsilon') \right\}
$$

$$
U(s,\varepsilon) = \ln(b(s,\varepsilon)) + \beta \mathbb{E}_{s',\varepsilon'|s,\varepsilon,u} \left\{ \lambda^u \int_{x \in \Theta^u(s',\varepsilon')} \kappa \left(W^{max}(s',x,L) - U(s',\varepsilon') \right) dG(x) + U(s',\varepsilon') \right\}
$$

B Additional Simulation Results

B.1 Further Simulation Results

In Section 5.1 of the main text, I briefly discussed how the estimated model performs in matching the empirical results from Section 2. In this section, I discuss how the model performs outside the regression context stressed in the main text.

While the results in Section 5.1 of the main text focus on implications of the model that were generated using the same estimation method as in the data, one can also generate similar implications using a so-called direct counterfactual. After all, the estimation methods used in the data are used in large part because a direct counterfactual is not observable in the data, but such a counterfactual can easily be created in the model simulation. In practice, this amounts to simulating the model twice, where the (initial) displacement is prevented from happening in the second simulation but all other random variable draws are the same between the two simulations. In Figures B.1 and B.2, I show the accompanying results and compare them with the regression-based results. As can be seen from the figures, the direct counterfactuals tend to predict a more persis-

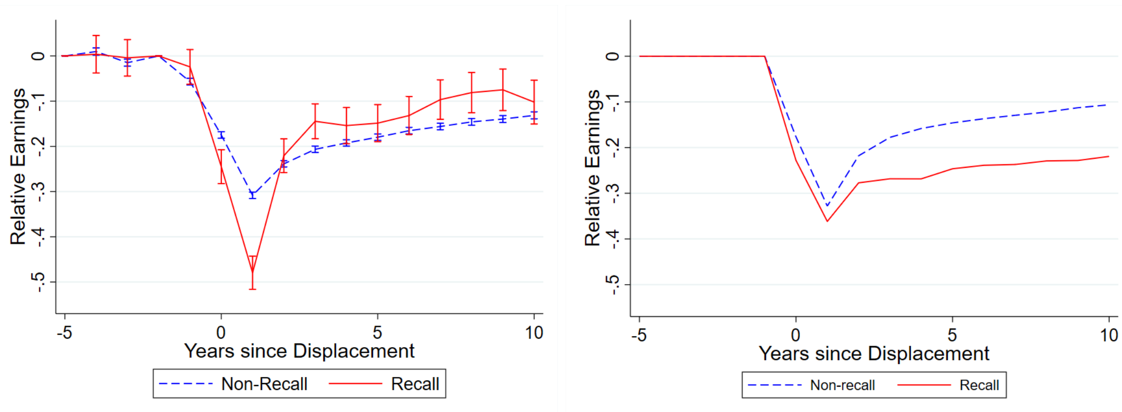


Figure B.1: *The effect of displacement on earnings relative to the control group, by ex-post recall status (materialization of recall within 5 years), using model simulation data and using a regression-based approach (left, corresponding to Figure 9) or a direct counterfactual (right).*

tence in the scarring effect for the recalled workers, especially for earnings, and less persistence for the non-recalled worker.

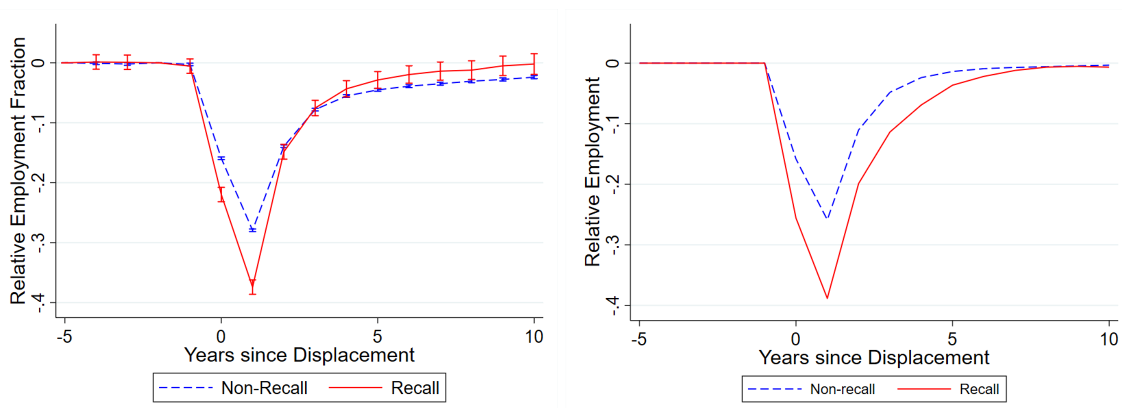


Figure B.2: *The effect of displacement on employment fraction relative to the control group, by ex-post recall status (materialization of recall within 5 years), using model simulation data and using a regression-based approach (left, corresponding to Figure 9) or a direct counterfactual (right).*

The difference between the two methods partially reflects a difference in counterfactuals. In the regression-based method, the counterfactual is the same for both the recalled and non-recalled workers, and based off workers who are never displaced. In contrast, the direct counterfactual method uses observations from the same worker, and therefore uses a different counterfactual observation for the recalled and non-recalled worker. If the two groups are different, for example in terms of their earnings trajectory prior to displacement, this can lead to the large observed difference in persistence between the two methods. Furthermore, since the direct counterfactual only removes the first displacement, the counterfactual is no longer based on "never-displaced" workers, which explains the eventual full recovery in the direct counterfactual results for employment. If I instead remove all future displacements from the counterfactual simulations, the estimated pattern shows more persistence for employment as well as earnings, as shown in Figure B.3

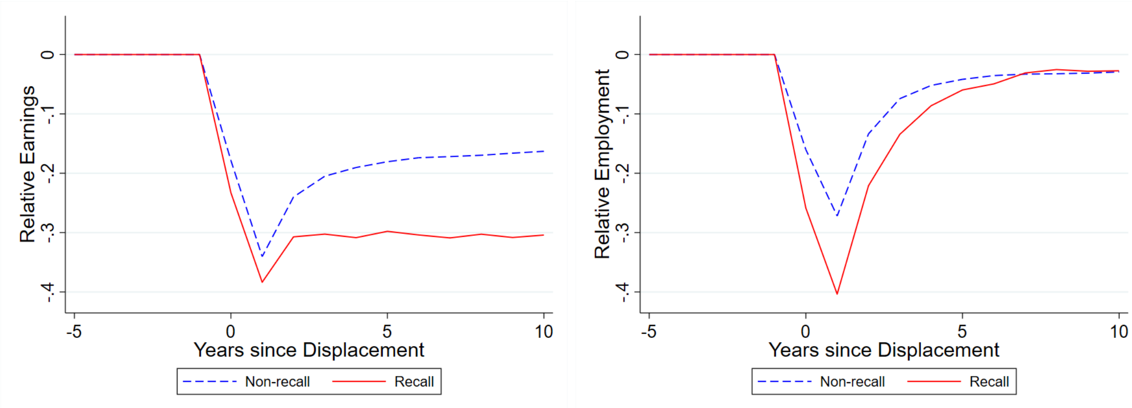


Figure B.3: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status (materialization of recall within 5 years), using model simulation data and using a direct counterfactual in which the current and all future displacements are removed.*

B.2 Further Decomposition Results

In the main text, in Section 5.1, I used the calibrated model to decompose the difference in the scarring effects of displacement (on earnings and employment) between recalled and non-recalled workers into the channels through which the two groups are potentially different according to the model, using the regression-based model implications. However, as I stressed in the previous section, these results can also be obtained using a direct counterfactual. In Figures B.4 and B.5, I compare the results of this alternative decomposition (in the right panel) with those from the decomposition discussed in the main text (re-printed in the left panel). The two decomposition methods generally yield similar conclusions, although the negative drivers are more persistent in the direct counterfactual method, thus leading to the negative difference in earnings losses.

Figure B.6 repeats the regression-based decomposition exercise with a focus on wages instead. As in the data, wages should be interpreted as earnings conditional on employment.[45] The main positive differences for wages are coming from the match preservation channel. This reflects

[45] As wages are only defined conditional on employment, this exercise is less meaningful in the direct counterfactual method, as it would require the worker to be employed both in the main simulation and in the counterfactual. The results of the wage decomposition using the direct counterfactual method are therefore omitted from this Appendix.

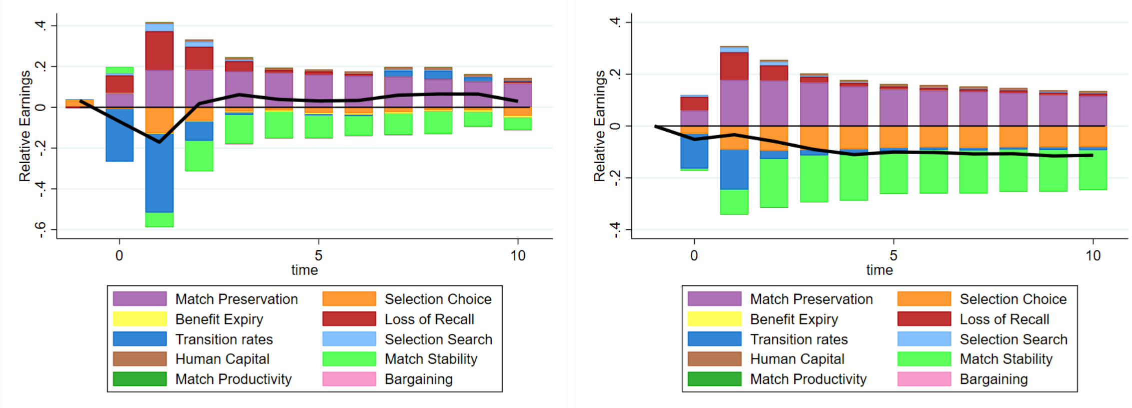


Figure B.4: *A decomposition of the difference in the scarring effect of displacement on earnings between (ex-post) recalled and non-recalled workers, using either a regression-based approach (left) or a direct counterfactual (right). The black line represents the total difference, calculated as the difference between the red and blue lines in Figure B.1. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Table B.1.*

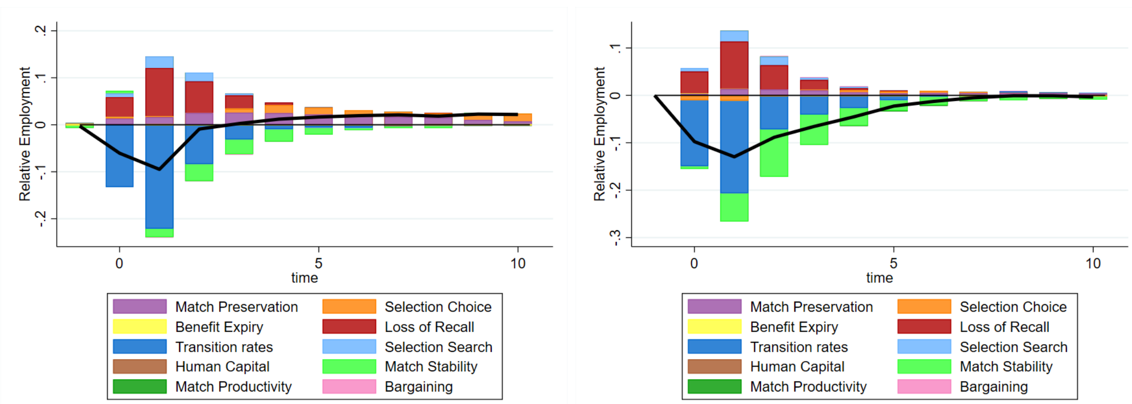


Figure B.5: *A decomposition of the difference in the scarring effect of displacement on employment between (ex-post) recalled and non-recalled workers, using either a regression-based approach (left) or a direct counterfactual (right). The black line represents the total difference. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Table B.2.*

that the worker does not fall off the job ladder when recalled (only accepting a competing offer if it dominates the potential recall). This positive effect is strengthened by selection into the potential recall state ("Selection Choice"), especially in the short run. In the long run, however, the positive effect is increasingly offset by the negative influence of higher separation rates.

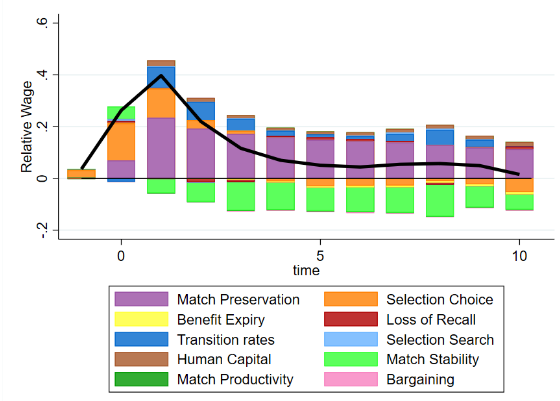


Figure B.6: *A decomposition of the difference in the scarring effect of displacement on wages between (ex-post) recalled and non-recalled workers, using a regression-based approach. The black line represents the total difference. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Tables B.3.*

In Tables B.1 (earnings), B.2 (employment), and B.3 (wages) I show the numerical values used to construct a selection of the bars in the corresponding Figures B.4, B.5 and B.6. In these tables, I separate the differential effects of displacement on earnings, employment, and wages (denoted "Total") between (ex-post) recalled and non-recalled workers into the 10 channels that could potentially drive these differences in the model. The contribution of these channels is calculated by switching them off one by one (cumulatively) in the model simulation, after which I estimate the earnings losses in the new simulation. I do so while keeping the model solution, and therefore the wages and decisions for given values of the state variables, constant. Below, I discuss the exact interpretation of these channels and how they are switched off in the model simulation.

The first listed channel, the "Bargaining" channel, corresponds to the difference in earnings driven by the potential difference in outside options. This contribution is therefore calculated by setting the calculated piece-rates for the newly recalled worker equal to those of a newly hired worker, for a given set of other state variables. The next two channels listed in the tables correspond directly to the two penalty parameters c^f ("Match Productivity Penalty") and c^δ ("Match Stability

	Regression-Based				Direct Counterfactual			
Channel	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 0$	$k = 1$	$k = 5$	$k = 10$
Bargaining	-0.001	-0.000	-0.003	-0.003	$0x$	$-0.000x$	$0.001x$	$0.000x$
Match Productivity Penalty	-0.000	-0.001	-0.002	-0.002	-0.000	-0.000	-0.001	$-0.001x$
Match Stability Penalty	0.031	-0.070	-0.108	-0.058	-0.004	-0.055	-0.094	$-0.084x$
Human Capital	0.001	0.007	0.011	0.015	0.000	0.003	0.005	$0.004x$
Selection Search	0.010	0.037	-0.004	-0.001	0.002	0.009	-0.006	$-0.006x$
Transition Rates	-0.258	-0.387	-0.004	0.003	-0.167	-0.316	-0.169	$-0.150x$
Loss of Recall	0.085	0.189	0.015	0.009	-0.001	-0.027	-0.089	$-0.082x$
Benefit Expiry	0.002	-0.002	-0.006	-0.009	-0.000	-0.000	0.000	$0.000x$
Selection Choice	-0.008	-0.127	-0.027	-0.041	-0.028	-0.090	0.084	$-0.081x$
Match Preservation	0.068	0.183	0.160	0.116	0.059	0.174	0.138	$0.114x$
Total	-0.070	-0.171	0.031	0.029	-0.140	-0.302	-0.299	$-0.285x$

Table B.1: *Summary of the decomposition of the difference in the scarring effect of displacement on earnings between (ex-post) recalled and non-recalled workers. The total difference is calculated as the difference between the red and blue lines in Figure B.1. The decomposition is generated by turning off the indicated channels one by one (in the presented order), thus generating counterfactuals. The numbers reflect the contribution of each channel to the difference in the scarring effect of displacement on earnings, k years after displacement, as depicted in Figures 10 and B.4.*

Penalty"). The contribution of these channels is calculated by setting c^f to zero and by setting c^δ equal to δ^{nh}. The "Human Capital" channel corresponds to the difference between human capital depreciation rates ψ_u and $\psi_r\psi_u$, and its contribution is calculated by setting $\psi_r = 1$.

In the model, a non-employed worker holding a potential recall generally transitions back into employment slower than other unemployed workers. In order to calculate the impact of these "Transition Rates", I set the transition rates from the two states equal by setting $\phi^r = \lambda^u$, and $\phi^{rg} = \lambda^{ug}$.[46] The "Loss of Recall" channel refers to the possibility that workers are exogenously forced from the potential recall state into the general unemployment state, and its impact is calculated by setting the corresponding probability ξ^r equal to zero. Similarly, the impact of the possibility that workers may lose eligibility to receive unemployment benefits (denoted "Benefit Expiry") is calculated by setting the corresponding probability ξ^ε equal to zero.

Finally, the model contains three explicit selection channels, which can be referred to as selection into displacement, selection into potential recall, and selection out of potential recall. The selection into potential recall, denoted by "Selection Choice" in the tables and figures, refers to the worker being able to choose whether or not move into the state of non-employment with a

[46]Note that I shut down this "Transition Rates" channel after shutting down the "Selection Search" channel, so at this point I already have $\lambda^r = 0$.

	Regression-Based				Direct Counterfactual			
Channel	$k=0$	$k=1$	$k=5$	$k=10$	$k=0$	$k=1$	$k=5$	$k=10$
Bargaining	-0.000	-0.001	0.000	0.000	$0x$	$0x$	$0.001x$	$0x$
Match Productivity Penalty	0.000	-0.000	0.000	0.000	0	0	-0.001	$0x$
Match Stability Penalty	0.006	-0.018	-0.014	0.001	-0.003	-0.033	-0.012	$-0.002x$
Human Capital	-0.000	-0.000	-0.000	0.000	0	0	0	$0x$
Selection Search	0.008	0.025	-0.001	-0.000	0.002	0.020	-0.000	$-0.000x$
Transition Rates	-0.132	-0.221	-0.004	-0.000	-0.152	-0.284	-0.028	$-0.004x$
Loss of Recall	0.042	0.103	0.000	-0.001	0.000	-0.000	-0.001	$-0.003x$
Benefit Expiry	0.001	-0.000	-0.001	-0.000	0.001	0.001	-0.001	$0.000x$
Selection Choice	0.002	0.002	0.014	0.016	-0.010	-0.010	0.006	$-0.001x$
Match Preservation	0.013	0.016	0.022	0.007	0.002	0.010	0.000	$0.004x$
Total	-0.061	-0.095	0.016	0.022	-0.161	-0.296	-0.035	$-0.006x$

Table B.2: *Summary of the decomposition of the difference in the scarring effect of displacement on employment fraction between (ex-post) recalled and non-recalled workers. The decomposition is generated by turning off the indicated channels one by one (in the presented order), thus generating counterfactuals. The numbers reflect the contribution of each channel to the difference in the scarring effect of displacement on employment fraction, k years after displacement, as depicted in Figures 11 and B.5.*

potential recall upon being offered as such (which happens at rate ϕ^f) and whether to (voluntarily) move from this potential recall state to the regular unemployment state. The contribution of this channel is calculated by removing this choice, thus forcing a worker into the potential recall state with probability ϕ^f and forcing the worker to stay in that state until they are re-employed. The selection out of potential recall, "Selection Search", is incorporated into the model by allowing the non-employed worker with potential recall to search for a new job, which arrives at a rate $\lambda^r\lambda^u$. I shut down this model element by setting $\lambda^r = 0$. Finally, the selection into displacement, "Match Preservation", refers to the fact that (in the model) displaced workers are coming from jobs with lower productivity and higher separation rates, due to the negative correlation between those two job characteristics. The contribution of this final channel is calculated as a residual. After all, if all other channels are shut down, the only remaining difference between the two states is that the workers holding a potential recall eventually move back to their previous job, whereas the other non-employed workers draw a new job from the distribution $G(\theta)$.

Naturally, one might expect that the order in which the aforementioned channels are shut down might matter for the results. In principle, one could use many of the $10! = 3,628,800$ permutations of the 10 channels to perform the decomposition, as long as two restrictions are satisfied. First, since the "Match Preservation" is calculated as a residual, this is necessarily the

	Regression-Based			
Channel	$k = 0$	$k = 1$	$k = 5$	$k = 10$
Bargaining	-0.001	0.000	-0.003	-0.003
Match Productivity Penalty	-0.000	-0.002	-0.002	-0.002
Match Stability Penalty	0.047	-0.055	-0.088	-0.058
Human Capital	0.001	0.023	0.011	0.015
Selection Search	0.007	0.002	-0.002	-0.001
Transition Rates	-0.013	0.082	0.011	0.002
Loss of Recall	0.008	0.002	0.010	0.011
Benefit Expiry	0.001	-0.002	-0.006	-0.008
Selection Choice	0.144	0.112	-0.031	-0.053
Match Preservation	0.069	0.234	0.149	0.113
Total	0.263	0.397	0.051	0.015

Table B.3: *Summary of the decomposition of the difference in the scarring effect of displacement on wages between (ex-post) recalled and non-recalled workers. The decomposition is generated by turning off the indicated channels one by one (in the presented order), thus generating counterfactuals. The numbers reflect the contribution of each channel to the difference in the scarring effect of displacement on wages, k years after displacement, as depicted in Figure B.6.*

last channel to be accounted for. Furthermore, since the "Selection Search" channel affects one of the transition rates for workers holding a potential recall, it should be shut down before the "Transition Rates" channel. Among the remaining $181,440$ permutations, the chosen order starts with the channels that have the most direct impact on the worker's wages after being recalled, and groups together related channels. In particular, I group the "Selection Search" and "Transition Rates" channel as they both affect the probability of moving out of the potential recall state, and I group the two penalties and the "Human Capital" channel as they each affect the direct impact of the recall on the state variables.

B.3 Policy Implications

In Section 5.2, I discussed a counterfactual exercise to illustrate the policy relevance of the findings in this paper. In this subsection, I provide some of the further underlying details, and illustrate how the result changes when I use the direct counterfactual method rather than the regression-based method.

First of all, Figure B.7 shows the results of the decomposition exercise from Figure 13 using the direct counterfactual method instead. Figure B.7 illustrates that using the direct counterfactual gives very comparable result, with the exception of the human capital appreciation no longer appearing as having a positive impact. This is because the counterfactual simulation also has workers potentially increasing their human capital in the period of "displacement", whereas

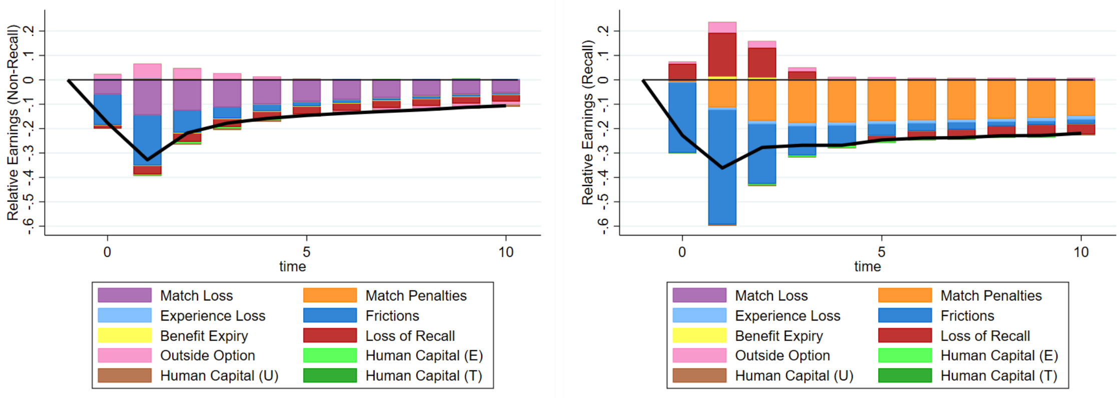


Figure B.7: *A decomposition of the scarring effect of displacement on earnings for (ex-post) non-recalled workers (left) and recalled workers (right), using a direct counterfactual approach. The black line represents the total earnings loss, and corresponds to the blue and red lines in the right panel of Figure B.1. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals. Corresponding numerical values for selected time periods (0,1,5, and 10 years after displacement) can be found in Tables B.4 and B.5.*

this increase is not picked up in the regression-based method, as argued in Section 5.2.

In Tables B.4 and B.5, I show some numerical values used to construct the decomposition in the corresponding Figures 13 and B.7. As discussed in Section 5.2, the method I use to calculate these contributions is identical to the one I used to decompose the differences in earnings and employment losses in Section 5.1 and Appendix B.2. However, since I am now interested in decomposing the magnitude of each of the losses rather than the difference between them, I use a number of different channels, on which I elaborate below.

The first three channels I shut down in the decomposition correspond to the transitions in human capital. I start by shutting down human capital depreciation for potentially recalled workers ("Human Capital (T)") by setting the corresponding parameter ψ_r to zero. Next, I additionally remove human capital depreciation for regularly unemployed workers ("Human Capital (U)") by setting ψ_u equal to zero. Finally, I shut down human capital appreciation on the job ("Human Capital (E)") by setting ψ_e equal to zero.

The next channel I shut down is the "Outside Option" channel. This captures any losses caused by the worker losing their outside option, and thus using the value of their relevant nonemployment state rather than whichever outside option they may have used before. In order to remove

	Regression-Based				Direct Counterfactual			
Channel	$k = 0$	$k = 1$	$k = 5$	$k = 10$	$k = 0$	$k = 1$	$k = 5$	$k = 10$
Match Loss	-0.028	-0.027	-0.058	-0.121	$-0.000x$	$-0.002x$	$-0.002x$	$-0.002x$
Match Penalties	-0.021	-0.159	-0.147	-0.130	-0.008	-0.111	-0.167	$-0.146x$
Experience Loss	0.064	0.061	0.005	0.018	-0.003	-0.010	-0.013	$-0.015x$
Frictions	-0.418	-0.798	0.010	0.100	-0.290	-0.478	-0.057	$-0.019x$
Benefit Expiry	0.006	0.004	-0.000	-0.001	0.000	0.004	-0.008	$-0.009x$
Loss of Recall	0.081	0.292	0.005	-0.051	-0.000	-0.017	-0.164	$-0.170x$
Outside Option	0.046	0.099	0.006	0.021	0.004	0.032	0.011	$0.004x$
Human Capital (E)	0.026	0.049	0.028	0.048	-0.001	-0.002	-0.009	$-0.002x$
Human Capital (U)	-0.002	-0.001	0.003	0.014	-0.001	-0.004	-0.006	$-0.006x$
Human Capital (T)	-0.000	-0.000	-0.001	-0.000	0	-0.000	-0.001	$-0.000x$
Total	-0.245	-0.480	-0.149	-0.102	-0.299	-0.588	-0.413	$-0.365x$

Table B.4: *Summary of the decomposition of the scarring effect of displacement on earnings for (ex-post) recalled workers. The total difference corresponds to the solid red line in the left panel of Figure 9. The decomposition is generated by turning off the indicated channels one by one (presented here in reversed order), thus generating counterfactuals. The numbers reflect the contribution of each channel to the difference in the scarring effect of displacement on earnings, k years after displacement, , as depicted in Figures 13 and B.7.*

this channel, I shut down any variation in outside options for workers of a given match by setting their benchmark values of s, θ and a equal to their current values: $\hat{s} = s$, $\hat{\theta} = \theta$, and $\hat{a} = a$. The "Loss of Recall" and "Benefit Expiry" channels are the same as the identically named channels in the other decomposition exercise, and their contributions are therefore again calculated by setting ξ^r and ξ^ε equal to zero respectively.

In order to shut down any remaining effects driven by the worker spending some time in nonemployment ("Frictions"), I shut down any channel that would lead the worker to move into either of the nonemployment states by setting λ^{ug} and $\phi^{rg}\phi^r$ equal to 1 while additionally removing the possibility for the worker to reject the corresponding offers. In order to remove the effect of the worker moving to a low attachment state upon being displaced and re-employed in a new job ("Experience Loss"), I remove the low attachment state, thus immediately transitioning workers into the high attachment state upon being hired.

The final channel I shut down corresponds to the match penalties c^f and c^δ ("Match Penalties"). I shut down the productivity penalty c^f by setting it to zero. In order to shut down the additional separation risk c^δ I take a similar approach as under the Experience Loss channel above, by shutting down the attachment state corresponding to a recently recalled worker, thus immediately transitioning the recalled worker back into the high attachment state. Any earnings

	Regression-Based				Direct Counterfactual			
Channel	$k=0$	$k=1$	$k=5$	$k=10$	$k=0$	$k=1$	$k=5$	$k=10$
Match Loss	-0.111	-0.207	-0.136	-0.075	$-0.059x$	$-0.144x$	$0.090x$	$0.053x$
Match Penalties	-0.000	0.000	-0.001	-0.002	-0.000	0.000	-0.001	$-0.002x$
Experience Loss	0.022	0.033	0.026	0.022	0.002	0.005	-0.001	$0.000x$
Frictions	-0.087	-0.153	-0.039	-0.023	-0.125	-0.204	-0.015	$-0.005x$
Benefit Expiry	-0.001	-0.001	-0.002	-0.001	-0.000	0.000	-0.000	$-0.000x$
Loss of Recall	-0.013	-0.037	-0.031	-0.030	0	0.000	-0.002	$-0.002x$
Outside Option	0.017	0.061	0.022	0.002	0.024	0.064	0.003	$-0.013x$
Human Capital (E)	-0.002	-0.004	-0.013	-0.016	-0.000	-0.005	-0.002	$0.003x$
Human Capital (U)	-0.000	-0.001	-0.005	-0.009	-0.001	-0.002	-0.007	$-0.009x$
Human Capital (T)	0.000	0.000	-0.000	-0.000	0	0	-0.000	$0.000x$
Total	-0.174	-0.308	-0.180	-0.131	-0.160	-0.286	-0.114	$-0.080x$

Table B.5: *Summary of the decomposition of the scarring effect of displacement on earnings for (ex-post) non-recalled workers. The total difference corresponds to the solid blue line in the left panel of Figure 9. The decomposition is generated by turning off the indicated channels one by one (presented here in reversed order), thus generating counterfactuals. The numbers reflect the contribution of each channel to the difference in the scarring effect of displacement on earnings, k years after displacement, as depicted in Figures 13 and B.7.*

loss that remains after switching off these 9 channels is treated as a consequence of the worker losing their match and thus drawing a new θ from $G(\theta)$. This residual channel is referred to as the "Match Loss" channel.

In Figure B.8, I confirm that the decomposition of the average scarring effect of displacement largely resembles the decomposition for non-recalled workers, reflecting that the group of non-recalled workers is larger than the group of recalled workers. Since the decomposition of the average scarring effect of displacement (on earnings) points towards human capital depreciation as one of the drivers of the long-run losses, it is natural to expect that a policy aimed at helping displaced workers may be targeted at bringing down the human capital depreciation probability for non-employed workers. In Figure 14, I showed that such a policy would disadvantage the recalled worker. In Figure B.9, I show that using the direct counterfactual leads to a similar conclusion, as non-recalled workers are seen to decrease their earnings loss, whereas there is barely any improvement visible for the recalled workers.

B.4 Alternative Calibrations

In this section, I analyze the robustness of the simulation results in Section 5 to an alternative calibration of the model. In this alternative calibration, I reduce the weight of the moments from the baseline estimation to approximately 10% of their original weight, and instead

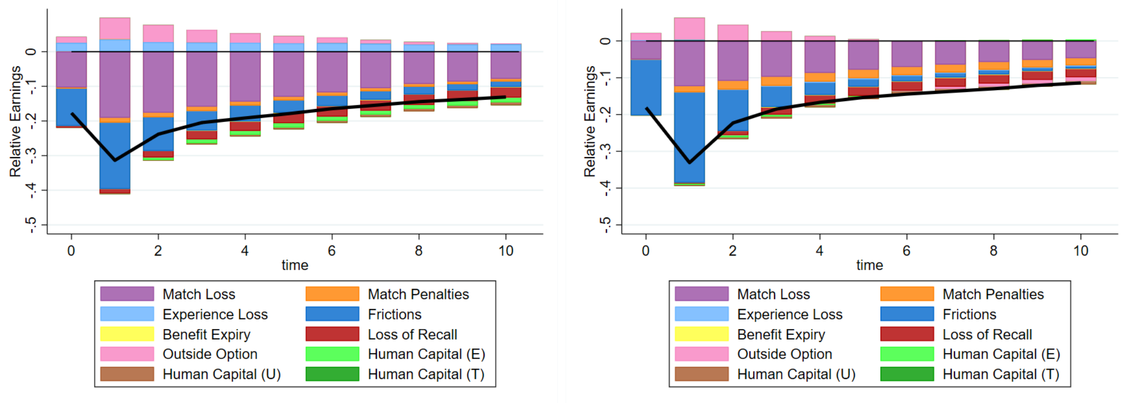


Figure B.8: *Decomposition of the average scarring effect of displacement on earnings, using the regression-based approach (left) or a direct counterfactual (right). The black line represents the total earnings loss. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals.*

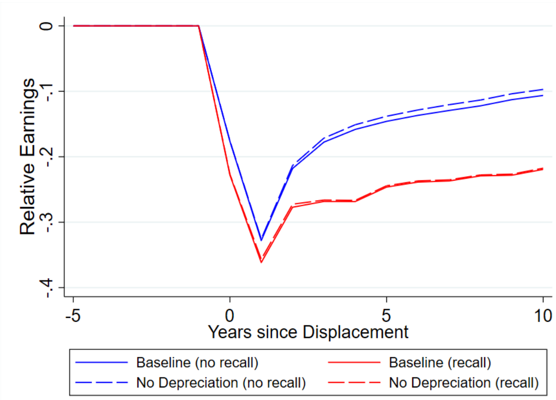


Figure B.9: *The effect of displacement on earnings relative to the control group, by ex-post recall status, estimated using the direct counterfactual method, using model simulation data from the baseline model (solid) and a counterfactual in which non-employed workers do not lose human capital during non-employment ($\psi_u = \psi_r \psi_u = 0$, dashed).*

directly target the scarring effects of displacement by (ex-post) recall status that were estimated in Section 2.3 of the main text. In other words, I target the outcome of the estimation of the following equation:

$$e_{it} = \alpha_i + \gamma_t + \beta X_{it} + \sum_{r=\{0,1\}} \sum_{C \neq 0} \sum_{\substack{k=-4 \\ k \neq -2}}^{K} \delta_k^{C,r} D_{it}^{C,r,k} + u_{it} \tag{B.10}$$

In the data, this equation can be estimated using standard fixed effects estimation. Given the number of individuals in the simulation (and therefore the number of individual fixed effects), however, this is too computationally intensive to execute in each iteration of the calibration. Fortunately, the structure of the model and its simulation allow me to make several simplifications. First, the different cohorts in the data are mainly meant to pick up effects of differences in economic conditions at the time of displacement, but there are no such differences in the model. Therefore, I do not allow for different estimates by cohort in the model equivalent. Then, to get around having to estimate the fixed effects explicitly (since they are not the parameters of interest), I use a within transformation. In particular, this means that for both the dependent and independent variables in equation (B.10), I calculate $\ddot{X}_{it} = X_{it} - \bar{X}_i - \tilde{X}_t + \bar{X}$, where $\bar{X}$ is the average variable over all individuals and time periods, $\tilde{X}_t$ is the average over individuals within a time period t, and $\bar{X}_i$ is the average over all time periods for an individual i. Using this transformation, the equation to be estimated reduces to the following equation:

$$\ddot{e}_{it} = \sum_{r=\{0,1\}} \sum_{\substack{k=-4 \\ k \neq -2}}^{10} \delta_k^r \ddot{D}_{it}^{r,k} + \ddot{u}_{it} \tag{B.11}$$

The above equation can be estimated fairly easily using OLS, which thus yields the model equivalent of the moments (with one moment for every k). Note that the model estimation is not exactly identical to the data equivalent, because the panel in the simulation is not completely balanced, and there do exist some time dependent differences (e.g. due to accumulated human capital). Therefore, the targeting of the scarring effect is not as precise as it would be if I were to estimate (B.10) directly, but the transformation does make this (imperfect) targeting feasible, and therefore allows me to use this for an alternative calibration. Using this alternative calibration, I then analyze whether targeting these empirical results directly might change results.

In Table B.6, I present the equivalent of Table 3 for the calibration using these additional moments (adding in the baseline results from Table 3 for comparison). The additional moments are not included in the table explicitly, as they are more naturally described in an event study graph (as done in Figure B.10)

		Baseline		Alternative Calibration	
Description of Moment(s)	**Data**	**Model**	**Parameters**	**Model**	**Parameters**
Average rate of job loss, tenure 1-3.5y	0.030	0.014	$\eta_\delta = 3.82$	0.069	$\eta_\delta = 5.87$
Average rate of job loss, tenure 3.5-6y	0.016	0.009	$\mu_\delta = 68.6$	0.057	$\mu_\delta = 13.8$
Average rate of job loss, tenure 6-9y	0.012	0.008	$c^\delta = 0.255$	0.051	$c^\delta = 0.639$
Average rate of job loss, tenure>9y	0.008	0.007	$\delta^{nh} = 0.054$	0.045	$\delta^{nh} = 0.065$
Average rate of job loss	0.023	0.012		0.073	
Relative subsequent separation, recall	0.109	0.076		0.136	
p75-p25 ratio of wages	1.644	1.975	$\eta_y = 8.71$	1.747	$\eta_y = 30.3$
Median-p25 ratio of wages	1.263	1.386	$\mu_y = 3.24$	1.343	$\mu_y = 1.07$
Job-to-job transition rate	0.028	0.020	$\lambda^e = 0.053$	0.067	$\lambda^e = 0.084$
Displacement among job-to-job transitions	0.424	0.311	$\lambda^{ug} = 0.544$	0.734	$\lambda^{ug} = 0.729$
Average job finding rate	0.140	0.221	$\lambda^u = 0.226$	0.173	$\lambda^u = 0.158$
Replacement rate	0.6	0.652	$b = 1.278$	1.674	$b = 1.456$
Yearly wage growth	0.013	0.006	$\psi_e = 0.033$	-0.008	$\psi_e = 0.003$
Pre- to post-layoff wage,	-0.058	-0.051	$\psi_u = 0.048$	-0.067	$\psi_u = 0.305$
duration gradient	-0.027	-0.024	$\psi_r = 0.144$	-0.083	$\psi_r = 0.252$
Pre- to post-recall wage, duration 0.25-0.5y	-0.019	-0.020	$c^f = 0.194$	-0.094	$c^f = 0.735$
Pre- to post-recall wage, duration gradient	0.010	0.011		-0.024	
Recall rate	0.058	0.042	$\phi^f = 0.542$	0.088	$\phi^f = 0.284$
Recall materialization rate	0.299	0.341	$\phi^r = 0.154$	0.314	$\phi^r = 0.164$
	0.296	0.369	$\phi^{rg} = 0.714$	0.360	$\phi^{rg} = 1.022$
New job finding rate, workers expecting a recall	0.293	0.207	$\lambda^r = 0.174$	0.259	$\lambda^r = 0.851$
New job finding rate, before expected recall	0.25	0.301	$\xi^r = 0.152$	0.990	$\xi^r = 0.007$
Wage of newly hired worker	0.682	0.748	$\kappa = 0.852$	0.799	$\kappa = 0.842$
Coefficient $\hat{\gamma}$ in equation (16)	-0.020	0.001	$\rho = -19.1$	-0.005	$\rho = -21.6$

Table B.6: *A summary of calibration moments, their values in the data and in the calibrated model, and corresponding parameter values, from the baseline calibration and an alternative calibration that directly targets empirical results.*

Comparing the estimated parameters in the final column of Table B.6 to those in the baseline calibration, a few parameter values stand out. First of all, the marginal distribution of separation rates is much less concentrated towards low values for δ, as indicated by the values of μ_δ and η_δ being closer together, resulting in a very high unconditional job loss rate. When it comes to match productivity y, the opposite is true, as indicated by the much higher value of η_y. The narrower productivity distribution is likely one of the main reasons why, despite a fairly similar value of b, the replacement rate is substantially above 100% in the alternative calibration. The human capital still increases very slowly during employment, but the depreciation is much faster than in the baseline, as indicated by the substantially higher value of ψ_u. Non-employed workers holding a potential recall also face a higher depreciation probability of $\psi_r \psi_u \approx 0.077$ rather than 0.007. In terms of transition rates, it is worth pointing out that the alternative calibration suggests a higher job-to-job offer arrival rate for the employed worker (0.084 instead of 0.053), which together with the higher job-to-job offer rate upon displacement (λ^{ug}) translates into a much higher simulated job-to-job transition rate.

When comparing the parameters directly related to the recall possibility, it can be seen that the recall offer probability is lower than it was in the baseline calibration. At the same time, the arrival rate of new job offers while holding a potential recall is much higher in the alternative calibration, leading the transition rate back into employment to be much higher in the potential recall state than in the regular unemployment state, while reducing forced transitions between the two (through the loss-of-recall shock) due to the very low value of ξ^r. The resulting realized recall rate in the alternative calibration is much higher than in the baseline calibration. Finally, it is worth noting that the post-recall penalties on the separation rate and productivity are much higher in the alternative calibration.

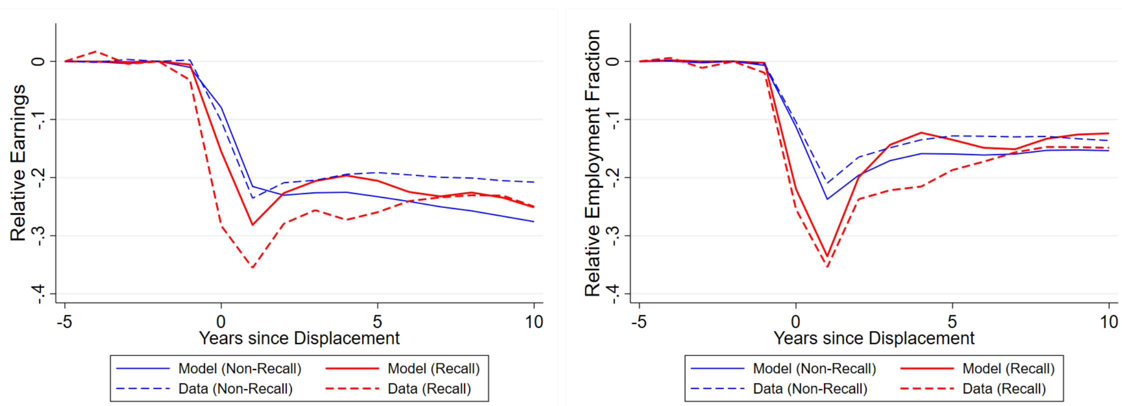


Figure B.10: *The effect of displacement on earnings (left) and employment (fraction of the year spent in an employment spell, right) relative to the control group, by ex-post recall status (materialization of recall within 5 years), using model simulation data from the alternative calibration model (solid) and using the data (dashed, corresponding to Figure 4).*

In Figure B.10, I show the estimated effect of displacement on earnings and employment fraction by ex-post recall status, compared to the results in Figure 4. As can be seen by comparing Figure B.10 to its equivalent from the baseline estimation, Figure 9, the alternative calibration also does much better in matching the persistence of the losses, although the differences between recalled and non-recalled workers in the long run are similar.

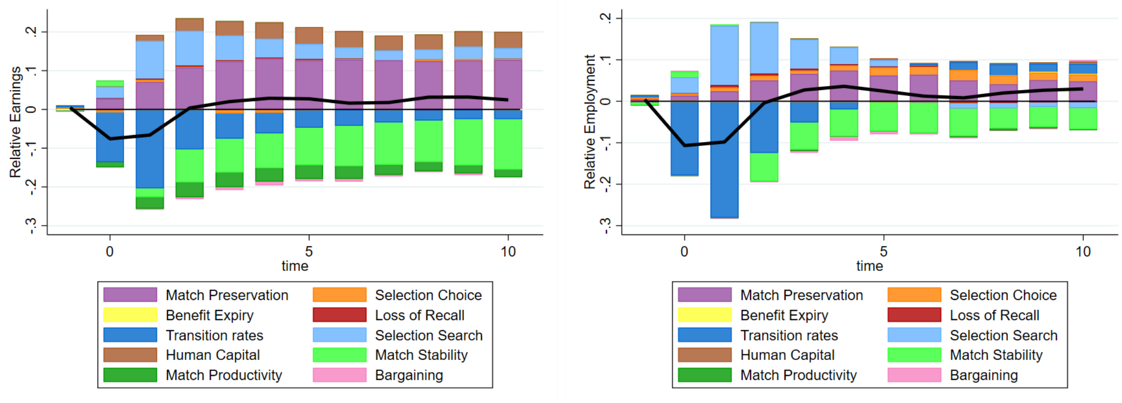


Figure B.11: *A decomposition of the difference in the scarring effect of displacement on earnings (left) and employment (right) between (ex-post) recalled and non-recalled workers, using the alternative calibration of the model. The black line represents the total difference, calculated as the difference between the solid red and blue lines in the left panel of Figure B.10. The decomposition is generated by turning off the indicated channels one by one (starting with those depicted on the outside), thus generating counterfactuals.*

In the left panel of Figure B.11, I fully decompose the differences in estimated post-displacement earnings between recalled and non-recalled workers (as shown in the left panel of Figure B.10), just like I did for the main calibration in Figure 10. Comparing the decomposition in Figure B.11 to the decomposition in Figure 10 reveals a fairly similar picture. In particular, it still holds that the main negative drivers for the recalled worker are the higher probability of subsequent separation and the differences in transition rates back into employment. In addition, the larger magnitude of the productivity penalty makes it clearly visible as one of the persistently negative drivers. The main positive driver is still the preservation of the match in the case of a recall, although the human capital channel is also shown to have a clear positive impact. Finally, the low value of the loss-of-recall shock probability in the alternative calibration has lead it to largely disappear from the decomposition figure. In its place, the “Selection Search” channel emerged as a positive driver, driven by the higher arrival rate of new jobs in the potential recall state.

In the right panel of Figure B.11, I further decompose the employment differences into the same 10 channels used for the earnings decomposition above. The decomposition of the employment differences is in line with the left panel of Figure B.5, primarily highlighting the importance of the “Transition Rates” channel, which accounts for the full impact in the short run, but quickly decreases in importance and is eventually overtaken by the “Match Stability” channel. Just like in the earnings decomposition, the role of the loss-of-recall shock is substantially reduced, mostly offset by the increased contribution of the “Selection Search” channel.

C Data Appendix

C.1 Individual Summary Statistics

	Frequency	Mean	(Std.Dev.)
Age	24.4m	41.25	(9.93)
Primeage (aged 35–60)	22.8m	0.6906	(0.462)
Gender (female)	24.4m	0.4636	(0.499)
Education (university)	23.0m	0.1555	(0.362)
Location (east)	19.6m	0.1892	(0.392)
Self-employed	7.8m	0.0192	(0.137)
Establishment size	20.1m	1,147	(4,610)
Establishment tenure (days)	20.5m	2,241.8	(2,253)
Job tenure (days)	20.5m	2,122.8	(2,202)
Yearly earnings (2015 Euros)[47]	22.6m	27,349.4	(20,621)
Separation	20.1m	0.1033	(0.304)
Displacement	19.7m	0.0145	(0.119)
Recall (conditional on separation)	1.5m	0.1379	(0.344)

Table C.1: *Summary statistics from the yearly sample. The table shows the estimated mean and standard deviation of a number of important variables, using the main sample from SIAB (as defined in Section 1, without any of the further restrictions imposed for the estimation).*

Table C.1 presents summary statistics for a number of worker-related variables mentioned throughout the main text. The summary statistics show that the data substantially undersamples self-employed workers. This is because the structure of the social security system is such that self-employed workers would often not be recorded in the administrative data my analysis is based on. Second, both workers residing in East Germany and female workers are slightly undersampled. Separation, displacement, and recall (conditional on separation) rates are in line with those discussed in the main text in Section 2.1.

C.2 Establishments in the Sample

In Section 2.1, I highlighted some of the differences between recalled, non-recalled, and non-displaced workers. However, one could imagine that some of these differences may also be driven

[47] In these yearly earnings, only earnings from employment are taken into account.

by the establishments these workers separated from. In Table C.2, I compare recalling, displacing and non-displacing establishments in the data on a number of variables. As can be seen in the table, establishments that recall workers generally tend to recall a large fraction (roughly 76% of observed displacements) of their displaced workers. Similarly, conditional on a mass layoff taking place, the size of the layoff tends to be very substantial, with 79% of the workers observed at that establishment being separated. In line with the observations in Section 2.1, recalling establishments tend to be larger, and tend to have a lower median wage. Displacing establishments, in turn, tend to be larger and have lower median wages than the average establishment in the data.

	Recalling (N=11,600)		Displacing (N=142,940)		All (N=1,485,816)[48]	
	Mean	(Std.Dev.)	Mean	(Std.Dev.)	Mean	(Std.Dev.)
Establishment size	177.7	(1,026)	96.33	(421)	34.65	(171)
Median wage (percentile)	43.4	(27.8)	48.2	(31.8)	50.8	(30.9)
Displaced (fraction)	0.761	(0.31)	0.793	(0.30)	0.793	(0.30)
Recalled (fraction of displaced)	0.780	(0.31)	0.063	(0.23)	0.063	(0.23)

Table C.2: *Summary statistics for establishments in the yearly sample, by displacement and recall status. The table shows the estimated mean and standard deviation of a number of important variables, using the main sample from SIAB (as defined in Section 1, without any of the further restrictions imposed for the estimation) and taking out duplicate observations for establishments.*

The high fraction of workers being displaced (conditional on displacement taking place) can partially be explained by the fact that establishment closures are included in my definition of a displacement. However, it should be noted that in order for an establishment exit to be denoted as a closure, a few more restrictions need to be satisfied. Since the dataset provides information on what happens to the majority of an establishment's former employees after an establishment exits, it is possible to distinguish between several exit types. Using the definitions from Hethey and Schmieder (2010), I define three exit types. Type A exits are interpreted to be a consequence of an establishment ID change, a takeover, or a spinoff. In practice, this means that the exiting establishment had at least 4 employees, and either at least 80% of the (newly entered) establishment at which the majority of workers are re-employed consists of workers from the exiting establishment, or at least 80% of the workers from the exiting establishment find work at the same (existing) establishment but do not make up more than 80% of the employment at their new establishment. An exit is classified as type B (establishment death) if either the exiting establishment had 3 employees

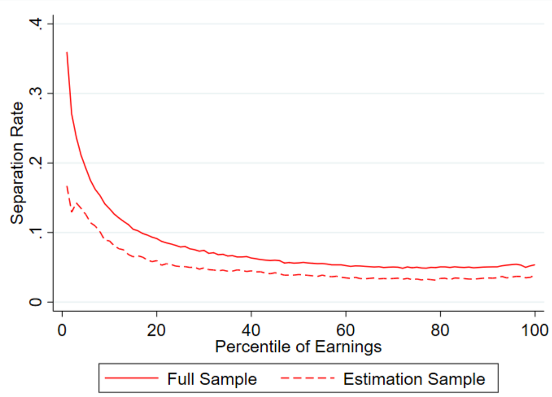
[48]Note that the statistics on "Displaced (fraction)" and "Recalled (fraction of displaced)" require an establishment to have displaced some workers. As such, these statistics are based on less observations in the rightmost column, and are instead calculated using 142,940 observations.

or less, or no more than 30% of the former employees of the establishment find employment at the same establishment (and if that establishment is an entrant, the former employees of the exiting establishment do not make up more than 80% of the new establishment's employment). Finally, an exit is classified as type C if it does not satisfy the conditions for type A and B. When defining displacement, I use only exits of type B.

C.3 Further Empirical Results

C.3.1 Further observations on the incidence of recall

In this subsection, I further analyze the incidence of separation, displacement, and recall (conditional on displacement), beyond the analysis in Section 2.1. Just like in Section 2.1, the graphs and tables in this section are generated (unless specified otherwise) using the sample in which I have not applied the restrictions on pre-displacement establishment tenure and establishment size used to generate the sample for the regression-based analysis (in Sections 2.2 and 2.3, as well as Appendices C.3.2 to C.3.4).



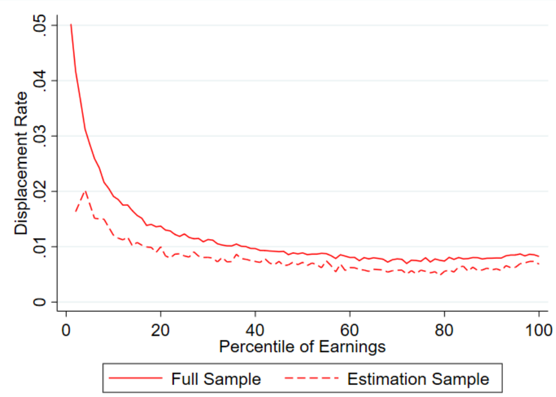


Figure C.1: *The incidence of separation (left) and displacement (right) over the recent earnings distribution, plotted for the full sample (solid) and the regression estimation sample (dashed).*

In Figure C.1, I show that the separation and displacement rates (for both the full sample and the regression estimation sample) tend to be higher for individuals located lower on the (recent) earnings distribution.[49] This can be thought of as supporting the idea of a job ladder

[49]The construction of the recent earnings distribution largely follows Guvenen et al. (2017). Recent earnings are calculated as the average earnings of an individual between years $y-5$ and $y-1$. When deriving these recent earnings, I condition on the individual having earnings available in the data for at least three of the years between $y-5$ and $y-1$, which must include year $y-1$. I then use these recent earnings to generate the recent earnings distribution, which is generated separately for each year, age group (below 35 and 35 to 60), gender, and location (East and West).

with slippery bottom rungs, where higher productivity matches are also more stable, as implied by the estimated model in Jarosch (2023). However, the pattern is not quite monotonic throughout the distribution: above the 70th percentile of the distribution, the displacement rates are slightly increasing again.

Figure C.2: *The incidence of recall within 5 years of displacement (as a fraction of total displacements), by percentile of the recent earnings distribution, and plotted separately for the full sample (solid) and the regression estimation sample (dashed).*

Figure C.2 shows the incidence of recall (within 5 years) after displacement, over the recent earnings distribution. The recall rate conditional on displacement is consistently above 1.6% across the recent earnings distribution (1.4% in the regression estimation sample), and much higher towards the bottom of the distribution. This indicates that, while recall is more prevalent for low earning workers, it is not a phenomenon exclusive to these workers. The recall rate itself may be a relatively low fraction, but given that workers may follow a very different path after job loss if they expect to be recalled, it is important to consider these workers separately.

Figure C.3 shows how the incidence of recall (within 5 years), conditional on displacement, changes if establishment closures are excluded from the definition of a mass layoff. As workers who are laid off from a closing establishment cannot be recalled, these closures mechanically drive down the recall rate. Indeed, as can be observed in the figure, excluding these establishment closures leads to an increase in the recall rate of more than 2 percentage points (from an average of 7.3% to an average of 9.6%).

Figure C.4 shows how the recall rates are affected by including so-called “indirect” recalls, that is, counting all workers who return to their former establishment (within 5 years) after working somewhere else in between, rather than only including such workers if they started their

Figure C.3: *The incidence of recall within 5 years of job loss over time, conditional on displacement, with and without including establishment closures in the definition of a displacement.*

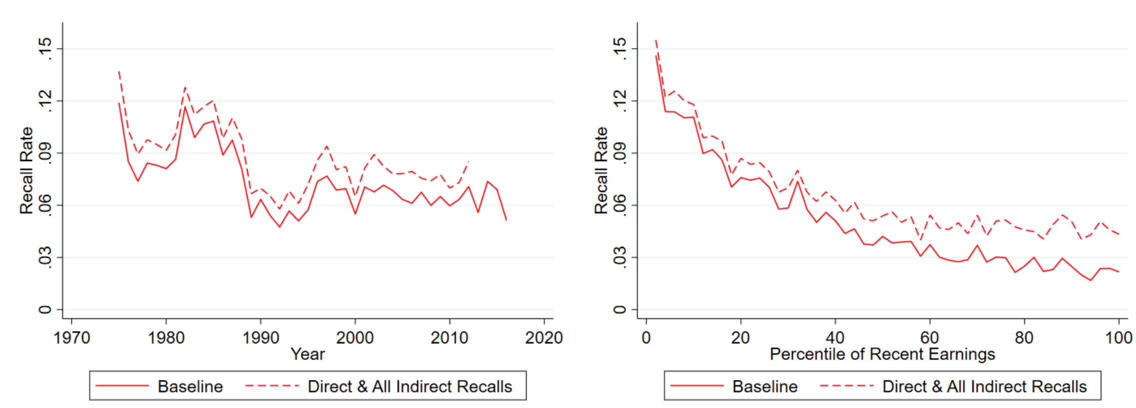


Figure C.4: *The incidence of recall within 5 years of job loss over time (left) and the recent earnings distribution (right), conditional on displacement, with and without including all indirect recalls.*

other job within 31 days of their initial displacement and returned to their original establishment in the same year or in the next year (as in the baseline definition). Including these recalls naturally increases the recall rate. However, as the figure shows, the additional recalls are generally spread out over time and over pre-displacement earnings (though slightly more prevalent at the top), so that excluding these indirect recalls from the estimation does not seem to strongly bias the sample towards a certain year or towards workers with higher or lower pre-displacement earnings.

	Time to recall				
Time to re-separation	$<$ 1mo	1-3mo	3-6mo	6-12mo	$>$ 12mo
$<$ 1 month	0	0.027	0.021	0.026	0.012
1-3 months	0	0.052	0.046	0.063	0.026
3-6 months	0	0.046	0.080	0.073	0.032
6-12 months	0	0.047	0.175	0.092	0.051
$>$ 12 months	0	0.002	0.027	0.049	0.055

Table C.3: *Distribution of displaced and (ex-post) recalled workers over different combinations of time to recall and time to re-separation (in months). The table lists the fraction of workers falling into the relevant combination of categories, relative to the total number of displaced and recalled workers for whom both recall and re-separation is observed.*

In Table 2 in the main text, I showed how recalled workers are distributed along different combinations of time to time to re-employment and re-separation. Table C.3 repeats the analysis from Table 2, but uses the time to recall instead of the time to re-employment, noting that the two may be different for indirect recalls. Indeed, due to the definition of recall discussed in Section 1, the time to recall is always at least 1 month. Using the table, it can be calculated that 82.4% of recalls (81.9% if I also incorporate recalls for whom I do not observe re-separation) materialize within a year of the displacement. Following the main text in flagging workers as potentially seasonal if their time to recall is between 3 and 12 months and their time to recall and time to re-separation add up to approximately a year, Table C.3 flags up to 41.9% of the recalled workers as potentially seasonal. In Table C.4, I show what this distribution looks like for non-recalled workers (using time to re-employment as in Table 2). Based on this table, and the same reasoning as above, 9.2% of the non-recalled workers are flagged as potential seasonal workers.

It is important to note that the high fraction of potentially seasonal workers calculated above is partially driven by the coarseness of the categories depicted in the accompanying tables. Instead, one could also directly calculate the fraction of workers who satisfy the pattern I would expect from a seasonal worker, which is separation from the same (or a similar) job around the same time every year. Correspondingly defining potential seasonality as the time to recall (re-

	Time to re-employment				
Time to re-separation	$<$ 1mo	1-3mo	3-6mo	6-12mo	$>$ 12mo
$<$ 1 month	0.023	0.007	0.006	0.008	0.017
1-3 months	0.057	0.012	0.012	0.016	0.031
3-6 months	0.092	0.018	0.017	0.020	0.039
6-12 months	0.172	0.032	0.028	0.028	0.058
$>$ 12 months	0.165	0.013	0.017	0.032	0.081

Table C.4: *Distribution of displaced and (ex-post) non-recalled workers over different combinations of time to re-employment and time to re-separation (in months). The table lists the fraction of workers falling into the relevant combination of categories, relative to the total number of displaced and non-recalled workers for whom both re-employment and re-separation is observed.*

employment for non-recalled workers) and time to re-separation adding up to a number between 11 and 13 months, I flag 27.1% of the recalled and 9.2% of the non-recalled workers in the full sample of displaced workers as potential seasonal workers. In the regression estimation sample, I find comparable shares of 26.9% and 9.1% respectively. In Appendix C.3.4, I show that taking these workers out of the estimation does not weaken the main result from Section 2.3 of the main text.

Finally, the results in the main text as well as in this Appendix may leave one to wonder whether the workers I observe being recalled to their displacing establishment are simply more likely to separate (compared to non-recalled workers) regardless of whether they were recalled. Alternatively, one might think that jobs at recalling establishments are generally more unstable than jobs at non-recalling establishments. In order to investigate this, I perform an exercise in the spirit of the evidence supplied by Jarosch (2023) in favour of establishment-level heterogeneity in job security. Since separations (and especially repeated separation) are likely to occur multiple times a year, I use the quarterly dataset for this exercise. On the worker level, I regress a worker separation indicator on a set of individual and time fixed effects, and plot the distribution of the resulting individual fixed effects separately for (ex-post) recalled and non-recalled workers (although they are estimated jointly). On the establishment level, I do a similar exercise, using the establishment-wide separation rate among observed workers rather than an indicator. In order to avoid contamination by the separations that occur after the initial displacement, I only consider job separations prior to the worker's (or establishment's) initial observed displacement.

Figure C.5 shows the resulting distributions of worker- and establishment-level fixed effects. These results provide some suggestive evidence that it does indeed seem to be the case that

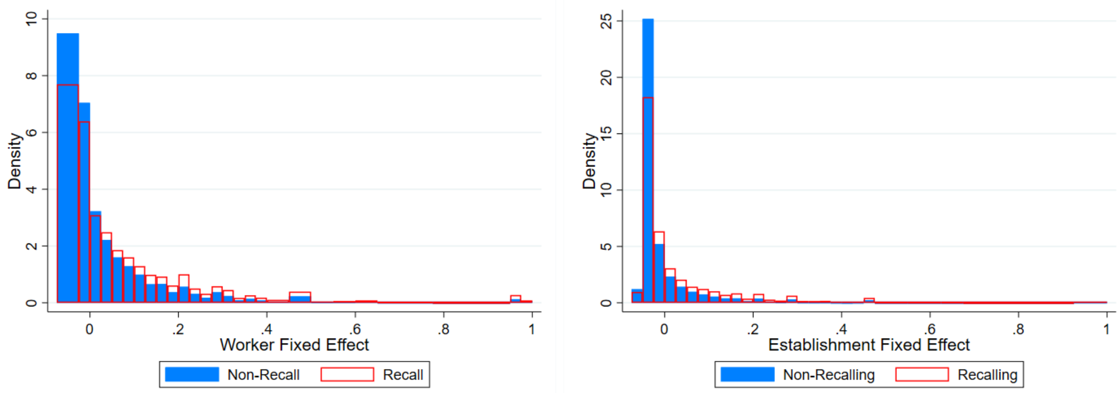


Figure C.5: *The distribution of estimated worker fixed effects (left) and establishment fixed effects (right), estimated from a regression of separation indicators (for the worker) or separation rates (for the establishment) on unit and time fixed effects. The distribution is plotted separately for recalled (red) and non-recalled (blue) workers and for recalling (red) and non-recalling (blue) establishments, although the fixed effects are estimated jointly.*

recalled workers are generally slightly more likely to separate. Similarly, recalling establishments generally tend to have a slightly higher separation rate. However, for both workers and establishments the differences appear quite small, and it is not possible to definitely conclude whether either recalled workers or recalling establishments (or both) are generally more unstable.

C.3.2 Methodological Considerations

In Section 2.2, I pointed out that my estimates of the average scarring effect of displacement are fairly large and persistent compared to estimates from the existing literature. In this subsection, I show that these differences can be partially explained by differences in empirical methodology. In particular, I show how the results from Sections 2.2 and 2.3 change when I use a different method instead of the interaction-weighted estimator from Sun and Abraham (2021).

The alternative method, which I will refer to as the stacking method in what follows, resembles the empirical methodology in recent work (e.g. Flaaen et al. (2019), Jarosch (2023), and Schmieder et al. (2023)) in that I create a separate dataset for each displacement year y. Each dataset consists of a group of displaced workers (displaced in y) and a group of control workers (not separated in y), both of whom are employed at an establishment with at least 50 workers in year y, with an establishment-level tenure of at least 6 years. For these workers, I keep only their observations for years $y-5$ to $y+10$. The resulting datasets are then stacked, after which I estimate the following equation:

$$e_{it} = \alpha_i + \gamma_t^C + \beta X_{it} + \sum_{\substack{k=-5 \\ k\neq -2}}^{K} \delta_k D_{it}^k + u_{it} \tag{C.1}$$

The notation in Equation (C.1) is similar to the notation introduced in Section 1. In particular, I include individual fixed effects α_i and a quartic in age βX_{it} and focus primarily on the coefficient on a set of indicator variables D_{it}^k which indicate that individual i was displaced in period $t-k$. Note that the coefficient δ_k is now no longer estimated separately for each displacement year (cohort c). However, I do allow for each cohort to have different time fixed effects γ_t^C. Note that an individual may appear more than once in the stacked dataset, as they may satisfy the restrictions on establishment-level tenure and size in multiple years, and they may even appear as a treated worker in one cohort and a control worker in other cohorts (such as those corresponding to the years before their displacement). In the estimation of Equation (C.1), these multiple instances of the same person are treated as if they are separate individuals (i.e. different instances correspond to a different i).

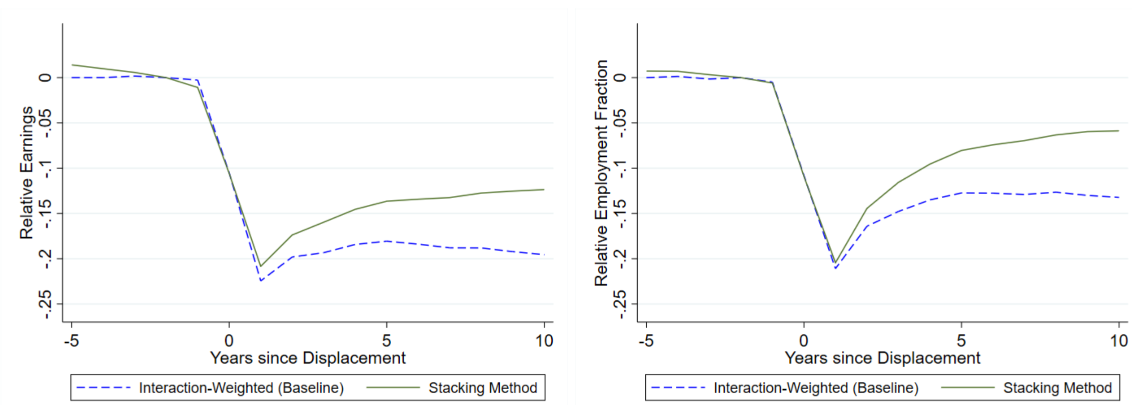


Figure C.6: *The effect of displacement on earnings (left) and employment fraction (right), relative to the control group. The dashed line mirrors the estimates from Figure 3, whereas the solid line estimates the effect using the stacking method.*

In Figure C.6, I compare the estimates of the average scarring effect of displacement obtained from using the stacking method to those obtained from the interaction-weighted method. The estimates obtained using the stacking method indicate that the effect of displacement on earnings and employment is much less persistent than suggested by the interaction-weighted method. This is in line with the discussion in Krolikowski (2018), who shows that using a control group that does not condition on future displacement status yields a much smaller long-term earnings loss than using a control group of never-displaced workers. Indeed, the estimates obtained by the stacking method are roughly in line with the most comparable estimates from the recent literature using German data. In particular, Schmieder et al. (2023) find that male displaced workers earn about 15% less after 10 years, whereas their employment loss after 10 years is about 10%. Their estimated losses upon impact (30% for earnings and 25% for employment) are slightly larger than what I find despite using data from the same source, but this may also be partially attributable to their main sample consisting of men from West Germany. Furthermore, they use a matched control group whereas the results in the stacking method displayed here include all non-separated workers in the year of interest as the control group (as long as they satisfy the aforementioned restrictions). The empirical estimation in Jarosch (2023) also uses data from the same source, and he finds similar long-run earnings losses (roughly 15% after 10 years) in his mass layoff sample. However, his estimated initial losses are much larger (larger than 50%), which is likely to be a consequence of the sample of displaced workers being restricted to workers who are observed to move into unemployment.

Given that the stacking and interaction-weighted methods yield such different estimates, the natural question is whether one should be preferred over the other. The main difference between the two methods lies in the control group. In the interaction-weighted method used in the main text of this paper, the control group consists of workers who are never observed to be displaced, but could have been in the treatment group if they were, i.e. they satisfy restrictions on establishment-level tenure and size at least once. As such, the control group consists of a group of individuals with a very stable employment history. In the stacking method, the control group contains workers who may be displaced in a different year, as long as they satisfy the restrictions on tenure and establishment size in a specific year. For that specific year, these workers are more likely to be similar to the treatment group (prior to the displacement) than the group of never-displaced workers. As such, one might reasonably prefer using the stacking method based on the composition of the control group.

On the other hand, by estimating Equation (C.1) on the stacked sample, and in particular by forcing coefficient δ_k to be the same for different cohorts, the stacking method does not allow the effect of displacement to be different for workers who are displaced in different years, which could be the case if displaced workers in different years are not comparable (e.g. due to different macroeconomic conditions). The interaction-weighted method not only allows for this, but does so simultaneously (by estimating all coefficients in a single estimation). Furthermore, the comparison between a displaced worker and a control worker who may be displaced in the future may make the interpretation of the coefficients unclear. After all, an apparent recovery in earnings may not actually reflect the displaced worker recovering their earnings, but rather the control worker being displaced. One might therefore prefer the interaction-weighted estimator, since it does not suffer from these drawbacks.

Given that I show that the estimated average earnings and employment losses are affected substantially by the method, one might wonder whether the method may differentially affect recalled and non-recalled workers, and thus affect the main conclusion in Section 2.3. In order to use the alternative method to re-estimate these results as well, I make an adjustment to Equation (C.1) in line with the adjustment I made to Equation (1). In particular, I allow for two sets of indicators $D_{it}^{r,k}$, one for each post-displacement recall status r, keeping everything else the same:

$$e_{it} = \alpha_i + \gamma_t^C + \beta X_{it} + \sum_{r=\{0,1\}} \sum_{\substack{k=-5 \\ k\neq -2}}^{K} \delta_k^r D_{it}^{r,k} + u_{it} \tag{D.1'}$$

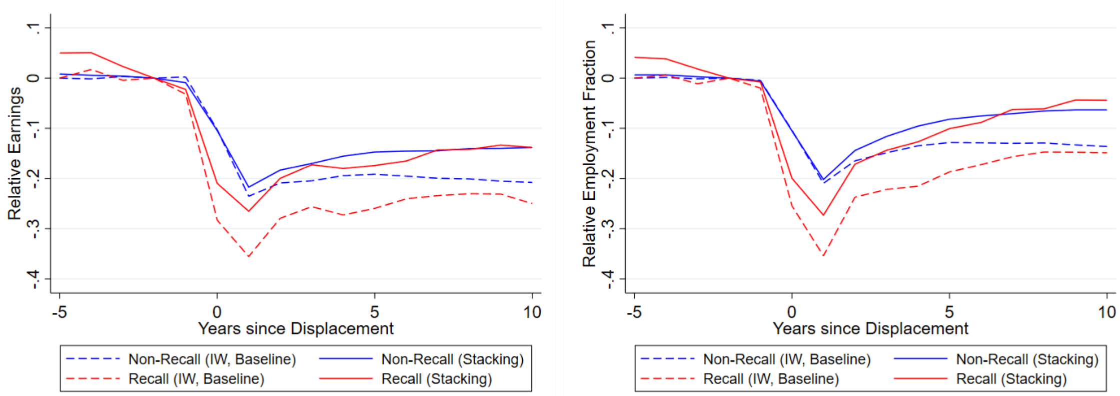


Figure C.7: *The effect of displacement on earnings (left) and employment fraction (right) by ex-post recall status, relative to the control group. The dashed lines mirror the estimates from the Figure 4, whereas the solid lines estimate the effect using the stacking method.*

Figure C.7 shows how the estimated earnings and employment losses by ex-post recall status change if I use the stacking method instead of the interaction-weighted method. In line with the discussion of the average effect, I find that losses are estimated to be less persistent when using the stacking method. In the case of the recalled workers, the loss upon impact is also slightly smaller. However, the comparison between recalled and non-recalled workers remains largely intact: the stacking method also leads me to conclude that recalled workers suffer from larger earnings and employment losses in the short run.

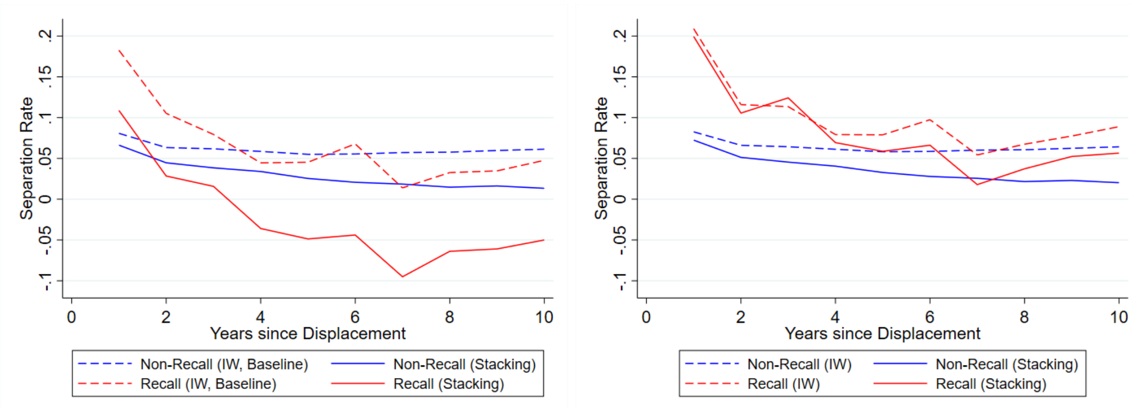


Figure C.8: *The effect of displacement on separation rates by ex-post recall status, relative to the control group. The dashed lines depict the effect estimated using the interaction-weighted method, whereas the solid lines estimate the effect using the stacking method. The left panel shows the comparison for the baseline sample, whereas the right panel conditions on workers not having been separated in the two years prior to displacement*

When I do the same comparison between the two methods for the estimated effect of displacement on subsequent separation rates, as shown in the left panel of Figure C.8, an interesting pattern arises. In particular, the estimates from the stacking method suggest that the long-run effect on separation rates is negative for recalled workers. Given the different composition of the control group, this is not infeasible, but it nevertheless raises some questions about the employment patterns of the group of recalled workers. Indeed, it is possible that the separation rate in the baseline period (2 years prior to displacement) is not zero for recalled workers if they were separated (but not displaced) and recalled before. This is possible because the establishment tenure variable does not reset upon separation and subsequent recall, and so the workers may still satisfy regression restrictions as long as they have worked 6 years at the establishment, even if those 6 years may have been spread out over a period of more than 6 years. As I show in the right panel of Figure C.8, conditioning on workers not having separated in the two years prior to displacement largely restores the patterns seen in the main text, and in this case the comparison between ex-post recalled and non-recalled workers is once again similar between the two methods. I show in Appendix C.3.3 that the estimated earnings and employment losses are not affected by conditioning on workers not having been separated in the two years prior to displacement.

C.3.3 Recalled Workers' Displacement Scars: Robustness

In Section 2.3, I showed how the scarring effect of displacement for workers who are recalled to their previous employer compares to that of workers who are not recalled. In this subsection, I highlight the robustness of these results.

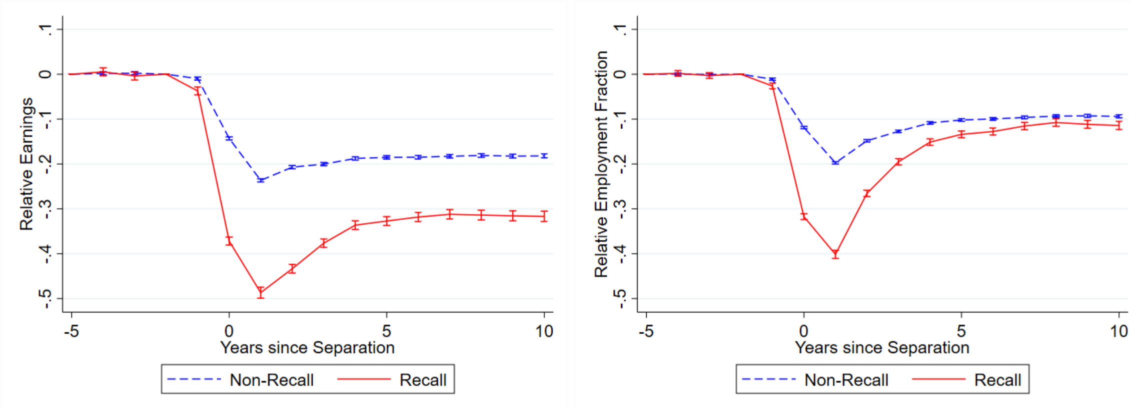


Figure C.9: *The effect of separation on earnings (left) and employment fraction (right) by ex-post recall status, relative to the control group of never-displaced workers, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals.*

In Figure C.9, I show how the baseline result changes if I include all separations rather than only displacements, noting that the recall rate conditional on separation is higher than the recall rate conditional on displacement (see Figure 2). Compared to the estimated effect of displacement discussed in Section 2.3, it can be seen that the magnitude of the effects on earnings and employment are generally slightly larger, especially for recalled workers. As a result, it remains true that recalled workers suffer from larger earnings losses than non-recalled workers.

In Figure 6, I showed that earnings losses are worse for recalled workers than for non-recalled workers when conditioning on experiencing at most 30 days of nonemployment (and vice versa if conditioning on being nonemployed for more than 30 days). A similar picture arises when looking at employment fraction, as seen in Figure C.10. This supports the narrative of pre-layoff search, thus suggesting that immediate transitions should be taken into account when modeling the earnings losses experienced by displaced workers, as I do in Section 3.

In Figure C.11, I show how the estimated post-displacement earnings and employment losses by ex-post recall status change when relaxing the sample restrictions on pre-displacement establishment tenure. Relaxing the restrictions on establishment tenure reduces the losses. This

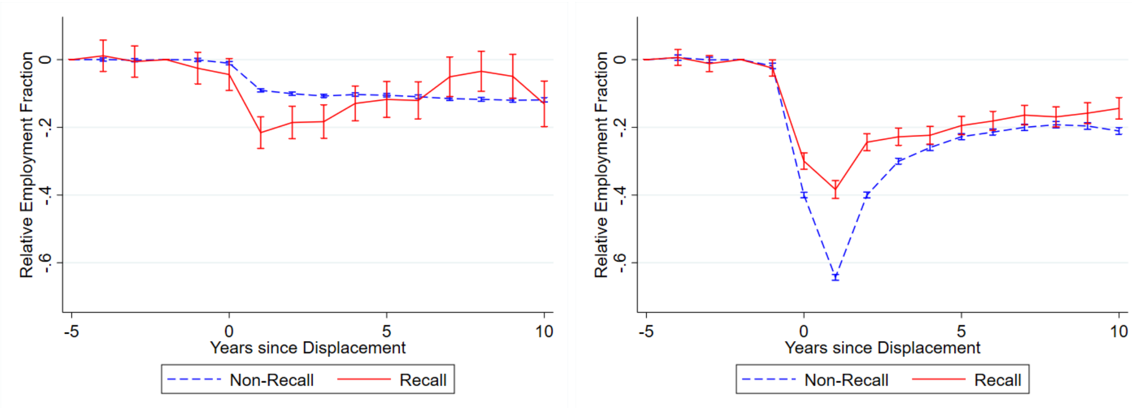


Figure C.10: *The effect of displacement on employment fraction by ex-post recall status, relative to the control group, using estimated coefficients from equation* (1'). *The effects are estimated separately for workers who transition within 30 days (left) and workers who spend time in nonemployment (right) after displacement. The error bars correspond to 95% pointwise confidence intervals.*

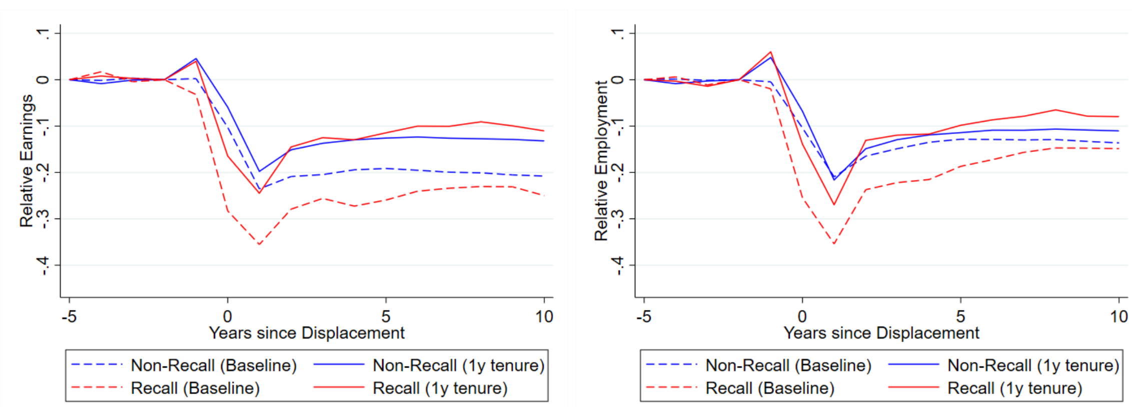


Figure C.11: *The effect of displacement on earnings (left) and employment fraction (right) by ex-post recall status, relative to the control group, using estimated coefficients from equation 1'. The dashed lines mirror the estimate from Figure 4, whereas the solid lines estimate the effect either after relaxing sample restrictions on pre-displacement establishment tenure.*

change is stronger for recalled workers than for non-recalled workers, so that in this sample recalled workers no longer face larger long-run losses than non-recalled workers.[50]

Figure C.12 shows that the comparison between recalled and non-recalled workers in Figure 4 of the main text is not largely affected by taking out potentially seasonal workers from the estimation sample, defining workers as potentially seasonal if their time to recall (or re-employment) and time to re-separation adds up to a number between 11 and 13 months.[51] If anything, removing these potentially seasonal layoffs and recalls strengthens my result that recalled workers experience larger losses than non-recalled workers.

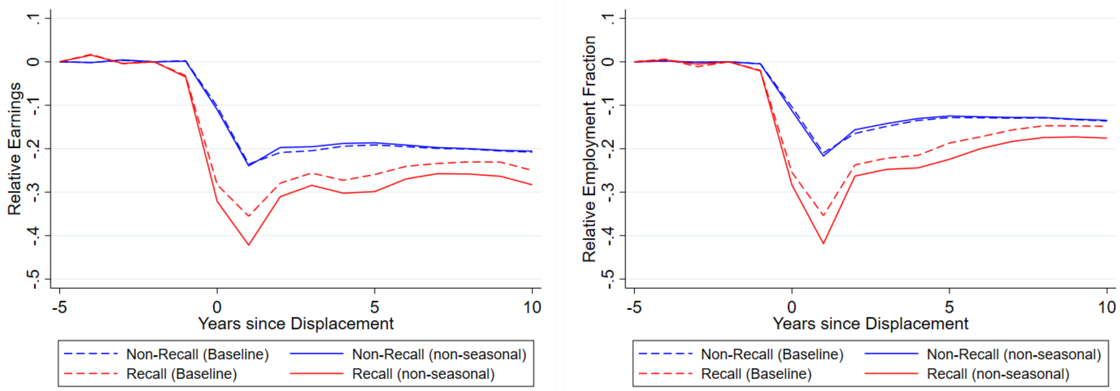


Figure C.12: *The effect of displacement on earnings (left) and employment (right) by ex-post recall status, relative to the control group, using estimated coefficients from equation 1'. The dashed lines mirror the estimate from the left panel of Figure 4, whereas the solid lines estimate the effect after excluding potentially seasonal workers, as measured by time to recall (or re-employment) and time to re-separation adding up to approximately a year.*

Next, I investigate the role of the recalling firm, by restricting the sample of displaced workers to workers who were displaced from an establishment that I observe recalling workers at any point in the sample. As can be seen by comparing the results in Figure C.13 with the baseline results in Figure 4, this slightly reduces the estimated earnings loss for non-recalled workers (while the estimated earnings loss for recalled workers naturally remains the same). As such, it can be concluded that the recalled workers suffer from larger earnings and employment losses even when restricting to workers displaced from recalling establishments.

[50]Note that the change in the results for earnings losses indicate that loss of firm-specific human capital may not play a large role in explaining earnings losses after displacement. After all, a large role for firm-specific human capital would imply that tenure is more important for non-recalled workers than for recalled workers (as recalled workers do not lose their firm-specific human capital), whereas Figure C.11 suggests that the opposite is true in the data.

[51]Following this definition, and as discussed in Appendix C.3.1, 26.9% of the recalls and 9.1% of the non-recall displacements in the estimation sample are marked as potentially seasonal and taken out of the estimation here.

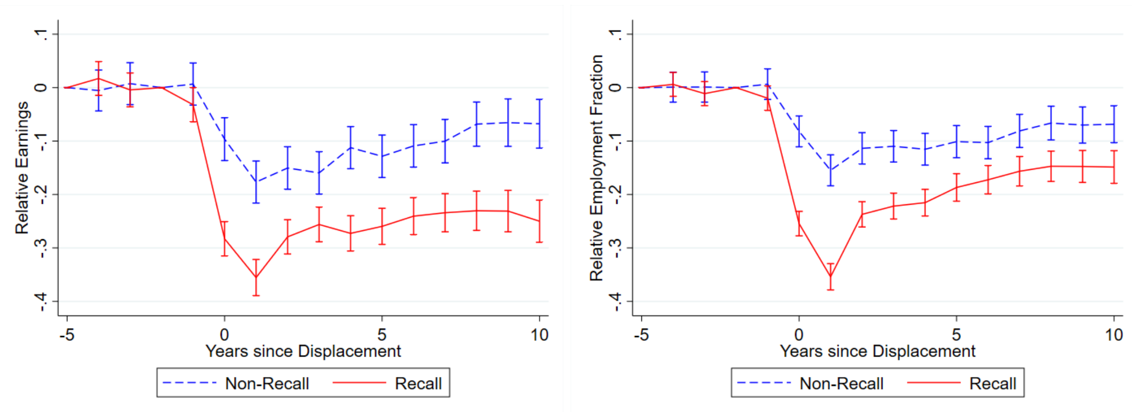


Figure C.13: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. The estimation displayed here includes displaced workers only if they were displaced from an establishment that recalled some workers.*

Another way to investigate the role of recalling firms is to replace the group of recalled workers with a group of non-displaced workers from an establishment that displaced and recalled some workers. As can be seen in the left panel of Figure C.14, these workers are revealed to be on a downward trend in subsequent years, thus supporting the interpretation of these workers working at an unstable establishment. If I restrict observations for both non-recalled and these "pseudo-recalled" workers to those made at the first employing establishment after (pseudo-)displacement, as I do in the right panel of Figure C.14, this downward trend largely remains, suggesting that the downward trend is not only driven by workers separating from their job, but also by earnings in the establishment decreasing (relative to the control group). Notably, implementing this restriction removes the recovery pattern among non-recalled workers.

Another potential concern with the comparison between recalled and non-recalled workers is that the group of non-recalled workers also includes workers that were displaced from an establishment that closed, thus leaving no room for a recall choice. However, the left panel of Figure C.15 shows that omitting workers displaced from a closing establishment does not affect the results highlighted in the main text. Similarly, removing indirect recalls from the sample of non-recalled workers does not substantially alter the results, as the right panel shows.

Finally, I noted in the previous subsection that while recalled workers in the regression sample have a pre-displacement establishment tenure of at least 6 years, it is not necessarily the

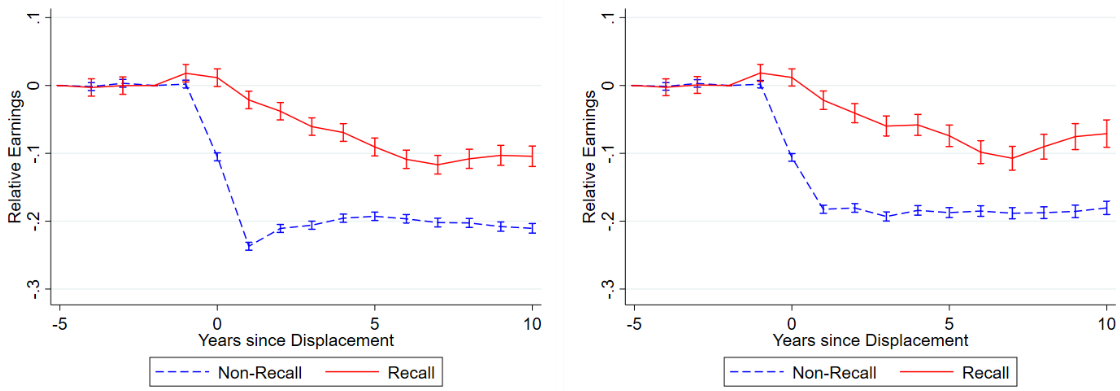


Figure C.14: *The effect of displacement on earnings, by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. The estimation here replaces recalled workers with workers who were not displaced from the establishment that recalled some workers. The right panel additionally only uses observations at the first employing establishment after displacement.*

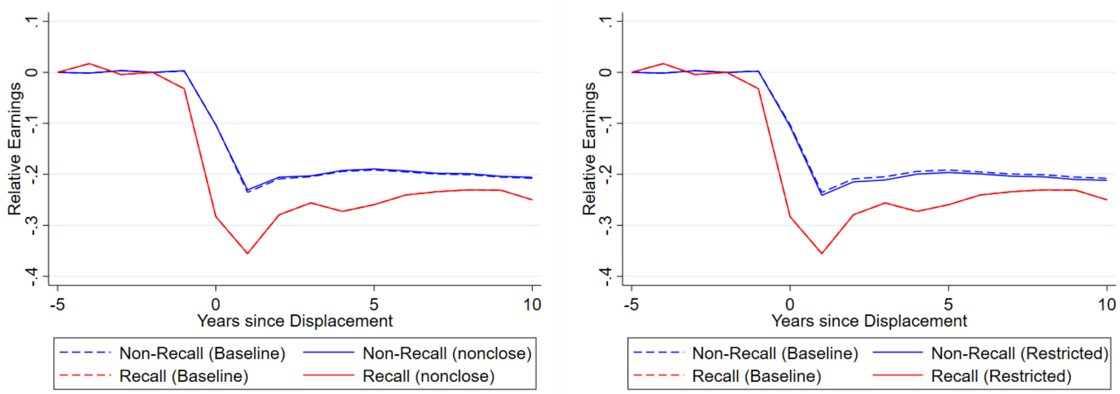


Figure C.15: *The effect of displacement on earnings, by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. Compared to Figure 4 in the main text (dashed lines here), the estimation excludes workers who were displaced from a closing establishment (left) or non-recalled workers who are eventually recalled (right).*

case that they were employed at that establishment continuously for the past 6 years, as the establishment tenure does not reset after a separation and subsequent recall. In Figure C.16, I show that conditioning on workers not having separated in the two years prior to displacement does not substantially change the results. If anything, the difference between the recalled and non-recalled workers' earnings and employment losses grows when making this restriction.

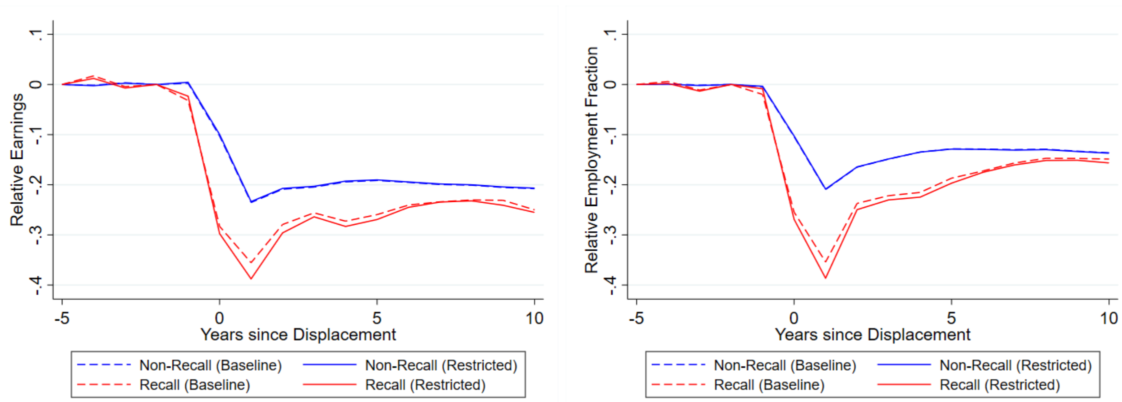


Figure C.16: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. Compared to Figure 4 in the main text (dashed lines here), the estimation excludes workers who were separated in either of the two years prior to displacement.*

C.3.4 Recalled Workers' Displacement Scars: Further Heterogeneity

In Section 2.3, I showed how the scarring effect of displacement for workers who are recalled to their previous employer compares to that of workers who are not recalled, focusing in particular on their earnings, employment status, and subsequent separation rates. In this section, I additionally highlight how the post-displacement experiences of recalled and non-recalled workers differ in a number of other dimensions. Furthermore, I show how the main results change when considering particular subsamples based on the dimensions I highlighted (in Table 1) when discussing how different recalled and non-recalled workers are.

The left panel of Figure C.17 shows how the effect of displacement on daily wages is different for recalled and non-recalled workers. As noted in the main text, it can be seen that recalled workers do better in terms of wages in the short run (1 to 2 years after displacement), compared to non-recalled workers, but this difference is not statistically significant (at the 5% level) and it is very short-lived. Generally, the evolution of the wage loss for recalled workers is

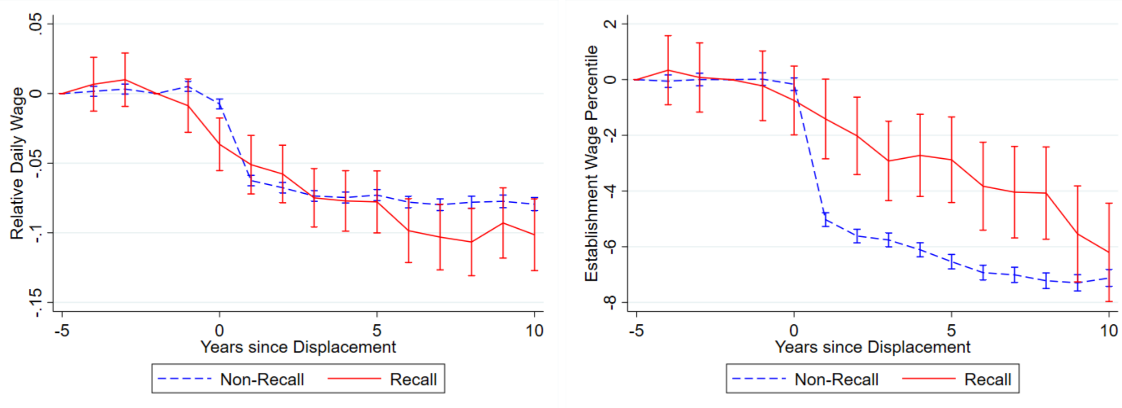


Figure C.17: *The effect of displacement on daily wages (left) and establishment median wage (percentile, right) by ex-post recall status, relative to the control group, using estimated coefficients from equation 1'. The left panel uses a sample that excludes the top and bottom 5% of the daily wages. The error bars correspond to 95% pointwise confidence intervals.*

quite gradual, whereas the non-recalled workers experience a sharp decrease upon impact. This can also be seen when examining the median wage of the workers' employing establishment, as done in the right panel of Figure C.17. Non-recalled workers move about 5 percentage points down the establishment median wage distribution upon impact, whereas recalled workers accumulate this loss over time.[52]

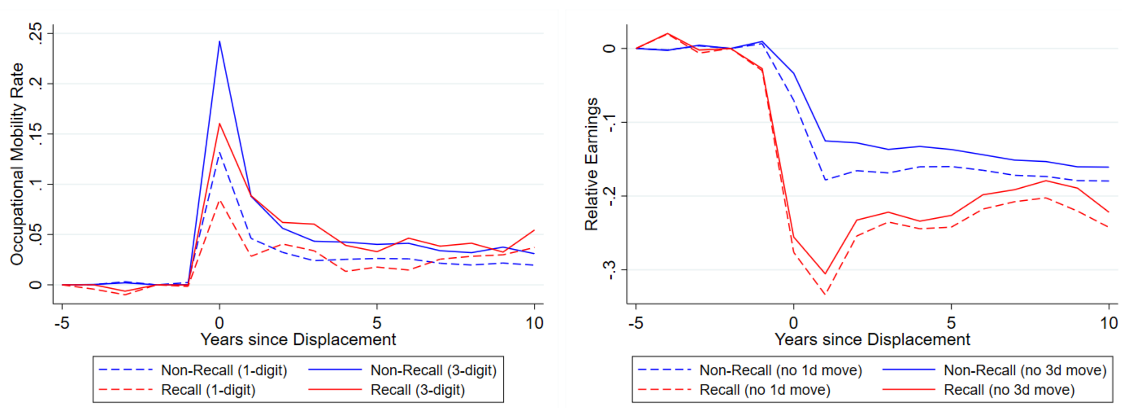


Figure C.18: *The effect of displacement on 1-digit and 3-digit occupational mobility (left) and earnings (right) by ex-post recall status, relative to the control group of never-displaced workers, using estimated coefficients from equation* (1'). *The dashed lines in the right panel estimate the effect on earnings excluding workers who switched 1-digit occupations, whereas the solid lines estimate the effect exclude workers who switched 3-digit occupations.*

[52]The establishment median wage distribution is constructed yearly among all establishments present in the data.

In the left panel of Figure C.18, I show how displacement differentially affects the yearly occupational mobility rates of (ex-post) recalled and non-recalled workers. In line with the descriptive evidence from Section 2.1, the occupational mobility rate upon impact is higher for non-recalled workers than for recalled workers, and this holds for the 1-digit as well as the 3-digit level. At the same time, the effect on the recalled workers' occupational mobility rate is also clearly different from zero, which indicates that workers may return to a different job if they return to their previous establishment (or re-separate and switch occupations within a year). Furthermore, the difference is only visible upon impact and disappears after a year. Importantly, this margin does not affect my conclusion that recalled workers tend to suffer from larger earnings and employment losses. In fact, conditioning on workers staying in the same 1-digit or 3-digit occupation widens the estimated difference in earnings losses between recalled and non-recalled workers, as shown in the right panel of Figure C.18. This is in line with the evidence from Huckfeldt (2022), who shows that occupational switchers tend to suffer from larger earnings losses than stayers (in the US). Given the larger occupational mobility rate among non-recalled workers, conditioning on occupational stayers then alleviates the earnings losses more for non-recalled workers than it does for recalled workers.

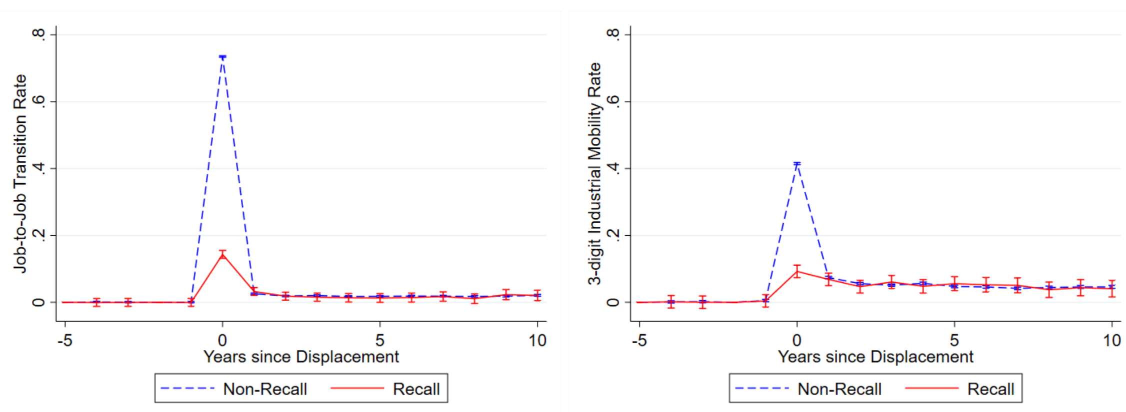


Figure C.19: *The effect of displacement on job-to-job transition rates (left) and 3-digit industry mobility (right) by ex-post recall status, relative to the control group of never-displaced workers, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals.*

In Figure C.19, I show that differences between (ex-post) recalled and non-recalled workers in terms of other transition rates are similarly short-lived. Non-recalled workers have a much larger (increase in) probability of making a job-to-job transition or switching 3-digit industries than recalled workers, but this difference only appears in the year of displacement.[53] Note that in Figures C.18 and C.19 I include nonemployed workers in the sample even though their transition indicator is zero by definition. Excluding these observations from the sample does not substantially alter the results.

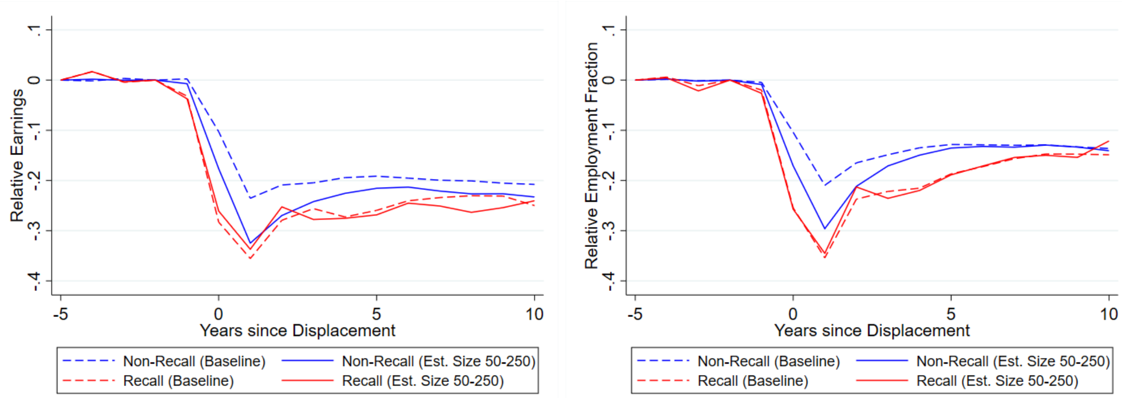


Figure C.20: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. Compared to Figure 4 in the main text (dashed lines here), the estimation excludes workers who were displaced from an establishment with more than 250 workers.*

In Section 2.1, I showed that recalled workers in the full sample tend to come from a smaller establishment and tend to have lower pre-displacement establishment tenure. In the regression estimation sample, where I impose a minimum establishment size and pre-displacement tenure, it remains true that the establishment size among non-recalled workers is double that of the recalled workers, whereas their pre-displacement establishment tenure remains roughly 16% higher (although neither difference is statistically significant). If one expects the earnings losses incurred after a job loss to be decreasing in establishment size or in tenure, this may lead to a concern that the results from Section 2.3 may be explained by this dimension. In Figure C.20 I show that losses are indeed larger for workers displaced from smaller establishments, but this only holds for non-recalled workers whereas the losses for recalled workers are comparable to the baseline. It is still true that recalled workers experience larger losses, but the difference is much smaller (and in some cases no longer statistically significant). For workers with lower pre-displacement tenure,

[53] In the estimation underlying the left panel of Figure C.19, I use a narrow definition of a job-to-job transition, requiring workers to transition within 5 days. Using the broader definition, which allows for 30 days (in line with Figure 6), yields very similar results.

Figure C.21 shows that the results goes in the opposite direction, as estimated losses are smaller for these workers than in the baseline (though more so for the recalled workers).

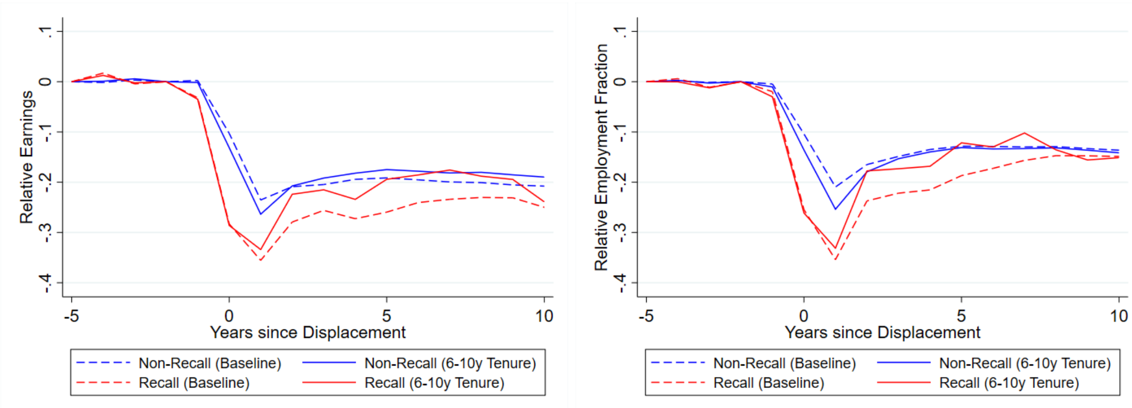


Figure C.21: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. Compared to Figure 4 in the main text (dashed lines here), the estimation excludes workers who had more than 10 years of pre-displacement establishment tenure.*

Table 1 (and Figure C.1) also shows that recalled workers tend to have lower earnings prior to displacement (and this continues to hold in the regression sample). Since it has been shown that relative earnings losses tend to be larger for workers with lower pre-displacement recent earnings, by Guvenen et al. (2017) in the US and Leenders (2025) in Germany, this could partially explain my results from Section 2.3. Indeed, Figure C.22 shows that conditioning on low (below the median) pre-displacement recent earnings, defined following Guvenen et al. (2017) as discussed in Appendix C.3.1, yields much smaller differences in earnings and employment losses than conditioning on pre-displacement recent earnings above the median. In fact, while earnings losses for non-recalled workers are decreasing in pre-displacement earnings, the opposite holds for recalled workers. This may point to the recall being a option of last resort for workers. Hypothetically, this could be explained by recalled workers coming back in a low rank in the establishment, which would be especially impactful for workers who previously held a higher-rank position (and therefore had higher earnings).

Appendix References

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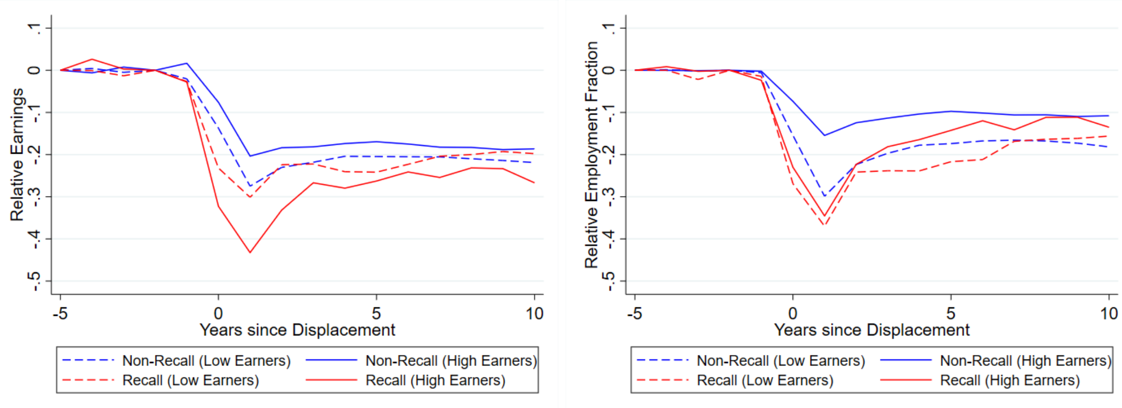

Figure C.22: *The effect of displacement on earnings (left) and employment fraction (right), by ex-post recall status and relative to the control group, using estimated coefficients from equation* (1'). *The error bars correspond to 95% pointwise confidence intervals. Compared to Figure 4 in the main text, the estimation only uses workers who had pre-displacement recent earnings below the median (dashed) or above the median (solid).*

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